

MONITORING REPORT

CAIARARA REDD+ PROJECT

Project title	Caiarara REDD+ Project
Project ID	5390
Crediting period	January 21, 2024 to January 20, 2064
Monitoring period	January 21, 2024 to June 30, 2025
(CCB) GHG accounting period	January 21, 2024 to January 20, 2064 – 40 years
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Version	01
VCS Standard Version	VCS Standard v4.7
CCB Standards Version	CCB Standard v3.1
Project location	Brazil, state of Pará, municipality of Breu Branco.
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CCB status history	First verification attempt.

Gold Level Criteria	<u>Gold Level Climate:</u> to assist communities and biodiversity in adapting to climate change, the Caiarara REDD+ Project has long-term planning. During this monitoring period, the project hired a new surveillance company to improve monitoring and activity recording processes to protect forest cover, carried out firebreaks maintenance, conducted community education on water and health, trained workers in firefighting, and conducted socioeconomic and biodiversity assessments. The project contributes to maintaining the local climate, minimizing global climate change, maintaining habitat connectivity, especially considering a larger spatial scale such as in the Project Zone, and protecting ecosystem services such as pollination, seed dispersal, gene flow, soil protection, and hydrographic basins. <u>Gold Level Biodiversity:</u> Four species classified as threatened according to the IUCN Red List were identified in the Caiarara REDD+ Project Area: Macaco-caiarara (<i>Cebus kaapori</i>) and Acapu (<i>Vouacapoua americana</i>) in the "critically endangered" (CR) category; Louro-canela (<i>Ocotea fragrantissima</i> Ducke) and Maçaranduba (<i>Manilkara elata</i>) in the "endangered" (EN) category. During this period, the project carried out diagnostics of the fauna and flora, initiating continuous monitoring of these species to track individuals and assist in the growth of their populations, and preserved natural habitats by reducing greenhouse gas emissions from deforestation and forest degradation, preventing extinction.
Prepared by	Biofilica Ambipar Environmental Investments S/A

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1 SUMMARY OF PROJECT BENEFITS

1.1 Unique benefits of the project

Result or Impact	Results obtained during the monitoring period	Section reference	Achievements during the project's lifetime
1) Climate: mitigation of climate change through the reduction of greenhouse gas emissions by preventing unplanned deforestation.	Total emissions avoided during the monitoring period: 199,208 tCO ₂ e Total deforestation avoided within the project area: 323 hectares	3	Total emissions avoided during the project's lifetime: 199,208 tCO ₂ e Total deforestation avoided within the project area: 323 hectares
2) Community and workers: promotion of sustainable development activities for stakeholders, seeking to improve their well-being and education. In addition, to improve communication, the project promotes activities that strengthen local governance among the parties.	Opening of communication channels between stakeholders and proponents. Start of planning sustainable development activities through meetings with the community and internal definition of activities and strategies.	4	As this is the first verification of the project, there are no figures to report yet. The achievements are the same as those presented in this report.
3) Biodiversity: protection, conservation, and monitoring of fauna and flora species and their habitats, including species with some degree of threat and endemic species, in addition to habitats considered threatened.	Diagnosis of fauna and flora conducted in the area. Number of species identified in the project area: - Avifauna: 124 species. - Mastofauna fauna: 24 species. - Herpetofauna: 23 species. - Flora: 172 species. Number of fauna species with some degree of threat and endemic species monitored: - With some degree of threat (IUCN): 10 species.	5	Number of species identified in the project area: - Avifauna: 124 species. - Mastofauna fauna: 24 species. - Herpetofauna: 23 species. - Flora: 172 species. Number of fauna species with some degree of threat and endemic species monitored:

	- Endemic: 2 species (Belém Endemism Center). Number of flora species with some degree of threat and endemic species monitored: - With some degree of threat (IUCN): 7 species. - Endemic: 1 species (Pará).		- With some degree of threat (IUCN): 10 species. - Endemic: 2 species (Belém Endemism Center). Number of flora species with some degree of threat and endemic species monitored: - With some degree of threat (IUCN): 7 species. - Endemic: 1 species (Pará).
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1.2 Standardized benefit metrics

Category	Metric	Achievements during the monitoring period	Section reference	Achievements during the project's lifetime
GHG emission reductions and removals	Estimated net removals of emissions in the project area, measured relative to the no-project scenario	Not applicable	-	Not applicable
	Estimated net emissions reductions in the project area, measured relative to the no-project scenario	199,208 tCO ₂ eq	3	199,208 tCO ₂ eq
¹ forest cover	For REDD projects ² : number of hectares of reduced forest loss in the project area measured relative to the no-project scenario	323 hectares	3	323 hectares
	For ARR projects ³ : Number of hectares of forest cover increased in the project area measured relative to the no-project scenario	Not applicable	-	Not applicable
Better land management	⁴ Number of hectares of existing production forest land where IFM practices occurred as a result of project activities, measured relative to the no-project scenario	Not applicable	-	Not applicable

¹ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO, or IPCC) of what constitutes a forest, which includes threshold parameters such as minimum forest area, tree height, and canopy cover, and may include mature, secondary, degraded, and wet forests (*VCS Program Definitions*)

² Reducing Emissions from Deforestation and Forest Degradation (REDD) – Activities that reduce GHG emissions by slowing or halting the conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (*VCS Program Definitions*)

³ Afforestation, reforestation, and revegetation (ARR) – Activities that increase carbon stocks in woody biomass (and, in some cases, soils) by establishing, increasing, and/or restoring vegetation cover through planting, seeding, and/or human-assisted natural regeneration of woody vegetation (*VCS Program Definitions*)

⁴ Improved forest management (IFM) – Activities that change forest management practices and increase carbon stocks in forest lands managed for wood products, such as lumber, pulpwood, and firewood (*VCS Program Definitions*)

	Number of hectares of non-forest land where improved land management has occurred as a result of project activities, measured relative to the no-project scenario	Not applicable	-	Not applicable
Training	Total number of community members who improved their skills and/or knowledge as a result of training provided as part of project activities	114 people	4	114 people
	Number of female community members who improved their skills and/or knowledge as a result of training provided as part of project activities	42 women	4	42 women
Employment	Total number of people employed in project activities, ⁵ expressed as number of full-time employees ⁶	Not carried out during this monitoring period.	4	Since this is the first monitoring period and there were no activities for this topic, there is no data to be disclosed.
	Number of women employed in project activities, expressed as the number of full-time employees	Not carried out during this monitoring period.	4	Since this is the first monitoring period and there were no activities for this topic, there is no data to be disclosed.
Livelihood	Total number of people with improved livelihoods ⁷ or income generated as a result of project activities	Not carried out during this monitoring period.	4	Since this is the first monitoring period and there were no activities for this theme, there is no data to be disclosed.

⁵ Employed in project activities means people who work directly on project activities in exchange for compensation (financial or otherwise), including employees, contract workers, subcontracted workers, and community members who are paid to perform project-related work.

⁶ Full-time equivalence is calculated as the total number of hours worked (by full-time, part-time, temporary, and/or seasonal personnel) divided by the average number of hours worked in full-time jobs in the country, region, or economic territory (adapted from paragraphs 17.14 [15.102] of the United Nations System of National Accounts (1993); [17.28])

⁷ Livelihoods are the capabilities, assets (including material and social resources), and activities necessary for a means of living (Krantz, Lasse, 2001. *The Sustainable Livelihoods Approach to Poverty Reduction*. SIDA). Livelihood benefits may include benefits reported in the employment metrics in this table.

	Number of women with improved livelihoods or income generated as a result of project activities	Not carried out during this monitoring period.	4	Since this is the first monitoring period and there were no activities for this theme, there is no data to be disclosed.
Health	Total number of people for whom health services have been improved as a result of project activities, measured in relation to the scenario without the project	Not applicable	-	Not applicable
	Number of women for whom health services were improved as a result of project activities, measured relative to the scenario without the project	Not applicable	-	Not applicable
Education	Total number of people for whom access to or quality of education has improved as a result of project activities, measured relative to the scenario without the project	Not applicable	-	Not applicable
	Number of women and girls for whom access to or quality of education has been improved as a result of project activities, measured relative to the scenario without the project	Not applicable	-	Not applicable
Water	Total number of people who experienced improved water quality and/or better access to drinking water as a result of project activities, measured relative to the scenario without the project	Not applicable	-	Not applicable
	Number of women who experienced improved water quality and/or better access to drinking water as a result of project activities, measured relative to the scenario without the project	Not applicable	-	Not applicable

Well-being	Total number of community members whose well-being has improved as a result of project activities	60	-	60
	Number of women whose well-being has improved as a result of project activities	40	-	40
Biodiversity conservation	Change in the number of hectares significantly better managed by the project for biodiversity conservation, ⁹ measured relative to the scenario without the project	32,625 ha	3	32,625 ha
	Number of globally critically endangered or endangered species ¹⁰ that benefit from reduced threats as a result of project activities, ¹¹ measured relative to the scenario without the project	4 species	5	4 species

2 PROJECT DETAILS

2.1 Summary description of project implementation status

2.1.1 Summary description of the project (VCS, 2.1, 3.6; CCB, G1.2)

⁸ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics in this table (e.g., Training, Employment, Health, Education, Water, etc.), but may also include other benefits such as empowerment of community groups, strengthened legal rights to resources, conservation of access to areas of cultural importance, etc.

⁹ Biodiversity conservation, in this context, means areas where specific management measures are being implemented as part of project activities with the aim of improving biodiversity conservation.

¹⁰ According to the IUCN Red List of Threatened Species

¹¹ In the absence of direct population or occupancy measures, the measurement of reduced threats can be used as evidence of benefit

The Caiarara REDD+ Project is the responsibility of Biofilica Ambipar Environmental Investments S/A and Dow Brasil and began on January 21, 2024, with the hiring of a new surveillance company to improve the monitoring and recording of activities to strengthen the property security of farms located in the municipality of Breu Branco, state of Pará, where the project is located. The activities were carried out simultaneously, from the project design process to the first verification, and some of the activities mentioned in this report will continue throughout its duration.

During the monitoring period, activities covered several actions related to the design of the project, including the signing of the contract between the proponents and the preparation of the project scope. In addition, initial studies were carried out, such as fauna and flora diagnostics, socioeconomic diagnostics, carbon stock diagnostics, and the creation of a baseline, in addition to activities to reduce greenhouse gas (GHG) emissions by containing unplanned deforestation. To this end, forest cover was monitored using satellite imagery, together with property surveillance actions.

The main activities of the Project implemented in the first monitoring period are listed below, with relevant dates.

1. Conducting biodiversity and socioeconomic diagnostics – December 2023 to May 2024.
2. Diagnosis of carbon stock estimates in the project area – December 2023 to May 2024.
3. Improvement of property surveillance, prevention of unplanned deforestation, and monitoring of forest cover – started in January 2024 and will continue throughout the duration of the Project.
4. Feedback event with stakeholders – July 2024.
5. Meeting with the community of Vila Mamorana – August 2024
6. Opening of official communication channels with stakeholders and sending of the first official project communications – started in December 2024 and will continue throughout the duration of the Project.
7. Activity carried out with the community: lecture on water and health for students – held in March 2025.

2.1.2 Audit History (VCS, 4.1)

Audit type	Period	Program	Name of VVB	Number of years
Validation	January 21, 2024	VCS/CCB	Earthood Services Private Limited	-
Verification	January 21, 2024 – June 30, 2025	VCS/CCB	Earthood Services Private Limited	1.5

2.1.3 Sectoral scope and project type (VCS, 3.2)

Sectoral coverage	14: Agriculture, forestry, and other land use
Project Category AFOLU	Reduction of Emissions from Deforestation and Degradation (REDD)
Project Activity Type	Avoided Unplanned Deforestation (AUD)

2.1.4 Project proponent (VCS, 3.7; CCB, G1.1)

Organization Name	DOW BRASIL INDÚSTRIA E COMÉRCIO DE PRODUTOS QUÍMICOS LTDA
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Organization Name	BIOFILICA AMBIPAR ENVIRONMENTAL INVESTMENTS S/A
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2.1.5 Other entities involved in the project

Organization Name	STA Soluções Técnicas e Serviços de Engenharia Ambiental Ltda.
Role in the project	Responsible for socioeconomic diagnosis, flora diagnosis, and carbon stock inventory estimation

Contact	Raniery Branco
Title	Partner Owner
Address	Rua 28 de Setembro, 1226 – Reduto ZIP Code 66053-355, Belém/PA – Brazil
Phone	+55 91 3355-4408 & +55 91 9161-2698
Email	raniery@stambiental.com

Organization Name	Ambiens Resolve Inovação Sustentabilidade e Meio Ambiente Ltda.
Role in the project	Responsible for fauna diagnosis
Contact	Fábio Bárbara
Title	Partner Owner
Address	Rua Marquês de Sabará, nº 30, Conj. 31 - Real Parque ZIP Code 05.684-020, São Paulo/SP - Brazil
Phone	+55 (11) 97309-7100
Email	fabio@ambiens.com.br

2.1.6 Project start date (VCS, 3.8)

Project start date	January 21, 2024
Justification	The start date of the Caiarara REDD+ Project was set for January 21, 2024, represented by the hiring of a new property surveillance company with improved process of monitoring and recording of activities, ensuring greater traceability and control of the information collected. For example, the creation of "vulnerability checklists," which report non-conformities found during patrols, allowing for the referral of incidents and better management of the surveillance operation. Thus, improvements in the security of the Project Area are considered the first activity that led to the generation of GHG emission reductions, as established by the VCS Standard v4.7, since the improvement in surveillance of the area contributed to the actions to contain deforestation in the Caiarara REDD+ Project.

2.1.7 Project Benefit Assessment and Credit Period (VCS, 3.9; CCB, G1.9)

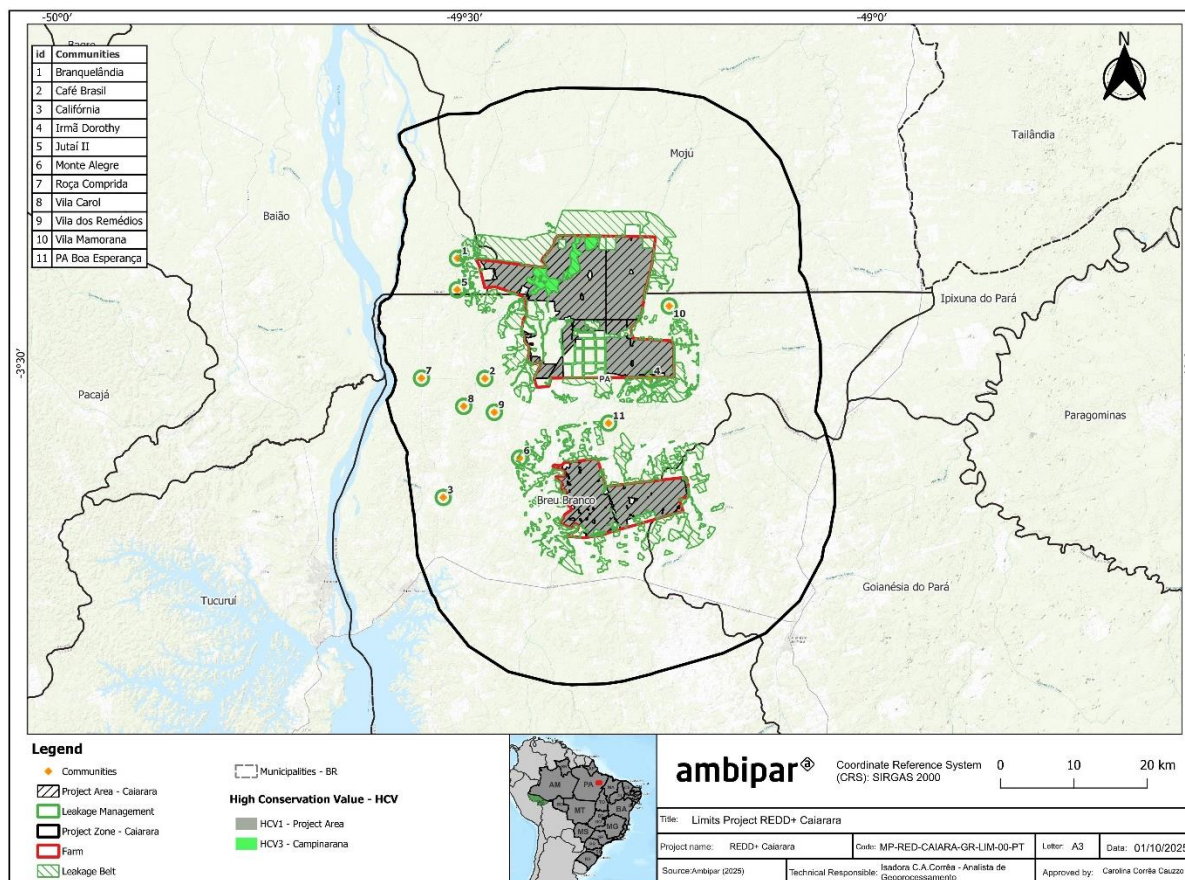
Credit Period	The credit period for the Caiarara REDD+ Project will be for the full 40-year period, considering the VCS v4.7 standard.
Start date of the first credit period or fixed credit period	January 21, 2024 to January 20, 2064
Total number of years of the credit period	40 years
CCB Benefit Assessment Period	There will be ongoing monitoring of benefits to the climate, communities, and biodiversity, subject to CCB verification, ideally every two years, throughout the duration of the Project.

2.1.8 Project Location (VCS, 3.11; CCB, G1.3)

Indicate the project location and geographic boundaries, including a separate KML file.

☒ KML file provided

Figure 1. Map of Project location.



2.1.9 Methodology Title and Reference (VCS, 3.1)

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Methodology	VM0048	VM0048 - Reduction of emissions from deforestation and forest degradation.	1.0
Tool	VT0001	Tool for demonstrating and evaluating additionality in agriculture, forestry, and other land use (AFOLU) project activities under the VCS - VT0001.	3.0
Tool	AFOLU NPRT	AFOLU Non-permanence risk tool	VERRA HUB
Tool	VMD0055	Estimation of Emission Reductions from Avoiding Unplanned Deforestation	1.1

2.1.10 Double Counting and Participation in Other GHG Programs (VCS, 3.23; CCB, G5.9)

2.1.10.1 Without double emission

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program, or any other form of community, social, or biodiversity unit or credit?

Yes ☐

No ☒

2.1.10.2 Enrollment in other GHG programs

Has the project been registered or is it seeking registration in any other GHG program?

Yes ☐

No ☒

2.1.10.3 Projects rejected by other GHG programs

Has the project been rejected by any other GHG program?

Yes ☐

No ☒

2.1.11 Double claim, other forms of credit, and Scope 3 emissions (VCS, 3.24)

2.1.11.1 No Double Claim with Emissions Trading Programs or Binding Emission Caps

Are the project's reductions and removals or project activities also included in an emissions trading program or binding emissions cap? See the VCS Program Definitions for definitions of emissions trading program and binding emissions cap.

☐ Yes

☒ No

2.1.11.2 No Double Claim with Other Forms of Environmental Credit

Has the project activity sought, received, or plan to receive credit from another GHG-related environmental credit system? See the VCS Program Definitions for the definition of GHG-related environmental credit system.

☐ Yes

☒ No

2.1.11.3 Supply chain emissions (scope 3)

Do the project activities affect the emissions footprint of any products (goods or services) that are part of a supply chain?

☐ Yes☒ No

2.1.12 Contributions to Sustainable Development (VCS, 3.17)

The activities of the Caiarara REDD+ Project aim to prevent unplanned deforestation and promote sustainable development in the region. As such, the activities implemented by the project are related to the protection of forests and biodiversity, as well as social activities involving local communities and farm workers to raise environmental awareness and support sustainable production alternatives, protecting the natural environment and bringing sustainable development to the region. The activities of this monitoring and their respective SDGs are described in the table below.

Table 1. Contributions to Sustainable Development.

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current contributions of the project	Contributions over the project's lifetime
1)	12.8	12.8.1 - Extent to which global citizenship education and education for sustainable development are integrated into national education policies; school curricula; teacher training; and student assessment	Implementation of activities to increase	The Caiarara REDD+ Project held a lecture on water and health, which provided information on the importance of drinking water consumption, water purification methods, and water protection against pollution. The activity was carried out with students, family members, and teachers from the Mamorana Community, and was an important action on citizenship and awareness.	The project carried out a total of one educational activity to ensure that people have relevant information and awareness for sustainable development and lifestyles in harmony with nature. The value is equal to 'current project contributions' as this is the first monitoring period.
2)	13.2	13.2.2 - Total greenhouse gas emissions per year	Implementation of activities to reduce	The Caiarara REDD+ Project reduced greenhouse gas emissions by 199,208 tCO ₂ e during this monitoring period.	The project has reduced a total of 199,208 tCO ₂ e of greenhouse gas emissions since the project began. The value is equal to 'current project contributions' as this is the first monitoring period.
3)	15.1	15.1.1 - Forest area as a proportion of the total area of the territory	Implementation of activities to reduce	The Caiarara REDD+ Project prevented the deforestation of 323 ha of Amazon rainforest during this monitoring period.	The project has prevented the deforestation of a total of 323 ha of Amazon rainforest since the project began. The value is equal to 'current project contributions' as this is the first monitoring period.

4)	15.5	15.5.1 - Red List Index	Implementation of activities to increase	Through the project's activities aiming to reduce the degradation of natural habitats and halt biodiversity loss, the Caiarara REDD+ Project has protected four species of native fauna and flora threatened with extinction, according to the Global Red List (IUCN).	Through the Project's activities aiming to reduce the degradation of natural habitats and halt biodiversity loss, the Caiarara REDD+ Project has protected four species of native fauna and flora threatened with extinction, according to the Global Red List (IUCN).
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2.2 Project implementation status

2.2.1 Implementation Schedule (VCS, 3.2; CCB, G1.9)

Date	Project implementation activities
October 2023	Contract between project proponents.
November 2023	Workshop with suppliers and proponents to develop the project design.
December 2023 to May 2024	Conducting socioeconomic, environmental, and carbon stock diagnoses.
January 21, 2024	Project Start Date (hiring of the new property surveillance company)
July 2024	Feedback event with stakeholders
August 2024	Meeting with the Vila Mamorana community
August 2024	Hiring of the Validation/Verification Body
December 2024	Consolidation of the Communication Plan and opening of the Project's official Communication Channels.
March 2025	Lecture on water and health with students from Vila Mamorana
December 2025	Field audit for validation and verification.
Beginning in 2024 until the last year of the Project	Development and monitoring of socio-environmental management activities. Continuous monitoring of deforestation and emissions, with annual data analysis and compilation.
Forecast every 2 years	GHG and CCB verifications
January 21, 2024, to January 20, 2064	GHG emissions accounting period

2.2.2 Baseline reassessment (VCS, 3.2.6, 3.2.7)

Was the project subject to baseline reassessment during the monitoring period?

Yes ☐

No ☒

2.2.3 Methodology deviations (VCS, 3.20)

There were no methodology deviations during the monitoring period referred to in this report.

2.2.4 Minor changes to the project description (CCB Program Regulation, 3.5.6)

There were no minor changes to the project description during the monitoring period covered by this report.

2.2.5 Deviations from project description (VCS, 3.21; CCB Program Regulations, 3.5.7 – 3.5.10)

There were no deviations from the Project Description during the monitoring period referred to in this report.

2.2.6 Grouped projects (VCS, 3.6; CCB, G1.13-G1.15, G4.1)

Not applicable, the project is not grouped.

2.2.6.1 New Project Areas and Communities (VCS, 3.6; CCB, G1.13)

Not applicable, the project is not grouped.

2.2.6.2 Removed Project Areas and Communities (CCB, G1.13)

Not applicable, the project is not grouped.

2.2.6.3 Eligibility criteria for grouped projects (VCS, 3.6.14; CCB, G1.14)

Not applicable, the project is not grouped.

2.2.6.4 Risk Mitigation for Grouped Projects (VCS, 3.6; CCB, G1.15)

Not applicable, the project is not grouped.

2.2.6.5 Project Zone Map (CCB, G1.13)

Not applicable, the project is not grouped.

2.2.6.6 Changes in Management (CCB, G4.1)

Not applicable, the project is not grouped.

2.2.7 Risks to the Project (CCB, G1.10)

Using the AFOLU Non-Permanence Risk Tool v4.2, the probable natural and human-induced risks to climate benefits reported in the VERRA Project Hub of the Caiarara REDD+ Project were verified. The Non-Permanence Risk analysis using the aforementioned tool generated a buffer of 17%.

In conjunction with the tool, probable risks to the expected benefits for the climate, community, and biodiversity during the life of the Project were identified, as well as their respective mitigation measures, as described in the table below.

Table 2. Risks to the project with their potential impacts and mitigation actions.

Identified risk	Potential impact of the risk on climate, community, and/or biodiversity benefits	Necessary and planned actions to mitigate the risk
Forest fires	Forest fires occur naturally but are more often induced by humans. Slash-and-burn activities are common among communities in the region. Forest fires in the region can spread beyond the project area and cause forest degradation. In addition, this practice pollutes the air, directly affecting the health of people in the region, as well as contributing to a drier and hotter climate, intensifying the impacts of climate change.	The properties involved in the project are equipped to respond quickly to a forest fire. Both properties have water trucks, employees trained in firefighting, and PPE for the activity. Fire brigade training is conducted annually for farm employees. In addition, there are also preventive activities to prevent forest fires from occurring, such as the construction of firebreaks at strategic points in the forest. To influence the reduction of slash-and-burn practices in the region, the project will focus on environmental education to raise awareness of the impacts on the region's biodiversity and the well-being of the community itself.
Drought	Communities in the region may suffer from a lack of water for agricultural irrigation due to decreased rainfall caused by longer periods of drought, influenced by climate change. The adversities faced by agriculture directly impact the well-being of communities that use agricultural products for subsistence and as a source of income.	The project will work in coordination with interested agencies and institutions to offer technical assistance and workshops to help farmers adapt their practices to climate change. In addition, all project activities aim to conserve the forest and its biodiversity, contributing to reduce the impacts of climate change.
Entry of third parties into the project area in search of extractive products or to carry out illegal hunting and fishing activities.	Illegal entry by community members to extract forest products can cause deforestation or forest degradation, raising the possibility of timber extraction. Entry may also impact the biodiversity of the project area, as illegal hunting and fishing are still	To prevent third parties from entering the project area, property surveillance will be reinforced. In addition, educational activities will be carried out for the communities surrounding the project with the aim of raising awareness among community members about the negative impacts

	common in the region, even with signs along the perimeter of the properties prohibiting the activity.	of deforestation and predatory hunting and fishing activities on the region's biodiversity.
Lack of interest from stakeholders in participating in Project activities	The lack of interest in participating in Project activities directly impacts the benefits that would be generated at the community and farm workers level. Furthermore, the lack of interest on the part of public agencies may compromise potential partnerships for the development of the Project's activities.	Strengthening and encouraging greater involvement of all stakeholders in decision-making processes related to the Project's activities, such as capacity building and technical assistance to encourage sustainable and adaptive agricultural production practices and models. Another important measure is the project's communication plan, which involves communication channels in line with the reality of the stakeholders and procedures for redressing complaints.
Failure in financial resource management	Difficulty in implementing the activities proposed by the Caiarara REDD+ Project, in addition to the difficulty of allocating resources for conservation and sustainable development in the region.	Biofilica Ambipar Environmental Investments S/A has a financial team responsible for managing project resources

The Caiarara REDD+ Project has an Adaptive Management Plan that focuses primarily on identifying, preventing, and/or mitigating risks based on the adaptability of the implementation of actions and activities, considering the monitoring of indicators, evaluation of lessons learned, and the incorporation of these into future decision-making. The document is justified by the need to evaluate the implementation of a project, considering actions and/or reactions to adversities and unforeseeable scenarios, and ensuring good management, quality of results, actions directly linked to reality, and financial health. In addition to the Adaptive Management Plan, the project has a Management, Financial, and Monitoring Plan, which aims to demonstrate that the Caiarara REDD+ Project has a solid governance structure, financial capacity, and effective monitoring mechanisms to ensure environmental and operational integrity throughout its lifetime.

2.2.8 Permanence of Benefits (CCB, G1.11)

During this monitoring period, the Caiarara REDD+ Project implemented concrete measures that reinforce the maintenance and improvement of climate, community, and biodiversity benefits, ensuring their permanence beyond the project's lifetime. Among the actions developed during the period are the strengthening of property surveillance, the opening of communication channels to strengthen relations with communities and workers, and social activities.

For **property surveillance** activities, the measures implemented include:

- Daily Patrols: Daily patrols have enabled constant monitoring of the perimeter, identifying and preventing potential invasions, deforestation, and other activities that could compromise environmental integrity.
- Fire Brigade Training: Training the local team in fire prevention and fighting reinforces rapid response to risk situations, ensuring the integrity of forest areas and the continuity of climate benefits.
- Firebreak Maintenance: Regular maintenance of firebreaks contributes to forest protection by preventing the spread of fire in the event of a fire, ensuring the integrity of forest areas and the continuity of climate benefits.
- Improvements in the process of monitoring and recording activities: the new surveillance company hired has introduced new processes for monitoring and recording activities carried out, with the creation of "vulnerability checklists" that report any non-conformities found during patrols, allowing incidents to be forwarded and enabling greater control and traceability of information.

The **opening of communication channels** represents the beginning of the implementation of the communication plan, enabling a continuous flow of information between stakeholders and the project team. This integration is essential for resolving environmental and social issues, ensuring that project activities and stages are properly disclosed and that conservation and development initiatives remain aligned with the needs and expectations of the community. In addition, meetings between stakeholders and the project team were held to strengthen the relationship between the parties, such as the feedback event with communities and workers and the meeting with Vila Mamorana (see Section 2.3.9 for more details), where information about the project was shared and stakeholder demands were collected.

Finally, a **lecture** was held **with the community**, kicking off the project's social activities. The lecture on water and health was held with students, family members, and teachers from the Paraíso Municipal School of Early Childhood and Elementary Education, in the Mamorana Community, and aimed to inform about the importance of drinking water consumption, water cleaning methods, and water protection against pollution, which is an important action on health, citizenship, and awareness, promoting improvements in the well-being of communities.

It is worth noting that these measures represent the initial activities implemented by the project, which have already shown positive results in maintaining and improving climate, biodiversity, and community benefits. However, REDD+ Caiaara has a wide range of initiatives planned for subsequent periods, which will complement the actions already carried out. Throughout the project's duration, new activities will be implemented to reinforce environmental conservation, strengthen local governance, and promote sustainable development, ensuring that the gains achieved so far are expanded and consolidated, perpetuating long-term benefits.

2.3 Stakeholder engagement and safeguards

2.3.1 Identification of stakeholders (VCS, 3.18, 3.19; CCB, G1.5)

There have been no changes in stakeholder composition since project validation, as validation and first verification were conducted in the same period. Stakeholders are described below.

2.3.1.1 Communities and population centers

The communities or population centers were identified based on a Socioeconomic Diagnosis conducted by STA during the project validation. In total, 11 communities were identified as stakeholders in the Caiarara REDD+ Project, as shown in the table below.

Table 3. Communities identified as stakeholders in the Caiarara REDD+ Project.

No.	Communities	No.	Communities
1	Branquelândia	7	PA Boa Esperança ⁽²⁾
2	Café Brasil	8	Roça Comprida
3	California	9	Vila Carol ⁽³⁾
4	Irmã Dorothy ⁽¹⁾	10	Vila dos Remédios
5	Jutaí II	11	Vila Monte Alegre
6	Mamorana		

Having the surrounding communities as stakeholders makes it possible to mitigate risks to the protection of the area, especially in relation to deforestation and hunting, activities that have historically been present in the region. Thus, it is possible to create more effective conservation strategies, aligning the interests of the communities with the success of the Project. Integrating these communities into the planning and execution processes not only strengthens forest protection but also reduces the likelihood that they will become agents of deforestation, promoting sustainable coexistence and generating direct benefits for local populations.

2.3.1.2 Employees and third parties on the property

The proponent, Dow Brasil, carries out charcoal production and forestry activities on the property where this project is located, outside the project area. In addition to Dow Brasil's own employees, there are 15 outsourced companies that provide services in the development of these productive activities. In total, there are 361 workers.

Therefore, the proponent's direct employees and outsourced workers were considered stakeholders. The following table shows the list of companies and number of employees considered.

Table 4. Number of employees and outsourced workers considered as stakeholders.

Company	Total number of workers per company
Dow	71
Outsourced	290
Total	361

Farm workers were considered stakeholders in the REDD+ project because approximately 50% of them live in communities surrounding the project area. The project can bring benefits that positively impact on their daily work, such as improved well-being and training of interest. In addition, by being included as

stakeholders, these communities and workers can engage more consciously in forest preservation, contributing to local sustainability and harmonious coexistence between agricultural activities and environmental conservation.

2.3.1.3 Public agencies and other actors

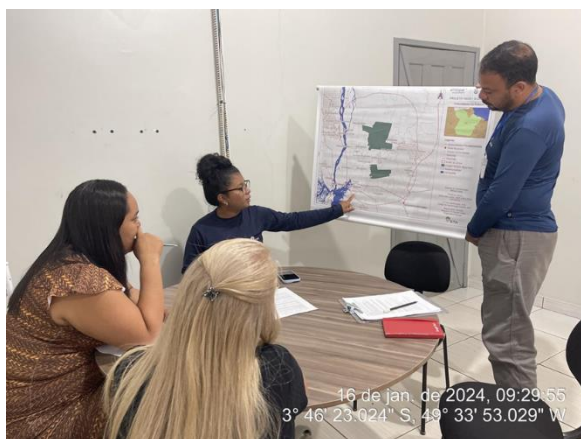
Among the public agencies, agencies in the municipality of Breu Branco were visited and interviewed, including the Municipal Secretariat of Agriculture, Fisheries, and Economic Development - Agriculture Sector (SEMAPEC), the Municipal Secretariat for Social Assistance and Development (SEMADS), the Municipal Secretariat for the Environment and Sanitation (SEMASA), the Municipal Secretariat for Education (SEMED), and the Municipal Secretariat for Health (SMS). In addition to the Union of Rural Workers and Family Farmers of Breu Branco (STRAAFBB).

As for state public agencies with offices in Breu Branco, visits were made to the Agricultural Defense Agency of the State of Pará (Adepará) and the Technical Assistance and Rural Extension Company of the State of Pará (Emater).

Figure 2. Right: visit to the Municipal Health Secretariat, Breu Branco, January 15, 2024. Left: visit to the Municipal Secretariat for the Environment and Sanitation, Breu Branco, January 17, 2024. Source: Socioeconomic diagnosis by STA.



Figure 3. On the right, Municipal Secretariat of Social Assistance and Development, Breu Branco, January 17, 2024. On the left, visit to the Breu Branco Rural Workers' Union, December 16, 2024. Source: Socioeconomic diagnosis by STA.



Public agencies were identified with a view to expanding the collection of information on the local reality, especially on the communities and productive, economic, and social activities in the vicinity of the project area.

2.3.2 Stakeholder access to project documents (VCS, 3.18, 3.19; CCB, G3.1)

To ensure continuous communication, consultation, availability of documents, and dissemination of information between proponents and stakeholders, the Caiarara REDD+ Project Communication Plan was established, ensuring that contributions are considered to adapt project management in an efficient and participatory manner throughout its implementation (more details on the Communication Plan in Section 2.3.10).

Access to the Caiarara REDD+ Project documents by stakeholders was provided through three main means: virtual, written, and oral, following the validated Communication Plan. The main objective was to ensure access to relevant documents and information related to the Project by all stakeholders.

Virtual communication: the project design document (PDD) and/or links to access it, relevant information to stakeholders, monitoring reports, and other pertinent documents are available virtually on the websites of Biofilica Ambipar Environmental Investments S/A and Verra. News and updates about the Project were also posted on Biofilica Ambipar's social media (LinkedIn: <https://www.linkedin.com/company/ambiparenvironment-br/mycompany/verification/> and Instagram: <https://www.instagram.com/ambiparenvironment.br/>), as well as via WhatsApp to stakeholders.

Written communication: a complete printed version of the Project Design Document (PDD), Monitoring Report, and other relevant documents was made available at the headquarters of the Reflorestamento Água Azul I (RAAI) and Reflorestamento Água Azul II (RAAI) farms for consultation by all stakeholders.

Oral/face-to-face communication: information, news, and updates on the Project were shared through meetings, events, and other face-to-face encounters with interested parties during the course of the Project activities.

The complete project documents will be updated in the three aforementioned media as new documents and new versions of these documents are generated throughout the project cycle, and summarized versions in accessible language will be made available to all interested parties.

In addition, the 30-day public consultation period on Verra was widely publicized (see Section 2.3.7 and 2.3.8) to stakeholders, who have access to the Project Design Document (PDD), where they can submit comments, criticisms, and contributions to the Project. The proponents collect, organize, and respond to the contributions received during the period, incorporating them where appropriate.

2.3.3 Disclosure of the Project's Summarized Documents (VCS, 3.18, 3.19; CCB, G3.1)

As described in the previous section, the summarized documents were disclosed and made available in the same way as the complete Project documents: summaries of the Project Design Document (PDD), Monitoring Report, and other important documents of the Caiarara REDD+ Project were made available and disclosed to all interested parties **virtually** via WhatsApp; and **in writing** through printed summaries at the farm headquarters.

Summary information on monitoring results and Project news was reported **virtually** via WhatsApp, orally through meetings, lectures, and other face-to-face meetings with stakeholders, and through banners displayed in high-traffic areas for employees and communities, in appropriate language to ensure accessibility for audiences with low levels of education or language barriers.

2.3.4 Informative meetings with stakeholders (VCS, 3.18, 3.19; CCB, G3.1)

Meetings, gatherings, and workshops between the proponents of the Caiarara REDD+ Project and stakeholders to discuss carbon project initiatives will take place throughout all phases of the design, development, implementation, and monitoring of activities.

Whenever necessary, informational meetings will be held between the project proponents and local actors, depending on local demand and availability. Information on the location and date of the meetings will be disclosed in advance through the communication channels established by the project (Section 2.3.11), which includes messages via WhatsApp (validated by the communities and workers as the preferred means of communication), the project's institutional email, and in-person visits to announce the meetings in the case of communities with limited internet access.

The informational meetings will always be conducted in accessible language, considering possible language or literacy barriers, in order to reach all stakeholders. Suggestions and criticisms received from stakeholders during the meetings will be recorded in a unified spreadsheet, analyzed, and responded to within defined deadlines depending on the complexity of each issue. The project's communications team will be responsible for managing these demands, with support from specialists in relevant areas, if necessary. When relevant and appropriate, the contributions received will be incorporated into the

project's methodologies and activities (see Section 2.3.12**Erro! A origem da referência não foi encontrada.** for more details).

The first contact between stakeholders and the Caiarara REDD+ Project initiatives was during the Socioeconomic Diagnosis conducted in the region by the company STA in early 2024. At that time, it was possible to explain the general concepts of a carbon project, provide specific information about the Caiarara REDD+ Project, and establish communication with communities and workers. More information on communication, planning, processes, and results is described in Section 3.2.3.

Furthermore, between July 11 and 14, 2024, the Biofilica Ambipar Environmental Investments S/A team, together with STA, held a feedback event with stakeholders of the Caiarara REDD+ Project. The event aimed to present the results of the socioeconomic diagnosis and provide information on the stages and initiatives of the Caiarara REDD+ Project. In addition, a group questionnaire was conducted with participants to ask about their preferred communication channels and collect additional information for the socioeconomic diagnosis for workers and surrounding communities.

There was a setback with the Boa Esperança Settlement, and the meeting could not be held in person, but an online meeting was organized on August 2, 2024. The table below contains information on each meeting held, including the date, stakeholders, and number of participants.

Community/ Institution	Date/Time	Number of participants
Palmyra Workers	July 11 at 8:30 a.m.	17
Palmyra workers	July 11 at 2:30 p.m.	53
Palmyra Workers	July 12 at 2:30 p.m.	23
Monte Alegre	July 13 at 8:00 a.m.	0
California		
Vila Carol		
Roça Comprida	July 13 at 11 a.m.	10
Vila dos Remédios		
Café Brasil		
Branquelândia	July 13 at 4 p.m.	9
Jutaí II		
Vila Mamorana	July 14 at 9 a.m.	11
Irmã Dorothy		
PA Boa Esperança	August 2 at 2 p.m.	3

Finally, on August 19, 2024, a meeting was held with the leaders of Vila Mamorana, the community that will be the focus of the project in its early years, to consult on the community's demands for the REDD+ Project.

2.3.5 Project Risks and No Net Harm (VCS, 3.18, 3.19)

	Identified risk(s)	Potential impact of the risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventive measures taken
Natural and human-induced risks to stakeholder well-being	No risks identified	-	The project activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. Therefore, no natural or human-induced risks related to the implementation of the project's activities were identified.
Risks to stakeholder participation	Low community involvement in the project's activities	Communities may not feel identified with and/or interested in the proposals and will not participate in the actions.	Consolidation of mechanisms that promote transparency and encourage the participation of all parties involved in the decision-making processes of the project's activities, as well as the implementation of communication tools to improve social relations.
	Low engagement of farm workers in the project's activities	Employees may not feel comfortable with the project and may not identify with the activities.	Promote activities that provide information about the project and its benefits, as well as value the participation of workers and their knowledge of the project area. Also consider actions that strengthen good practices and relations with workers through courses/training on their labor rights, occupational safety, and understanding of their duties and position. In addition, establish efficient communication and feedback mechanisms.
Working conditions	No risks identified		No risks were identified for existing and future working conditions in relation to the project's activities.

Safety of women and girls	No risks identified	-	No risks were identified for the safety of women and girls in the implementation of the project's activities, as these are linked to the intensification of surveillance activities on properties, biodiversity monitoring, and education and technical assistance actions for the community.
Safety of minority and marginalized groups, including children	No risks identified	-	No risks to the safety of minority and marginalized groups, including children, were identified in the implementation of the project's activities, since these are linked to the intensification of surveillance activities that take place on properties, biodiversity monitoring, and education and technical assistance actions for the community.
Pollutants (air, noise, water discharges, waste generation, and release of hazardous materials, pesticides, and chemical fertilizers)	No risks identified	-	The project does not intend to use products that cause pollution-related impacts.
Discrimination	No risk identified	-	The Caiarara REDD+ Project follows the codes of conduct of Biofilica Ambipar Environmental Investments S/A and Dow Brasil, which guarantee the promotion of an environment of mutual respect by prohibiting discrimination and harassment (including sexual harassment) and unprofessional behavior.
Sexual harassment	No risk identified	-	The Caiarara REDD+ Project follows the codes of conduct of Biofilica Ambipar Environmental Investments S/A and Dow Brasil, which guarantee the promotion of an environment of mutual respect by prohibiting discrimination and harassment (including sexual harassment) and unprofessional behavior.
Equal pay for equal work	No risk identified	-	The project's activities do not provide for the payment of salaries, but if salaries are offered in the future, there

			will be no distinction in pay for similar positions based on gender, color, religion, or other factors.
Gender equality at work	No risk identified	-	The project's activities can be carried out by anyone, regardless of gender.
Forced labor ¹²	No risk identified	-	The Caiarara REDD+ Project follows the codes of conduct of Biofilica Ambipar Environmental Investments S/A and Dow Brasil, which ensure compliance with all applicable laws on forced labor in their activities.
Child labor	No risk identified	-	The Caiarara REDD+ Project follows the codes of conduct of Biofilica Ambipar Environmental Investments S/A and Dow Brasil, which ensure compliance with all applicable laws on child labor in their activities.
Human trafficking	No risk identified	-	The Caiarara REDD+ Project follows the codes of conduct of Biofilica Ambipar Environmental Investments S/A and Dow Brasil, which ensure compliance with all applicable laws on human trafficking in their activities ¹³ .
Recognition, respect, and promotion of the rights of indigenous peoples, local communities, and customary rights holders	No risk identified	-	The properties of the Caiarara REDD+ Project are private areas and do not have areas with customary rights of third parties.
Preserving and protecting cultural heritage	No risk identified	-	Not applicable to the project, as no cultural heritage impacted by the project's activities has been identified, since the communities in the region are not considered traditional.

¹² The identified risks and proportionate mitigation or prevention measures for forced labor, child labor, and human trafficking should include employees and workers hired by third parties.

¹³ Available at: < <https://corporate.dow.com/content/dam/corp/documents/legal/473-00001-11-dow-code-of-conduct.pdf>> and < <https://ambipar.com/site2020/wp-content/uploads/2023/09/Codigo-de-Conduto-e-Compliance-2023.pdf>>

Protecting and preserving property rights, customary rights, or protecting legal or customary rights of ownership/access to territories, properties, and resources, including collective and/or conflicting rights	No risk identified	-	The properties of the Caiarara REDD+ Project are private areas, with no disputes over land or resources.
Impacts on biodiversity and ecosystems	No risk identified	-	The project's activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. As such, the project's activities aim to have positive impacts on biodiversity and ecosystem conservation.
Soil degradation and erosion	No risk identified	-	The project's activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. Therefore, the project's activities do not pose any risks to soil degradation and erosion.
Water consumption and stress	No risk identified	-	The project's activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. Therefore, the project activities will contribute to the conservation of water stocks.
Habitats (and areas necessary for habitat connectivity) for rare, threatened, and endangered species	No risks identified	-	The project's activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. As such, the project's activities will contribute to the conservation of these habitats.
Areas necessary for habitat connectivity	No risk identified	-	The project's activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. Therefore, no risk has been identified for the degradation of

			areas necessary for habitat connectivity.
Invasive species	No risk identified	-	The Project does not foresee the use of invasive species in its activities.
Ecosystem conversion	No risk identified	-	The project's activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. Therefore, no risk has been identified in terms of ecosystem conversion.
Loss of access to areas by community members with the definition of the Project Area and increased surveillance and monitoring of the forest area.	Loss of access to areas for extractive products in general, hunting, and subsistence fishing.	Illegal and uncontrolled entry into native forest areas of the project to search for extractive products in general, food, such as animals and fish, in order to ensure the food subsistence of some families.	Prioritize the involvement of the most affected communities in alternative activities, such as the promotion of diversified and sustainable agricultural production practices and models, the development and strengthening of value chains for agricultural or forestry products, which aim to ensure income and food security while minimizing possible negative impacts and reducing pressure for deforestation.

2.3.6 Community costs, risks, and benefits (CCB, G3.2)

A socioeconomic diagnosis was carried out involving the communities in the region and workers on the farms of the Caiarara REDD+ Project, in which the local scenario and context were analyzed through interviews (Section 2.3.1). During this process, the possible positive and negative impacts that the Project may have on the communities were identified, based on the social impact assessment method developed by Richards and Panfil (2011)¹⁴.

Based on this information, the activities within the social scope of the Project were designed for the social and economic development of local actors, aiming to protect natural resources and reduce deforestation in the region. To validate the proposed activities and present the Project again, a feedback event was held with stakeholders (Section 2.3.4), in which the potential costs, risks, and benefits relevant to the communities were presented transparently, using posters and slides with accessible language to ensure that the communities understood the information. It was made clear from the outset that participation in

¹⁴ Richards, M. and Panfil, S.N. 2011. Social and Biodiversity Impact Assessment (SBIA) Manual for REDD+ Projects: Part 1 – Core Guidance for Project Proponents. Climate, Community & Biodiversity Alliance, Forest Trends, Fauna & Flora International, and Rainforest Alliance. Washington, DC.

the Project is voluntary, and communities were advised that the decision to participate or not can be reviewed throughout the Project cycle without incurring penalties or restrictions.

Additionally, throughout the implementation of the Project, new relevant and appropriate information on the potential costs, risks, and benefits to the communities will be provided at follow-up meetings, consultations, and meetings on the development of activities.

In accordance with the established Communication Plan, described in the section ***Erro! A origem da referência não foi encontrada.***, suggestions and criticisms from stakeholders received during the meetings were recorded in a unified spreadsheet, analyzed, and responded to within defined deadlines, depending on the complexity of each issue. The management of these demands is the responsibility of the project's communication team, with support from specialists in relevant areas, if necessary. When relevant and appropriate, the contributions received were incorporated into the project's methodologies and activities. Stakeholders also have the opportunity to make suggestions and criticisms via institutional email, WhatsApp messages, and physical forms through suggestion boxes, which will be installed at strategic points to facilitate access to communication channels for communities and workers.

2.3.7 Information to stakeholders about the verification process (VCS, 3.18.6, 3.19; CCB, G3.3)

Palmyra workers, potentially participating communities, and other Project stakeholders were informed about the verification process through the communication channels already presented above (more details in Section 2.3.11), including institutional email, WhatsApp messages, banners, and printed posters distributed in high-traffic areas, as well as in-person meetings and gatherings as a means of directly exchanging information about the validation and verification process.

Virtual channels, such as the Biofilica Ambipar Environmental Investments S/A social media, and Dow Brazil communication channels were also used to inform stakeholders and the general public.

Information about the verification process was disclosed during the Rapid Participatory Rural Diagnosis (part of the socio-economic diagnosis) and the stakeholder feedback event held between July 11 and 14, 2024 (see Section 2.3.10 for more details). On these two occasions, stakeholders were informed about the Project's audit process and the presence of auditors. In addition, banners and posters were installed in locations with high traffic of workers and communities to inform them about the project's results during the monitored period, the verification process, the audit, and the arrival of the auditors on site.

Doubts and suggestions about the Project's verification process could be submitted via WhatsApp and the institutional email address. The requests were recorded in a unified spreadsheet, analyzed, and responded to within defined deadlines, depending on the complexity of each issue.

2.3.8 Information on site visits and opportunities to communicate with the auditor (VCS, 3.18.6, 3.19; CCB, G3.3)

The auditor's visit to the Project's site was communicated to the surrounding communities, Palmyra workers, proponents, partners, local institutions, and other stakeholders of the Caiarara REDD+ Project through emails, WhatsApp messages, poster distribution, and in person.

At the "feedback to stakeholders" event held in July 2024, the next steps of the Project were mentioned, explaining the verification process, the audit event, the auditor's visit, the possibility of direct communication, and the expected date.

In addition, the field audit event and the auditor's visit were reinforced by sending official letters to relevant local institutions in the state of Pará, among others that are directly and indirectly involved in the forest conservation and environment sector, such as unions, non-governmental organizations (NGOs), educational institutions, and state and federal government agencies, where updated information on the status of the project, the communication channels used, and an invitation to participate in Verra's public comment period were also presented.

Prior to the opening of the public comment period, a digital card was sent via WhatsApp to the communities and workers to inform them about the public consultation, the date of the audit, and the possibility of communicating with the auditor. In addition, banners and posters were placed in locations with high traffic of workers and communities to inform them about the verification process, the audit, and the arrival of the auditors on site. During the field audit, local residents, workers, and other stakeholders will have the opportunity to communicate with the auditor in person and independently, without the presence of the project's proponents, in order to ensure the independence of communication and a direct channel between the auditor and stakeholders.

2.3.9 Consultation with stakeholders (VCS, 3.18; CCB, G3.4)

Three events were held by the Caiarara REDD+ Project to consult local stakeholders (communities surrounding the Project's Area and farm workers): the first was the meetings and interviews for the socioeconomic diagnosis, carried out between December 2023 and January 2024; the second was the "feedback to stakeholders" event held in July 2024; and the third was the meeting with the Vila Mamorana community held in August 2024.

The consultations were conducted in language accessible to all, considering possible language or literacy barriers. All information collected was recorded on forms, reports, attendance lists, and/or photos, organized and archived.

In addition, Verra's public comment period ran from x to x, 2025.

Ongoing consultation	All consultations carried out were essential to validate the Communication Plan and begin its implementation. Considering its guidelines, all stakeholders were duly mobilized and consulted, supported by informative and educational materials with appropriate language that
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	publicized the communication channels that served the project at the time.
Stakeholder consultation dates	December 8, 2023, to January 16, 2024 – Initial contact with stakeholders, presentation of the project, and conduct of the socioeconomic <i>diagnosis</i>. July 11, 2024, to July 14, 2024 – Feedback with <i>stakeholders</i>. August 19, 2024 – Vila Mamorana meeting X to X/2025 – Public consultation on the VERRA platform.
Communication of monitored results	As a way of presenting the results during the monitoring period, the Feedback action stands out: With the completion of the technical studies, the “feedback to stakeholders” event took place between July 11 and 14, 2024, involving the surrounding communities and workers. During the meetings, the results of the socioeconomic diagnosis and proposals for activities to be conducted by Dow and Bioflica Ambipar throughout the Caiarara REDD+ Project were presented. Contact with the communities was made through personal invitations sent to community leaders, while workers were invited through company leaders. It should be noted that participation was voluntary, without any type of imposition or restriction for those who chose not to participate. The main objective of the event was to present the Project and provide an open space for participants to clarify doubts and offer suggestions about its design. As a result, participants showed interest in getting involved in the activities and committed to monitoring the Project's progress. No changes were suggested to the proposed Action Plan, which reinforces the efficiency of the diagnostics in portraying the social context and supporting developers in designing activities in an assertive manner. Another way of disseminating the project's results during the monitored period was through banners installed in high-traffic areas on farms and in communities and cards sent via WhatsApp, in order to present the activities carried out

	during this period to local stakeholders, publicize the REDD+ project and its impacts, and provide information on the status of the project and the audit process (Sections 2.3.7 and 2.3.8).
Consultation records	During the feedback events with stakeholders, participants filled out a form with questions about their satisfaction with what was presented, suggestions or criticisms for the activities and scope of the project, and other comments. The Communication Plan and preferred communication channels were also validated. Posters and brochures were also distributed and presented, expanding the dissemination of information about the project and bringing it, in a simple and visual way, to those who were unable to attend the meetings. The questionnaires were scanned and entered into the project's database. In addition to the feedback event, records resulting from other activities were collected, such as contracts signed between proponents and suppliers for initial studies; the Socioeconomic Diagnosis includes photos, graphs, and analyses of consultations, recording the context and information collected in the study; records of emails sent; audit report responding to questions raised in Public Comments, among others. All evidence was made available to the VVB.
Contributions from stakeholders	Feedback: Participants showed interest in getting involved in the activities and commitment to monitoring the progress of the Project. No suggestions were made to change the proposed Action Plan, reinforcing that the diagnoses were effective in reporting the social context and assisting developers in designing project activities in an assertive manner. Other actions: All evidence of information to contextualize the social parameters of the Project's Area and its surroundings, as well as to identify the main sensitivities for the preparation of the Project's Activities and the Theory of Change and the entire Section on Stakeholder and Community Engagement.

	Public Consultation on the VERRA Platform: All comments received will be evaluated and included in the second round of the audit process. All questions will be answered and, where appropriate, changes will be made to the PD.
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2.3.10 Continued Consultation and Adaptive Management (VCS, 3.18; CCB, G3.4)

As already mentioned in the Section 2.3.2, the Caiarara REDD+ Project has established a robust Communication Plan to ensure ongoing communication and consultation with communities and other stakeholders throughout its implementation. The goal is to ensure that perceptions, suggestions, and concerns are considered in order to adapt the project's management in an efficient and participatory manner.

The structure of the Communication Plan includes: Communication Channels for information exchange and collection of contributions, Social Mobilization Strategies to maintain engagement, Communication Procedure and Demand Log for organizing inputs, Incorporation of Suggestions for project adaptations, Conflict Management for dialogue and negotiation, and Adaptive Management focusing on changes and adjustments over time (more information in the PDD, Section 2.3.11).

Among the available channels are: the WhatsApp group, which facilitates quick exchanges of information and general notices; the project's email, used for more formal and specific requests; a physical form, which will be made available in strategic locations (after the implementation of suggestion boxes) to record suggestions, questions, and opinions in an accessible way; and a digital form accessed via QR Code, which offers a modern and practical alternative for those who have access to mobile devices connected to the internet.

At the time of writing this Monitoring Report, one request was received:

Summary of comments received	Actions taken
On May 26, 2025, a request was received via the project email (<i>projetocaiarara@dow.com</i>) from a teacher at the Paraíso Municipal Preschool and Elementary School in the Vila Mamorana community, requesting a visit by students to Dow's farms.	The request was received via the project email, recorded in the spreadsheet, and forwarded to those responsible for the area. The students' visit took place in October 2025. Receiving the demand demonstrates that the Project's communication channels are working and being useful to the local community.

The expectation is that, with continued dissemination and familiarization of communities and workers with these new means, their use will increase progressively, making communication more agile and

inclusive. It is important to note that the means of communication were created in accordance with the reality of the community and workers and validated by them.

Communication strategies will be continuously evaluated and adapted as necessary, especially if impediments or failures in communication between the parties involved are identified. This continuous monitoring aims to ensure that everyone's demands are heard and that the dialogue and consultation processes remain efficient and accessible.

2.3.11 Stakeholder Consultation Channels (CCB, G3.5)

The activities of the Caiarara REDD+ Project were planned and implemented considering the interests, characteristics, and needs of the surrounding communities and workers involved. This was made possible through interviews and workshops conducted during the Socioeconomic Diagnosis and at the meeting to validate the proposed activities with stakeholders ("feedback event").

The consultations and participatory processes were conducted directly with the communities and workers, with the support of local leaders and officials, who acted as legitimate representatives and facilitators of dialogue, formally recognized by the communities themselves during the participatory diagnosis. In addition, a feedback event was held, ensuring the sharing of more information and consultation with the communities and promoting greater understanding of the Project by the communities and other stakeholders.

During the meetings, the content was presented in simple language adapted to the local context, using visual and oral resources, such as posters, whenever necessary. All consultations were documented through attendance lists, reports, and photographic records, filed physically and/or digitally.

The proposed communication channels — including face-to-face meetings, WhatsApp messages, virtual forms, and institutional emails — are accessible to stakeholders and have been approved by communities and workers as effective tools for consultation and interaction, ensuring accessibility and transparency.

Communication and consultation will continue throughout the implementation of the Project, using the channels described and adapting them as necessary to ensure that information is adequately shared and that the voices of stakeholders are heard and respected.

2.3.12 Stakeholder participation in decision-making and implementation (VCS, 3.18, 3.19; CCB, G3.6)

Involvement in all phases, from design, implementation, monitoring, and evaluation of the Project, is strongly encouraged and occurs through available communication channels and informational meetings, in which all stakeholders have the opportunity to participate.

As shown in Section 2.3.9, participatory events were held with stakeholders, taking into account cultural and gender aspects, using accessible language, appropriate schedules, and encouraging the participation of women, youth, and other minority groups, while respecting local social dynamics. The participation of communities in these events during the design and development stage of the Project

contributed to incorporating their perceptions and suggestions for activities for the project, according to the local context and needs.

The processes related to decision-making and implementation, as well as the various activities related to the Project, are open to the participation of communities and workers. In addition, the project seeks equal participation of historically marginalized groups in decision-making and in the implementation of the Project's activities.

The contributions of stakeholders that are incorporated will be widely disseminated, mainly to participating communities and workers on the properties, through established communication channels, for continuous monitoring and validation throughout the Project's life cycle, ensuring transparency and a participatory approach.

2.3.13 Anti-Discrimination Guarantee (VCS, 3.19; CCB, G3.7)

Dow Brazil has a solid policy of respect and responsibility, as established in its code of conduct, which ensures the prevention of any form of discrimination and harassment, including sexual harassment. The company promotes an environment of mutual respect, prohibiting discrimination based on gender, race, religion, sexual orientation, or other habits, as well as any unprofessional behavior. The Respect and Responsibility Policy reinforce Dow Brazil's commitment to creating an inclusive and safe workplace, and this commitment will be strictly enforced in all activities of the Caiarara REDD+ Project, ensuring that no discrimination or harassment occurs during its implementation.

Dow Brazil also has an Ethics and Compliance Office ("Office") that reinforces the company's long-standing commitment to ethical conduct. The Office communicates company standards, provides guidance on issues related to ethical conduct, and oversees mechanisms for action. It is available to answer questions or address concerns and has the authority to take immediate action in the event of conduct that is incompatible with its Code of Conduct. In addition, Dow Brazil has the Dow EthicsLine for those who wish to ask questions about Dow's policy, seek guidance on specific situations, or report possible violations of its Code of Conduct. All those involved, including local employees and community representatives, will have access to the reporting channel, with guidance in accessible language and a guarantee of confidentiality and non-retaliation. Use of the channel will be reinforced during activities with workers and surrounding communities.

Similarly, Ambipar has policies that represent all companies in the Ambipar Group. The Code of Conduct and Compliance serve as a fundamental tool to guide the conduct of all parties involved in adopting best practices in relationships and business dealings, promoting attitudes of respect for diversity and combating any form of prejudice. In addition, the Diversity and Inclusion Policy guide respect for human dignity and the preservation of human rights, observing the parameters, precepts, and principles established in the Universal Declaration of Human Rights and the UN Guiding Principles on Business and Human Rights.

The project adopts an inclusive approach in its relationship with local communities, promoting respect for diversity in gender, culture, generation, and origin, with special attention to potentially vulnerable groups. To date, no incidents of harassment or discrimination have been reported within the scope of the

Caiaara REDD+ Project, and all activities are implemented with specific safeguards to ensure that no form of harassment, including sexual harassment, or discrimination occurs as a result of the project's actions.

The aforementioned policies are directly applied in the Caiaara REDD+ Project through contractual clauses, periodic training for all teams, and a formal requirement for partners and service providers to adhere to ethical guidelines. Dow raises awareness among direct and indirect employees on site through informational banners distributed on farms, through emails and campaigns, and lectures promoted by employee diversity networks, addressing topics such as diversity, ethical conduct, and harassment prevention, reinforcing the principles of the Code of Ethics in the context of carrying out activities. In this way, the project proponents ensure that they follow all the guidelines applied in their codes of ethics, policies, and contracts established throughout the Project.

2.3.14 Complaints (VCS, 3.18.4; CCB, G3.8)

Dow Brazil, in conjunction with Ambipar, has established an accessible procedure for conflict resolution, which describes the processes and means of communication for conflict management and the feedback procedure (as described in the Communication Plan*Erro! A origem da referência não foi encontrada.*). This procedure will be implemented throughout the life of the Caiaara REDD+ Project. It includes receiving questions, complaints, suggestions, and requests from stakeholders through available communication channels, which must be recorded on a standard form, analyzed, and resolved within a predetermined time frame with the parties involved.

During this monitoring period, no complaints were recorded, only a request for a visit by a teacher from the Vila Mamorana community school (described in Section 2.3.10).

2.3.15 Worker Training (VCS, 3.19; CCB, G3.9)

In August 2024, training was provided to farm workers, focusing on updating procedures for preventing and fighting forest fires, covering the RAAI and RAAll farms.

The training included practical and theoretical activities, where participants were instructed on how to act correctly in emergencies, prioritizing personal safety when moving to protected locations and activating firefighting teams. These teams, composed of workers trained annually, use specialized techniques, appropriate equipment, and effective methods to control and extinguish fires.

To ensure that local capacity is not lost, the project adopts a continuous training model, offering annual training courses that consolidate and expand technical knowledge among workers. This approach contributes to worker safety and environmental protection, reinforcing the project's commitment to sustainable and responsible management of forest resources.



Figure 4. Mosaic of photographs from the training held in August 2024 to update firefighting procedures. Source: Dow Brazil Archive.

The actions offered by the Caiarara REDD+ Project are fundamental to ensuring the active engagement of workers and communities, promoting the development of essential skills for the implementation and success of the Project. By aligning training with proposed activities and ensuring the continuity of knowledge, the project not only strengthens environmental protection but also improves living conditions in the communities involved. This investment in human capital contributes to the longevity of the environmental, social, and economic benefits generated by the project, creating a positive cycle of sustainable development.

2.3.16 Employment opportunities in the community (VCS, 3.19.13; CCB, G3.10)

The employment opportunities offered by Dow Brazil are equal for surrounding communities, including management positions if the requirements for the position are met, in addition to promoting specific training to prepare local candidates for these roles.

All job positions generated locally by the Project follow a recruitment and selection process that Dow Brazil will carry out according to the need and availability of vacancies.

Criteria such as race, gender, sexual orientation, color, religion, age, ethnic origin, physical or mental disability, or social class are not taken into account. All stages of the selection process, as well as the hiring of professionals, will be based on the criteria established in the job descriptions, and minimum qualifications are desirable. The project will develop active inclusion strategies, such as prioritizing

female candidates for administrative and operational roles, encouraging the hiring of young people and people with disabilities, and periodically monitoring diversity in hiring.

It is worth noting that Dow Brazil has residents from local communities on its permanent and temporary teams, and the Project will work to reinforce the actions already taken in this regard. Currently, approximately 50% of Dow Brazil's own or outsourced employees are from surrounding communities.

2.3.17 Occupational Safety Assessment (VCS, 3.19; CCB, G3.12)

To date, no substantial risk to worker safety has been identified that may arise from the implementation of the Project. However, Dow Brazil takes a rigorous and preventive approach to ensure a safe and healthy work environment. To this end, three key programs have been implemented: the Risk Management Program (PGR), the Occupational Health Medical Control Program (PCMSO), and the Technical Report on Environmental Working Conditions (LTCAT). These programs aim to anticipate, track, and diagnose early any occupational health problems that may arise due to the execution of activities.

Each of these programs supports workers through detailed operating procedures, specific work instructions, environmental procedures, and strategies for controlling and preventing illness and injury. Information is disseminated and communicated on an ongoing basis to ensure that all workers are aware of potential risks, even if they are minimal, and of best practices for safety in the workplace.

In addition, the project strictly follows internal safety regulations and complies with occupational safety and health standards defined by the federal and state governments. Knowing that processes and the work environment can potentially impact workers' health, the project adopts a series of measures aimed at promoting and maintaining the physical and social well-being of employees. These measures include the prevention of occupational diseases and protection against risks that may arise from factors adverse to health.

Risk assessments will be conducted frequently to ensure that the work environment remains safe and in compliance with established guidelines. If any occupational safety risk is identified in the future, it will be included in the project's risk management system, and mitigation and monitoring actions will be implemented immediately to ensure that workers are properly protected and informed.

Through these programs and continuous risk assessment, the project seeks to ensure that all workers are in a work environment that matches their physical and psychological abilities, promoting work experience that minimizes risks and preserves their health and safety.

2.4 Management Capacity

2.4.1 Required technical skills (VCS, 3.19; CCB, G4.2)

The main technical skills required for the implementation of the Caiarara REDD+ Project are knowledge of the methodology for preventing unplanned deforestation, knowledge of the development and management of forest conservation projects in the Amazon biome, experience in the implementation, development, and assistance of programs for communities, implementation of effective land security,

and property surveillance. All parties involved in the implementation and management of the project have the technical skills necessary for the successful completion of the Caiarara REDD+ Project.

The Ambipar Group operates throughout Brazil, offering integrated solutions for the entire business chain. Rapidly expanding worldwide, Ambipar respects compliance and social and environmental responsibility rules, valuing ethics and prompt response to its customers' demands.

As part of the Ambipar Group, within the Carbon Solutions pillar, and in line with its objectives, Biofilica Ambipar Environmental Investments S/A is a Brazilian company that promotes the management of forest areas in the Amazon and Atlantic Forest biomes. The company has a specialized team and is a reference in the development of forest conservation projects, ensuring the quality and effectiveness of the REDD+ activities developed. Biofilica Ambipar Environmental Investments S/A aims to make environmental conservation an economically attractive activity for forest owners, communities, and investors. Its goal is to reduce deforestation and carbon emissions into the atmosphere, conserve biodiversity and water resources, and promote social inclusion and the development of communities living in the Amazon biome through the sale of credits for environmental services, the development and financing of scientific research activities, and the development of sustainable business chains.

Dow is a global materials science company, present in more than 160 countries and active in various sectors, such as packaging, infrastructure, consumer goods, and transportation. In Brazil, the company also operates in silicon production, with an industrial unit in Breu Branco, Pará. This plant uses advanced technologies and responsible practices, contributing significantly to the global supply chain and promoting regional economic development.

In addition, Dow plays an important social role in the city where it operates, generating jobs, offering professional growth opportunities, and investing in training for local residents. The company also demonstrates a strong commitment to mitigating environmental impacts, strictly adhering to its code of ethics and prioritizing practices that reduce the effects of its operations on the environment. This stance contributes to strengthening the local economy and ensures operations aligned with values of social responsibility and sustainability.

In the context of the Caiarara REDD+ Project, Dow plays a crucial role as a proponent, bringing its solid governance and social experience acquired over years of operating in the region. Its contribution will be fundamental to the success of the project, especially when working in partnership with Ambipar, which has expertise in the development of REDD+ projects. This union between Dow's social governance and Ambipar's technical knowledge results in a solid and strategic partnership, prepared to generate lasting positive impacts on environmental conservation and community strengthening in the Amazon.

2.4.2 Management Team Experience (VCS, 3.19; CCB, G4.2)

Below is the experience of the members of the Caiarara REDD+ Project management team.

Developer: Biofilica Ambipar Environmental Investments S/A

BOARD OF DIRECTORS:

Plínio Ribeiro – Advisor

Plínio Ribeiro holds a degree in business administration from the Institute of Education and Research (INSPER) and a master's degree in public administration and the environment from Columbia University and the Earth Institute (USA). He has participated in several conservation projects in the lower Rio Negro region through the Institute for Ecological Research (IPÊ) since 2005 and was one of the producers of Jean Michel Cousteau's documentary "Return to the Amazon." He has worked at Biofilica Ambipar Environmental Investments S/A since 2008, where he has led projects, operations, and business management. He is currently an advisor and shareholder of the company.

Cláudio Pádua – Scientific Director

Cláudio Pádua has a degree in Business Administration and Biology, a master's degree in Latin American Studies, and a doctorate in Ecology from the University of Florida in Gainesville (USA). A retired professor from the University of Brasília, Pádua is currently dean of the School of Conservation and Sustainability and vice president of the Institute for Ecological Research (IPÊ). He is also a senior associate researcher at Columbia University's Center for Environmental and Conservation Studies (USA) and international director of conservation at the Wildlife Trust Alliance, as well as a consultant to the Brazilian Biodiversity Fund (FUNBIO) and WWF Brazil. Pádua represents Brazil on the International Advisory Group (IAG) of the G7 Pilot Program. In 2003, together with his wife, Suzana Pádua, he was named a "Hero of the Planet" by Time Magazine for his activities in support of biodiversity conservation. Between 1997 and 2007, he won six conservation awards, three national and three international. Pádua has published two books and more than 30 articles in national and international scientific journals. Since 2008, he has directed the scientific involvement and production of Biofilica Ambipar Environmental Investments S/A as scientific director and consultant.

Soraya Dias Pires – Head of Carbon Solutions

Soraya Dias has a degree in agronomy from the University of São Paulo (USP), two MBAs from USP, one in agribusiness and the other in strategic management, and an MBA from the Getúlio Vargas Foundation (FGV) in business administration. She has over 15 years of experience in agro-industrial business management and development, with a career developed in large multinationals in the sugar-energy sector: Adecoagro, BP Biocombustíveis, and BP Bunge Bioenergia. She has experience in agro-industries in activities associated with agro-industrial planning and control, agricultural operations, and negotiations. At Biofilica Ambipar Environmental Investments S/A, she serves as Director of Operations and New Business.

OPERATIONS & NEW BUSINESS TEAM**Caio Gallego – Intelligence and Operations Manager**

Caio Gallego is a Forest Engineer graduated from ESALQ-USP with an MBA in Corporate Sustainability from Fundação Getúlio Vargas. He is a specialist in geoprocessing and remote sensing focused on environmental conservation, mapping, and analysis of changes in land use. He has knowledge focused on Sustainable Forest Management, environmental modeling, and the use of alternative GIS for forestry and agribusiness. He has advanced knowledge in the use of GIS software and analysis of changes in

land use and coverage, such as ArcGIS, QuantumGIS, and DinâmicaEGO. He has experience in the areas of forest conservation and restoration, climate change, environmental modeling, and environmental project management. He works as an operations manager in projects developed for the carbon market.

Marcio Sales – Spatial Statistics Specialist

Marcio Sales is a statistician with a degree from the Federal University of Pará, a master's degree in Geography from the University of California, Santa Barbara, and a PhD candidate in Production Ecology and Resource Conservation at Wageningen University in the Netherlands. He is a specialist in data analysis and conducts research in geostatistical modeling of processes distributed in space and time. He works at Biofilica Ambipar Environmental Investments S/A producing GHG emission projections from future deforestation for project baselines and monitoring deforestation by satellite.

Paula Conde – Administrative Manager and Operational Support

Paula Conde holds a degree in Business Administration from São Luís - PUC and a postgraduate degree in Accounting and Financial Management from FAAP. She has extensive experience, most of it at one of the largest media and education groups in Latin America, Editora Abril, where she worked in Financial Control and Reporting, Treasury, Financial and Accounting Reconciliation, Accounts Payable and Receivable, and Royalties. At Biofilica Ambipar Environmental Investments S/A, she is responsible for administrative and financial activities and logistical support for the team and projects.

Luisa Cotrim – New Business Strategy Specialist

Economist graduated from UNICAMP, with experience in carbon markets, corporate decarbonization, and market intelligence. She leads the development of new business at Ambipar Carbon Solutions, responsible for structuring proposals and technical solutions in emissions management and carbon projects, with a growing focus on strategic partnerships and international expansion.

Carla Paulino – New Business Analyst

Graduated in Agronomic Engineering and Business Administration from ILES/ULBRA Itumbiara, she has a specialization in Compliance and Project Management. She worked in the development and execution of a project for Continuous Improvement of Origination processes and procedures. She has 16 years of experience in land analysis for partnership and supply contracts. She is currently a New Business Analyst III, focusing on prospecting and serving clients for the implementation of Carbon Projects. She has had the opportunity to develop teams and provide intercompany training.

Maria Claudia Martinelli Trabulsi Alves – Project Coordinator

Maria Cláudia is a Forest Engineer graduated from UNESP - Botucatu, with a master's degree in Energy in Agriculture from the same university and a specialization in Sustainability from UNESP - Rio Claro. She has over 10 years of experience in auditing forestry, agricultural, sustainability, and carbon credit projects.

Gabriella Hita Marangom Cesilio – Project Analyst

Gabriella Hita is a Forest Engineer and holds a degree in Agricultural Sciences, both from the University of São Paulo (USP/ESALQ). She also has a specialization in Environment and Sustainability from the Getúlio Vargas Foundation (FGV) and is pursuing an MBA in Project Management from the University of São Paulo (USP/ESALQ). At Biofilica Ambipar Environmental Investments S/A, she works on the development, implementation, and monitoring of REDD+ projects in accordance with VERRA standards, in addition to contributing to the establishment of procedures and the development of project intelligence.

Carolina Corrêa Cauzzo – Project Analyst

Environmental Manager graduated from the University of São Paulo (USP/ESALQ), with an interuniversity exchange at the Institut Agro Montpellier, in France, where she studied Sustainable Management of Natural Resources and Agroecology. She has experience with forest restoration projects in the Atlantic Forest, socio-environmental studies in rural settlements, and geoprocessing tools. At Ambipar, she works on the development, implementation, monitoring, and support of REDD+ projects, evaluation of supplier diagnostics, communication with stakeholders, and development of project intelligence.

Fábio Souza – Geotechnologies and Innovation Manager

Geographer, master's degree in natural resource use and conservation, specialized in climate analysis, big data technologist, and analytical intelligence, he has worked in environmental diagnostics, geotechnology service development, and geoprocessing team management. Specialist in PostGIS database modeling, webgis platform development, business intelligence, and geoprocessing process automation. Currently part of the operational team at Biofilica Ambipar Environmental Investments S/A as geotechnology coordinator.

SALES & MARKETING

Laion Pazian – Sales Manager

Laion Pazian holds a degree in economics from USP (ESALQ Campus) and an MBA in Commercial Management from Fundação Getúlio Vargas. He manages the carbon credit sales team, serves national key accounts, and supports relevant international sales. He leads part of the commercial strategies and manages the carbon credit portfolio of Biofilica Ambipar Environmental Investments S/A, reporting the results to the company's Board of Directors.

Ricardo Cordeiro – Marketing and Communications Coordinator

Ricardo Cordeiro has a degree in digital marketing and an MBA in Marketing, Branding & Growth from PUC in Rio Grande do Sul. He has over 16 years of experience in art direction (online and offline) and project management. A specialist in digital marketing strategies, SEO, Google Adwords, and Social Ads, throughout his career he has worked in digital agencies, trade, and live marketing. At Biofilica Ambipar Environmental Investments S/A, he leads the marketing team, planning and executing actions focused on Branding and Growth.

ADVOCACY

Gabriel Trivelino – International Relations Specialist

Gabriel Trivelino is an International Relations Specialist at Biofilica Ambipar Environmental Investments S/A, working on the company's relationship with the public sector and representative organizations. With a degree in Political Science from the University of Brasília (UnB) and an MBA in Government Relations from the Getúlio Vargas Foundation (FGV), he has over 10 years of professional experience representing interest groups in specific legislation or regulations in decision-making bodies such as the Brazilian Executive Branch and the country's Legislature.

Proponent: Dow Brasil

Vinicius Sambuc – Commercial Manager of Energy & Climate

Vinicius Sambuc holds a degree in Metallurgical Engineering from the Federal University of Minas Gerais (UFMG) and a specialization in Production Engineering from Uninter and an MBA in Project Management from the Getúlio Vargas Foundation (FGV). He has worked at Dow since 2008 and has experience in operations, continuous improvement, and project management. He assumed his current position in 2022 and in 2024 was invited to be the Manager of the Caiarara REDD+ Project.

Adryane Hermann – Senior Product Manager

Adryane holds a degree in Foreign Trade from Universidade São Judas Tadeu (USJT/SP), a master's degree in Supply Chain Management from BSP-Business School São Paulo, and is currently studying Law at UNIP-Universidade Paulista. She has worked at Dow since 2001 and has extensive experience in product, project, and supply chain management. She assumed her current position in December 2021 and was invited to co-lead the Caiarara REDD+ Project with Vinicius Sambuc.

Alexandre Forman Prata – Site Operations Leader

Alexandre Forman Prata has a diverse educational background, including law, mining, and an MBA in business management. Since 2001, he has contributed significantly to DOW, where he has accumulated extensive experience in several areas. His professional career spans supply chain, logistics, forestry operations, coal, and quartz mining.

Francisco Neto – Regulation and Compliance Specialist

Francisco Neto holds a degree in Forestry Engineering from the Federal Rural University of Amazonia (UFRA), with a specialization in Environmental Management from the Federal University of Pará and a specialization in occupational safety engineering (in progress). At Dow since 2008, he initially worked in forestry operations management and currently works in the area of regulation and compliance, focusing on environmental licensing in coal production, forestry, and mining. In 2024, he was invited to join the team of managers for the Caiarara REDD+ Project.

2.4.3 Project Management Partnerships / Team Development (VCS, 3.19; CCB, G4.2)

The Caiarara REDD+ Project has all the necessary partnerships for the construction and implementation of forest asset conservation activities. Currently, the partner institutions mentioned in the section 2.1.5

are responsible for preparing the Socioeconomic and Environmental Diagnostics and Carbon Stock, which comprise the Project Design Document, through service provision contracts. When other initiatives throughout the development of the Project require new technical knowledge and partners, those responsible for the Caiarara REDD+ Project will coordinate partnerships with governmental, non-governmental, and private sector organizations in order to enable the generation of net positive impacts on society and biodiversity.

2.4.4 Financial health of the implementing organization(s) (CCB, G4.3)

Biofilica Ambipar Environmental Investments S/A is a multinational company that operates in various segments, offering comprehensive services and products focused on environmental management. In full global expansion, Ambipar strictly adheres to compliance and social and environmental responsibility standards, always prioritizing ethics and efficiently meeting the demands of its customers. One of the company's main focuses is on generating and trading carbon credits through nature-based solutions (NBS). This business area was significantly strengthened with the acquisition of Biofilica, a Brazilian company with over 15 years of experience in conservation projects through carbon initiatives such as REDD+, ARR, and ALM. The diversification of its business lines and the support of solid investors contribute to Ambipar's financial stability.

Dow Brasil, with more than 64 years of experience in the country, is a multinational company that combines chemistry, biology, and physics to develop innovative technologies that drive progress in various sectors, such as agriculture, transportation, electronics, health, pulp and paper, among others. In the context of this project, the participating headquarters is Palmyra do Brasil, located in Breu Branco, which is part of the Dow Group and operates in the production of metallic silicon and natural resources.

Based on this, it can be stated that the project developers and proponents have sufficient financial strength to guarantee the necessary financial support throughout the entire project implementation period, ensuring the commitment of resources with long-term investment capacity and reinvestment of revenues in the REDD+ Project. The documents proving this financial health are classified as Commercially Sensitive Information and were made available to the audit team on a confidential basis.

2.4.5 Prevention of corruption and other unethical behavior (VCS, 3.19; CCB, G4.3)

Biofilica Ambipar Environmental Investments S/A maintains a legal team responsible for implementing and monitoring the best practices established in the "Ambipar Group Code of Conduct and Compliance," including, in particular, the "Anti-Corruption and Anti-Money Laundering Policy," which is in full compliance with the Brazilian Anti-Corruption Law (Law No. 12,846/13), Law No. 9613/98 (Money Laundering Law), as well as the International Anti-Corruption Conventions (UN, OECD, Transnational Convention against Money Laundering in the International Civil Service, Merida Convention, Palermo Convention, Convention against Organized Crime); and observes the precepts contained in the FCPA (Foreign Corrupt Practices Act) and the UK Bribery Act.

Therefore, Biofilica Ambipar Environmental Investments S/A clearly expresses its position against acts of corruption and/or conduct that seek personal gain to the detriment of the company, society, or the

government, strictly complying with national and international anti-corruption principles, as well as the main guidelines and positions of the Ambipar Group regarding the fight against money laundering, terrorist financing, and all forms of corrupt conduct, such as bribery (active and passive corruption), embezzlement, and granting of undue advantages (administrative impropriety), and curbing acts by its employees that impede investigation and inspection activities. It should also be noted that Biofilica Ambipar undergoes annual financial audits to ensure that its resources are used responsibly and free from corruption.

In order to promote compliance with these policies, Ambipar has an online reporting channel that can be used by interested parties in the Program for any complaints regarding: conflicts of interest; corruption involving public officials; fraud/theft or embezzlement; favoritism toward suppliers or customers; damage to company property; misuse of resources; leakage/misuse of information; violation of laws or non-compliance with policies/procedures, among others. Complaints should be made, whenever possible, through the channel on the Company's website, at the link <https://ambipar.com/denuncias/> or by email to canaldeetica@ambipar.com. All Ambipar Group policies are available on its website: <https://ambipar.com/politicas-estatuto/>.

Dow Brazil is committed to maintaining the highest ethical and legal standards in its relationships around the world. This includes relationships with governments, government authorities, and other companies, as well as with institutions representing communities. Corruption in any form, such as kickbacks, bribes, nepotism, favors, fraud, favoritism, extortion, money laundering, among others, is not tolerated. Dow Brazil also has an Anti-Bribery and Anti-Corruption Policy included in its Code of Conduct, which has a solid global risk-based Anti-Corruption Due Diligence process established to perform *due diligence* audits on third parties based on their risk profile. The results of due diligence are used to establish risk mitigation strategies, including, if necessary, the termination of the business relationship.

Dow Brazil has the *Dow EthicsLine* for those who wish to ask questions about Dow's policy, seek guidance on specific situations, and/or report possible violations of its Code of Conduct. Its Ethics and Compliance Office maintain a confidential, secure, and reliable helpline for reporting ethical concerns and allows anonymous reports when legally permitted. The *Dow EthicsLine* is operated by a third-party company, whose calls are answered by a trained communications specialist who documents the question or concern and forwards the report to the Ethics and Compliance Office for further analysis and handling. The channel for contacting the *Dow EthicsLine* is through the website www.dowethicsline.com, where it is possible to file a report online or by contacting the Dow Ethics and Compliance Office at +1-989-636-2544 or by email at ethics@dow.com. All Dow Brazil policies are available at <https://corporate.dow.com/en-us/about-dow/corporate-governance/living-our-values.html>.

2.4.6 Commercially Sensitive Information (VCS, 3.5.2-3.5.4; CCB Program Rules, 3.5.13 – 3.5.14)

No sensitive information was excluded from the report.

2.5 Legal statute and property rights

2.5.1 National and Local Laws (VCS, 3.1, 3.6, 3.7, 3.14, 3.18, 3.19; CCB, G5.6)

Compliance with laws, statutes, and other regulatory bodies significant to the Caiarara REDD+ Project is related to REDD+ activities in Brazil. With regard to these activities, there is a history of initiatives, despite the construction and negotiation of this concept through agreements and meetings at the United Nations Framework Convention on Climate Change (UNFCCC).

In December 2015, Brazil's National Strategy for REDD+ (ENREDD+) was established by MMA Ordinance No. 370, a document that formalizes to Brazilian society and the signatory countries of the UNFCCC how the Brazilian government has structured its efforts, contributing to the mitigation of climate change through the control of deforestation and forest degradation, the promotion of forest recovery, and the promotion of sustainable development. In this context, in Brazil, Decree No. 11,548, of June 5, 2023, established the National Commission for REDD+ (CONAREDD+) with the objective of coordinating, monitoring, and reviewing the National Strategy for REDD+ and issuing resolutions on various aspects of the implementation, financing, and accounting of actions to reduce emissions from deforestation and forest degradation in Brazil. Still regarding CONAREDD+, Ambipar has a seat on the Commission, representing the private sector, a designation consolidated by MMA Ordinance No. 765, dated October 2, 2023. The new Internal Regulations were made official by Resolution No. 16, dated March 21, 2024.

As for the carbon market in Brazil, on December 12, 2024, Law No. 15,042/2024 was enacted, establishing the Brazilian Greenhouse Gas Emissions Trading System (SBCE), creating the Brazilian Regulated Carbon Market. Among other points, greenhouse gas emission limits were established, based on which it will be possible to trade Brazilian Emission Quotas (CBE), in addition to clarifying aspects of REDD+ projects and programs that generate carbon credits based on reducing emissions from deforestation, degradation, and increasing carbon stocks in native vegetation.

Below are a list and detailed description of the main relevant legislation and regulations at the federal and state levels. In addition, a brief analysis was made of the international climate agreements that have been guiding the creation and development of REDD+ initiatives around the world.

Topic: Brazilian Federal Constitution of 1988

The Brazilian Constitution of 1988 addresses the protection of ecosystems in Article 225, establishing the following main points:

- Everyone has the right to an ecologically balanced environment, which is a common good of the people and essential to a healthy quality of life.
- The government and the community have a duty to defend and preserve the environment for present and future generations.

Applicability to the project: Regulates and guides all national legislation that influences the project. The project also reinforces the provisions of Article 225, which imposes on the community the duty to preserve the environment.

Theme: Land regulationRelated legislation:

Law 601/1850 - Vacant Land Law: Provides for vacant land in the Empire, and for land owned by title of sesmaria without fulfilling the legal conditions, as well as by simple title of peaceful possession; and determines that, once the former have been measured and demarcated, they may be transferred for consideration, both to private companies and for the establishment of colonies of nationals and foreigners, with the Government authorized to promote foreign colonization in the manner declared.

Law 4,504/1964 - Land Statute: Provides for the land statute and other measures. This Law regulates the rights and obligations concerning rural real estate, for the purposes of implementing Agrarian Reform and promoting Agricultural Policy.

Law 9,393/1996 - ITR Law - Rural Land Tax: Provides for the ITR Tax, on the payment of debt represented by Agricultural Debt Securities, and other measures.

Law No. 8,629/1993 - Law on Agrarian Reform - regulates and disciplines provisions relating to agrarian reform, as provided for in the Federal Constitution.

Law 10,406/2002 - Brazilian Civil Code - On the rights and duties in private life, of persons, property, and legal facts. Divided into two parts, (1) General Part, referring to persons, property, and legal facts, and (2) Special Part, referring to the law of obligations, companies, things, family, and succession.

Applicability to the project: The project complies with the laws listed, which guarantees the right to property that includes the Project Area and compliance with taxes related to rural properties.

Theme: EnvironmentRelated legislation:

Decree No. 58,054/1966: Promulgates the Convention for the Protection of Flora, Fauna, and Scenic Beauty of the Countries of America.

CONAMA Resolution No. 16, dated December 7, 1989 - Establishes the Integrated Program for Environmental Assessment and Control of the Legal Amazon.

Law 7,347/1985 - Law on Public Civil Action that regulates public civil action for liability for damage caused to the environment, consumers, and property and rights of artistic, aesthetic, historical, and tourist value, and makes other provisions.

IBAMA Ordinance No. 438, dated August 9, 1989 - Amends the wording of Article 4 of Ordinance No. 218, dated May 4, 1989.

Law 5,887/1995 - State Law that provides for the State Environmental Policy and other measures.

State Decree 1,523/1996 - Approves the Regulations of the State Environmental Fund (FEMA), created by Law No. 5,887, of May 9, 1995.

Law 5,977/1996 – State Law providing for the protection of wildlife in the State of Pará.

Decree 2,661/1998 - Regulates the sole paragraph of Article 27 of Law No. 4,771, of September 15, 1965 (Forest Code), by establishing precautionary rules regarding the use of fire in agricultural, pastoral, and forestry practices, and makes other provisions.

Law 9,605/1998 - Federal Law on environmental crimes that provides for criminal and administrative penalties for conduct and activities harmful to the environment and makes other provisions.

MMA Normative Instruction No. 02, of May 10, 2001 - Provides for the economic exploitation of forests on rural properties located in the Legal Amazon, including Legal Reserve areas and excepting those areas of permanent preservation established in current legislation, which will be carried out through sustainable multiple-use forest management practices.

Legislative Decree No. 143/2002 - Approves the text of Convention No. 169 of the International Labor Organization on indigenous and tribal peoples in independent countries.

Law 6.506/2002 - State Law establishing basic guidelines for the implementation of Ecological-Economic Zoning (ZEE) in the State of Pará and other measures.

Law 6,462/2002 - State Law that provides for the State Policy on Forests and other forms of vegetation.

Law 6.671/2004 – State Law that provides for the State Environmental Policy of Pará and other measures, and amends Article 122 of State Law No. 5.887, of May 9, 1995.

Law 6,745/2005 – State Law establishing the Ecological-Economic Macro-Zoning of the State of Pará and other measures.

Decree 5,975/2006 - Regulates Articles 12, final part, 15, 16, 19, 20, and 21 of Law No. 4,771, dated September 15, 1965, Article 4, item III, of Law No. 6,938, of August 31, 1981, Article 2 of Law No. 10,650, of April 16, 2003, amends and adds provisions to Decrees No. 6,514/08 and No. 3,420/00, and makes other provisions.

CONAMA Resolution No. 378, dated October 19, 2006 - Defines projects that may cause national or regional environmental impact for the purposes of item III, § 1, Article 19 of Law No. 4,771, dated September 15, 1965, and makes other provisions.

CONAMA Resolution No. 379, dated October 19, 2006 - Creates and regulates a data and information system on forest management within the scope of the National Environmental System - SISNAMA.

State Decree 2,141/2006 - Regulates provisions of State Law No. 6,462 of July 4, 2002, which provides for the State Policy on Forests and other Forms of Vegetation and makes other provisions, with the objective of encouraging the recovery of altered and/or degraded areas and the restoration of legal reserves, for energy, timber, fruit, industrial, or other purposes, through forest and agroforestry reforestation with native and exotic species, and makes other provisions.

COEMA Resolution No. 54, dated October 24, 2007 (ANNEX 1): Approves the list of endangered flora and fauna species in the State of Pará.

State Decree 1,148/2008 and subsequent amendments - Provides for the Rural Environmental Registry (CAR-PA), Legal Reserve area, and other measures.

Law 12,187/2009 - Federal Law establishing the National Policy on Climate Change (PNMC) and other measures.

State Law 7,389/2010 - State Law defining activities with local environmental impact in the State of Pará and other provisions.

State Law 7,381/2010 - State Law that provides for the restoration of vegetation cover and riparian forests in the State of Pará.

State Decree 2,099/2010 - Provides for the maintenance, restoration, natural regeneration, compensation, and composition of the Legal Reserve area of rural properties in the State of Pará and other measures.

Law 12,651/2012 - Brazilian Forest Code establishing general rules on the protection of vegetation, Permanent Preservation Areas, and Legal Reserve Areas; forest exploitation, the supply of forest raw materials, the control of the origin of forest products, and the control and prevention of forest fires, and providing for economic and financial instruments to achieve its objectives.

Law 12.727/2012* amends Law No. 12.651/2012 - Federal Law establishing general rules on the protection of vegetation, Permanent Preservation Areas, and Legal Reserve Areas; forest exploitation, the supply of forest raw materials, the control of the origin of forest products, and the control and prevention of forest fires, and provides economic and financial instruments to achieve its objectives.

MMA Ordinance No. 370, dated December 2, 2015 - Establishes the National Strategy for Reducing Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conservation of Forest Carbon Stocks, Sustainable Forest Management, and Increase in Forest Carbon Stocks (REDD+) in Brazil-ENREDD+.

State Decree 254/2019 - Establishes the Pará Forum on Climate Change and Adaptation (FPMAC).

State Decree No. 59/2019 - Regulates the Pará State Environment Council (COEMA/PA) and provides other measures.

MMA Ordinance No. 544, dated October 26, 2020 - Publishes the Internal Regulations of the National Commission for the Reduction of Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conservation of Forest Carbon Stocks, Sustainable Forest Management, and Increase in Forest Carbon Stocks - REDD+.

Law 14,119/2021 - Federal Law establishing the National Policy on Payment for Environmental Services; and amends Laws 8,212/1991, 8,629/1993, and 6,015/1973, to bring them into line with the new policy.

MMA Ordinance No. 148, of June 7, 2022 - Recognizes the following official lists: "Official National List of Endangered Flora Species," "Official List of Endangered Brazilian Fauna Species," and "Official List of Extinct Brazilian Fauna Species."

State Ordinary Law 9,781/2022 - Amends State Law No. 9,048, of April 29, 2020, which establishes the State Policy on Climate Change in Pará (PEMC/PA).

State Decree No. 2,596/2022 - Regulates the registration of forestry activities, the State Environmental Information Management System, and the license for the transport of forest products and by-products in the State of Pará.

Decree No. 11,548/2023 - Establishes the National Commission for the Reduction of Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conservation of Forest Carbon Stocks, Sustainable Forest Management, and Increase in Forest Carbon Stocks - REDD+.

Decree No. 11,550/2023 - Provides for the Interministerial Committee on Climate Change.

Ordinary Law No. 10,750/2024 - Establishes the Amazon Now State Plan (PEAA).

Law 15,042/2024 - Establishes the Brazilian Emissions Trading System and provides other measures.

Applicability to the project: The farms where the Caiarara REDD+ Project is located have Permanent Preservation Areas and Legal Reserve Areas in accordance with current legislation. In addition, according to the diagnoses carried out in the area, it was possible to identify endangered species of Brazilian fauna and flora, leading the project to recognize protection and preservation activities in accordance with the aforementioned legislation.

It is worth noting that the Ambipar Group has a commitment to the environment, recorded in its Code of Conduct and Compliance, which makes it clear that all its activities and those of its collaborators must strictly comply with environmental legislation and standards, seeking to optimize natural resources.

For its part, Dow is committed to complying with environmental legislation, standards, and Dow requirements in all regions where it operates. In cases where local or national laws, codes, or regulations impose additional requirements beyond Dow standards, the affected facilities comply with the legislation and Dow requirements, whichever is more stringent.

In addition to complying with legislation, Dow Brazil adopts voluntary measures and is committed to its environmental performance, with a long-standing commitment to achieving Sustainability Goals, circular economy, and climate protection, among others. To demonstrate its commitment, Dow also publishes Periodic Progress Reports, such as the one found at the link: [2023 INtersections Progress Report](#). An important highlight is its adherence to the most established environmental/climate reporting format, called CDP - Carbon Disclosure Project, whose reports can be found at the link: [Dow's CDP Scores and Submissions](#).

Theme: Combating corruption

Related legislation:

Law 12.846/2013 – Anti-Corruption Law: Provides for the administrative and civil liability of legal entities for acts against the public administration, national or foreign, and other measures.

Applicability to the project: Any action by the Caiarara REDD+ Project is contrary to acts of corruption or conduct aimed at personal, societal, or governmental benefit, and all those involved are responsible for refraining from such practices.

Theme: International agreements

CITES, dated March 3, 1973: “Convention on International Trade in Endangered Species of Wild Fauna and Flora,” signed in Washington, D.C., on March 3, 1973, amended in Bonn on June 22, 1979.

FCCC/CP/2005/Misc.1: Reducing emissions from deforestation in developing countries: approaches to stimulate action. Submission from Parties. (COP 11, Montreal, 2005.)

FCCC/CP/2007/6/add.1: Report of the Conference of the Parties on its thirteenth session, held in Bali from December 3 to 15, 2007. Addendum. Part two: Action taken by the Conference of the Parties at its thirteenth session. (“Action Bali Plan”. COP 13, Bali, 2007.)

FCCC/CP/2009/Add.1: Report of the Conference of the Parties on its fifteenth session, held in Copenhagen from December 7 to 19, 2009. Addendum. Part Two: Action taken by the Conference of the Parties at its fifteenth session. (“Copenhagen Accord.” COP 15, Copenhagen, 2009.)

FCCC/CP/2010/7/Add. 1: Report of the Conference of the Parties on its sixteenth session, held in Cancun from November 29 to December 10, 2010. Addendum. Part Two: Action taken by the Conference of the Parties at its sixteenth session. (“Cancun Agreement.” COP 16, Cancun, 2010.)

FCCC/CP/2011/9/Add. 1: Report of the Conference of the Parties on its seventeenth session, held in Durban from November 28 to December 11, 2011. Addendum. Part Two: Action taken by the Conference of the Parties at its seventeenth session. (COP 17, Durban, 2011.)

FCCC/CP/2012/8/Add.1: Report of the Conference of the Parties on its eighteenth session, held in Doha from November 26 to December 8, 2012. Addendum. Part Two: Action taken by the Conference of the Parties at its eighteenth session.

FCCC/CP/2013/Add.1: Warsaw Framework for REDD-plus, held in Warsaw, Poland, from November 11 to 22, 2013, in particular the following decisions:

Decision 9/CP.19: Work program on results-based finance to progress the full implementation of the activities referred to in decision 1/CP. 16, paragraph 70.

Decision 10/CP.19: Coordination of support for the implementation of activities in relation to mitigation actions in the forest sector by developing countries, including institutional arrangements.

Decision 12/CP.19: The timing and frequency of presentations of the summary of information on how all the safeguards referred to in Decision 1/CP.16, Appendix I, are being addressed and respected.

Decision 13/CP.19: Guidelines and procedures for the technical assessment of submissions from Parties on proposed forest reference emission levels and/or forest reference levels.

Decision 14/CP.19: Modalities for measuring, reporting, and verifying.

Decision 15/CP.19: Addressing the drivers of deforestation and forest degradation.

FCCC/CP/2015/Add.1: Report of the Conference of the Parties on its twenty-first session, held in Paris from November 30 to December 13, 2015. Addendum. Part two: Action taken by the Conference of the Parties at its twenty-first session.

FCCC/CP/2015 Paris Agreement: Global, legally-binding agreement that sets out a global framework to avoid dangerous climate change by limiting global warming to well below 2°C and pursuing efforts to limit it to 1.5°C. Entry into force on November 4, 2016.

FCCC/CP/2016 Decisions adopted by the Conference of the Parties (COP): Especially decisions 1 (preparation into force of the Paris Agreement), 3 (Warsaw International Mechanism for Loss and Damage associated with Climate Change Impacts), 6 (National adaptation plans) and 7 (Long-term climate finance).

FCCC/CP/2017, FCCC/CP/2018, FCCC/CP/2019 Decisions adopted by the COP: Especially decision 1 reporting on developments of the implementation of the Paris Agreement.

Article 6 of the Paris Agreement (2021): Decision 1/CP.21 mandated the SBSTA to operationalize the provisions of this Article through recommending a set of decisions to the COP serving as the meeting of the Parties to the Paris Agreement at its first session. At COP26, the Parties to the Paris Agreement at its third session (CMA 3) adopted three main decisions related to Article 6: decision 2 (on Article 6.2), decision 3 (on Article 6.4) and decision 4 (on Article 6.8).

Glasgow Leaders' Declaration on Forests and Land Use (2021): Signatories (including Brazil) promise to reverse and end deforestation by 2030.

Brazilian Nationally Determined Contribution (NDC): First Brazilian NDC submitted in September 2015 to the UN Framework Convention on Climate Change for mitigation, adaptation, and means of implementation, in a manner consistent with the purpose of contributions to achieve the ultimate objective of the Convention, pursuant to Decision 1/CP.20, paragraph 9. The updated Brazilian NDC was presented at COP26 on December 8, 2022.

Applicability to the project: The Caiarara REDD+ Project is in line with international agreements, aiming to achieve objectives related to climate change, combating deforestation, protecting biodiversity, sustainable land use, and safeguards, such as the Paris Agreement and other reports from the Conferences of the Parties (COPs).

2.5.2 Relevant laws and regulations related to worker rights (VCS, 3.18, 3.19; CCB, G3.11)

The Caiarara REDD+ Project ensures that all employees of Biofilica Ambipar Environmental Investments S/A, Dow Brasil, and service providers are legally hired in compliance with Brazilian labor legislation. In addition, international agreements ratified by Brazil and issues related to worker welfare are respected.

Compliance with labor standards and laws applied by Biofilica Ambipar Environmental Investments S/A is verified annually by an audit, due to the fact that it is a publicly traded company. In turn, Dow Brazil recognizes and respects all applicable labor and employment laws, including those addressing freedom of association, privacy, and equal employment opportunities. As a publicly traded company, Dow issues detailed financial statements and other reports on best governance practices, including labor issues.

After hiring and before the employee begins work, training and capacity building on technical procedures are provided, and empowerment regarding their rights and applicable laws is promoted. In addition, employees are encouraged to join the institution responsible for defending their rights, the unions respective to their area of work.

The Caiarara REDD+ Project is committed to complying with current labor laws, providing quality of life to its employees. Listed below are the relevant laws and regulations that protect workers' rights in Brazil, as well as international agreements ratified by Brazil on labor issues.

Federal legislation and regulations

Ambipar, in accordance with the Federal Constitution, with special attention to item IV, art. 3, art. 5, and art. 7, as well as current legislation and the Ambipar Group's compendium of policies and statutes, is against any type of discriminatory act based on the individual characteristics of each person. Thus, we value equality and diversity among all Ambipar Environment employees and stakeholders.

In addition to the above regarding attention and respect for the provisions of the Constitution and the policies and statutes already in force by the Ambipar Group, we emphasize special respect for legislation and infra-legal diplomas concerning the issue of equal treatment, such as:

- Decree-Law No. 5,452/1943, which approves the Consolidation of Labor Laws;
- Law No. 14,611/2023, which provides for equal pay and remuneration criteria between women and men; and amends the Consolidation of Labor Laws, approved by Decree-Law No. 5,452, of May 1, 1943;
- Law No. 7,716/1989, which defines crimes resulting from discrimination or prejudice based on race, color, ethnicity, religion, or national origin;
- Law No. 12,288/2010, which establishes the Statute of Racial Equality;
- Law No. 10,741/2003, which provides for the Statute of the Elderly;
- Law No. 13,146/2015, which establishes the Brazilian Law for the Inclusion of Persons with Disabilities;
- Law No. 8,069/1990, which provides for the Statute of Children and Adolescents;

- Law No. 11,788/2008, which provides for student internships, and Law No. 10,097/2000, which provides for Young Apprentices; and
- Law No. 9,029/1995, which provides for the prohibition of requiring pregnancy and sterilization certificates, and other discriminatory practices, for admission purposes.

We emphasize that any violation of the above-mentioned legislation and/or policies and statutes already in force by the Ambipar Group may result in the termination of the employment contract or commercial relationship, without excluding any applicable civil, administrative, and criminal measures.

International agreements ratified by Brazil

International Labor Organization Convention No. 87 of 1940 - Provides for freedom of association.

International Labor Organization Convention No. 98 of 1949, ratified by Brazil on November 18, 1952 - Provides for the right to unionize and collective bargaining.

International Labor Organization Convention No. 29 of 1930, ratified by Brazil on April 25, 1957 - Provides for the abolition of forced labor.

International Labor Organization Convention No. 100 of 1951, ratified by Brazil on April 25, 1957 - Provides for equal remuneration between men and women.

International Labor Organization Convention No. 111 of 1958, ratified by Brazil on March 1, 1965 - Provides for discrimination in employment and occupation.

International Labor Organization Convention No. 97 of 1949, ratified by Brazil on June 18, 1965 - Provides for migrant workers.

International Labor Organization Convention No. 105, ratified by Brazil on June 18, 1965 - Provides for the abolition of forced labor.

International Labor Organization Convention No. 143 of 1975 - Provides for immigration under abusive conditions and the promotion of equal opportunities for migrant workers.

International Labor Organization Convention No. 142 of 1975, ratified by Brazil on November 24, 1981 - Provides for the development of human resources.

International Labor Organization Convention No. 131 of 1970, ratified by Brazil on May 4, 1983 - Provides for the establishment of minimum wages, especially in developing countries.

International Labor Organization Convention No. 155 of 1981, ratified by Brazil on May 18, 1992 - Provides for the safety and health of workers.

International Labor Organization Convention No. 182, ratified by Brazil on February 2, 2000 - Provides for the prohibition of the worst forms of child labor and immediate action for its elimination.

International Labor Organization Convention No. 138 of 1973, ratified by Brazil on June 28, 2001 - Provides for the minimum age for admission.

International Labor Organization Convention No. 169 of 1989, ratified by Brazil on July 25, 2002 - Provides for indigenous and tribal rights.

In addition to international legislation and agreements, the Caiarara REDD+ Project is based on well-structured policies that guide the Program's management practices. For example, the Ambipar Group's Code of Conduct and Compliance, which aims to guide all parties involved in the adoption of good practices in relationships and business conduct, based on ethical and moral principles. Thus, the document provides guidelines on topics such as hiring, compensation and benefits, training, labor and union relations, modern slavery, moral harassment, and workplace quality, among others. Another example is the Dow Brazil Code of Conduct, which summarizes many of the ethical principles and policies created to address issues of equal employment opportunity and respect in the workplace, inclusion, diversity, labor practices, and human rights, as well as health and safety in the workplace. More information on ethics and workers' rights can be found in the section 2.4.5.

2.5.3 Human Rights (VCS, 3.19)

The Caiarara REDD+ Project implementation area does not cover any region where traditional communities reside, as it is implemented on private properties, whose responsible company has signed a long-term agreement with Biofilica Ambipar Environmental Investments S/A for the conservation of surplus vegetation areas within its properties. Although there are no Indigenous Peoples or traditional communities present, the principles of the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention No. 169 on Indigenous and Tribal Peoples are recognized and voluntarily incorporated as ethical guidelines. Should relevant IPs, LCs, or customary rights arise in the future, the Project will adopt processes based on Free, Prior, and Informed Consent (FPIC) in accordance with the aforementioned international instruments.

The Caiarara REDD+ Project is guided by the policies and codes of conduct of the Ambipar Group and the Dow Group, as described in the section 2.3.13, and is committed to the stakeholders located in its surroundings, acting on the following premises:

- Encourage respect for people, their traditions, and values;
- Adopt and maintain a transparent process for defining social actions;
- Respect the interests and needs of the communities in which it operates;
- Collaborate with knowledge and technical information that can bring public benefits;
- Preserve and guarantee the political freedom and freedom of expression of employees;
- Guide the relationship between representatives of the Caiarara REDD+ Project and authorities, politicians, and public officials through transparent, professional, and correct attitudes, and any form of pressure or request by public officials that does not correspond to this definition must be immediately rejected and communicated to the management of Biofilica Ambipar Environmental Investments S/A;

- Guide the relationship by respecting laws and conventions that deal with fundamental human rights and the protection of sustainability. In all their relationships, representatives of the Caiarara REDD+ Project must observe ethical, health, safety, and human rights standards and social and environmental responsibility.

2.5.4 Indigenous Peoples and Cultural Heritage (VCS, 3.18, 3.19)

Based on mapping in government databases and the socioeconomic diagnosis carried out in the project implementation region in 2024, in which all residents within a 20km buffer zone of the Project area were identified, it was concluded that the Caiarara REDD+ Project area does not encompass any areas inhabited by traditional communities or indigenous peoples, nor were any traditional or indigenous communities identified around the area or dependent on it.

However, the Project recognizes that it can play a crucial role in preserving and protecting the rich cultural heritage of the Amazon. In this context, the project's scope of activities is aligned with the socio-environmental and economic objectives focused on the identified stakeholders, such as activities to strengthen local governance, develop and strengthen value chains, improve the well-being of stakeholders, promote education and strengthen inclusion, diversity, and equity, aiming at forest conservation and sustainable development, consequently valuing local cultural heritage and preserving and protecting the culture in the region. In addition, the Project is committed to acting in accordance with protocols and competent bodies if cultural elements are identified in the future.

2.5.5 Recognition of Property Rights (VCS, 3.7, 3.18, 3.19; CCB, G5.1)

The Caiarara REDD+ Project recognizes and respects all property rights, complying with significant statutory and regulatory requirements, as well as having the necessary approvals from state and local authorities.

Dow Brasil owns the rights to use and exploit the properties involved in the Project economically, as well as obtaining the right to access the natural resources existing therein under the terms of the Brazilian Federal Constitution and Civil Code, by virtue of being the owner of the properties where the Caiarara REDD+ Project is located. All rights were guaranteed through the signing of a contract, in accordance with the relevant legal requirements, with the free, prior, and informed consent of all identified stakeholders. The documents proving ownership and the contracts formalized between the parties were shared with the audit team on a confidential basis.

There are no communities residing in the Project Area and surrounding areas that depend on it for their survival and/or reproduction of customs and traditions, and therefore there are no customary and/or informal rights. Furthermore, there was no need to support or strengthen third-party rights.

2.5.6 Benefit Sharing Mechanism (VCS, 3.18, 3.19)

The Caiarara REDD+ Project is carried out on duly titled private properties, without overlapping with areas belonging to traditional communities, indigenous peoples, or squatters. All areas involved have regular land documentation, and there are no conflicts of use or third-party claims on the land. Although there is

no impact on property rights, the project provides for the involvement of neighboring communities through socio-environmental and productive actions, as described in Section 2.3.9. The community engagement plan has been validated with participants and includes initiatives that promote social and environmental benefits in a transparent manner.

2.5.7 Free, prior, and informed consent (VCS, 3.18, 3.19; CCB, G5.2)

Free, prior, and informed consent will be obtained throughout the project's life cycle, always using an approach based on dialogue and consent between the parties involved. The proponents ensure that all actors who may be impacted in any way by the Caiarara REDD+ Project have been consulted. It is declared that there are no traditional communities, indigenous peoples, or customary rights holders in the Project Area or in the zone of direct influence, according to field checks (STA, 2024) and government databases. The table below summarizes the results of the CLPI process as part of the stakeholder consultation process.

Description of the process for obtaining consent	Consent was obtained from all key stakeholders, especially the landowner, who is the main stakeholder in terms of providing the land and facilitating the implementation of the project, who is fully informed and in agreement about their participation, role, and responsibilities within the project. This was achieved by the proponent through several rounds of discussions and negotiations with the aim of providing a general explanation of the project and its objectives, timeline, conditions, projections, and shared benefits. This step took place in parallel with all the aforementioned documentary analyses on ownership and cross-referencing with all public boundaries to mitigate any risk of conflicts or false information. The consent of the proponent and landowner was formalized through a contract between the project proponents. Workshops were held during the field implementation of the socio-environmental diagnostics in order to convey information about the Project to the communities neighboring the Project's area and to farm workers. Consultations and questionnaires were also conducted on their opinions regarding the Project and its activities, as described in Section 2.3.9. Project's information was presented in an accessible manner and in language adapted to the local reality. Future consultations related to stakeholder consent will continue to be held throughout the Project's life cycle and will be formally documented, in accordance with good traceability practices. All information about the Caiarara REDD+ Project can be obtained
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	through Biofilica Ambipar Environmental Investments S/A's virtual channels, such as its website, newsletter, and social media.
Outcome of the CLPI process	The Project is located in a private area (private property) with no residents or traditional communities within its boundaries. The Project does not intend to develop any activities on private properties belonging to third parties, traditional and indigenous communities, or the government without their request. Regarding social and biodiversity activities, it is ensured that no activity will be carried out without the free, prior, and informed consent of the stakeholders. No action related to the Project will result in the relocation of activities that are important to the culture or livelihoods of property rights holders, nor will it aim at the involuntary relocation or removal from their lands or territories. Any proposed removal or relocation must be carried out only after obtaining the free, prior, and informed consent of the appropriate Property Rights Holders.

2.5.8 Protection of Property Rights (VCS, 3.18, 3.19; CCB, G5.3)

The proponent of the Caiarara REDD+ Project holds the right to use the properties where the Project is located, as they are privately owned and have all the documentation proving land ownership. Additionally, as identified in the socioeconomic diagnosis, there are no communities or families residing anywhere on the property or in the project's area. Even though there are no communities present, the Project recognizes and respects the rights of the communities that live in the vicinity of the Project's Area.

As described in Section 2.5.7, no activity related to the Project has resulted in the involuntary removal or relocation of Property Rights Holders from their lands or territories, nor has it forced them to relocate activities important to their culture or livelihoods.

2.5.9 Identification of Illegal Activity (VCS, 3.19, CCB, G5.4)

Deforestation is the main illegal activity that can negatively impact the development of the Caiarara REDD+ Project, as well as the predatory exploitation of fauna and flora. The socioeconomic diagnosis identified the members of the communities surrounding the project area as the main perpetrators of this illegal deforestation, who enter the area to remove wood for fence construction and charcoal production, in addition to engaging in illegal hunting and fishing, even though the perimeter of the forest area is adequately marked prohibiting the entry of unauthorized persons and hunting and fishing activities.

The Project seeks to control and combat illegal activities commonly found in the Project region through mitigating and preventive measures such as strengthening property surveillance, as well as encouraging the involvement of other actors and stakeholders, social inclusion, and local socioeconomic development by encouraging economic activities that are alternatives to deforestation and discouraging hunting and predatory fishing.

With the implementation of these measures encouraged by the Project's activities, it is hoped that the well-being of communities will be improved without placing a burden on the native forest and local biodiversity. Property surveillance aims to curb illegal practices such as deforestation, the extraction of plant species, and the hunting and capture of wild animals by third parties. The mechanisms and procedures for preventing illegal activities are summarized in Table 5.

Table 5. Mechanisms and procedures for the prevention of illegal activities in the Caiarara REDD+ Project

MECHANISMS AND PROCEDURES FOR THE PREVENTION OF ILLEGAL ACTIVITIES	
General conditions	- Conduct regular patrols to ensure the protection of Dow Brazil's land assets; - Prevent deforestation, forest fires, or other acts of environmental aggression; - Prevent illegal logging and timber trade; - Maintain good relations with surrounding communities and other stakeholders; - Control entry and exit from the RAAI and RAAll farms; - Promote environmental education with local actors who practice predatory hunting and fishing and encourage the practice of alternative and sustainable economic activities; - Request support from police and enforcement authorities when necessary; - Activate the legal sector to take action; - Record incidents involving trespassing, property damage, and illegal extraction of forest products; - Incidents involving environmental damage must be reported to the responsible agencies (IBAMA, Environmental Police, etc.); - In all situations involving land conflicts, it is necessary to avoid confrontation between the parties, respecting the laws in force in the country.
Forms of registration	- Incident report; - Photographic record of incidents; - Monthly monitoring program;

- Report on property security activities.

2.5.10 Ongoing disputes (VCS, 3.18, 3.19; CCB, G5.5)

In the Caiarara REDD+ Project area, there have been no ongoing or unresolved conflicts or disputes over land, territory, and resource rights, and any disputes that have been resolved over the last twenty years, as verified through consultation with local registries and land surveys. Thus, the project's properties have as their legitimate owners and possessors the company Dow Brasil, the project's proponent. In addition, it is emphasized that no project activity that could prejudice the outcome of any dispute relevant to the project will be initiated in the event of an unresolved conflict.

In the unlikely event of future disputes over land ownership, the project will implement activities to assist in their resolution and to clarify emerging claims, in order to pacify the situation and allow for the consolidation of the projects developed, in accordance with the conflict management procedure implemented, described in Section 2.3.10 and 2.3.14

3 CLIMATE

3.1 Monitoring GHG emission reductions and removals

3.1.1 Data and parameters available for validation (VCS, 3.16)

Data/parameter	APA-UDef
Data unit	ha
Description	Project area in which unplanned deforestation is to be avoided
Data source	Calculated in GIS
Applied value	32,625
Justification for the choice of data or description of the measurement methods and procedures applied	The Project Area was delimited in accordance with the requirements established in both the VM0048 methodology and the VMD0055 module, ensuring compliance with eligibility criteria, forest integrity, and the definition of spatial boundaries.
Purpose of the data	Calculation of project's emissions
Comment	None

Data/parameter	DLF
Data unit	
Description	Leakage Displacement Factor
Data source	Determined by the proponent
Applied value	15
Justification for the choice of data or description of the measurement methods and procedures applied	In accordance with Section 3.15.14 (VCS Standard v4.7), for validation, projects may apply the standard optional deduction of 15% for leakage due to activity displacement on gross GHG emission reductions and/or removals
Purpose of the data	Leakage calculation
Comment	None

Data/parameter	EA _{EF,t}
Data unit	
Description	Ex-ante effectiveness of baseline emission reductions in year <i>t i</i>
Data source	Determined by the proponent
Applied value	83
Justification for the choice of data or description of the measurement methods and procedures applied	The Project decided to apply a conservative discount of 17% to consider the risk of the Project's ineffectiveness in preventing deforestation. This value was based on the risk of the Project's permanence
Purpose of the data	• Calculation of project's emissions
Comment	None

3.1.2 Monitored Data and Parameters (VCS, 3.16)

Data/parameter	AD _{BSL} ,PA-UDef
Data unit	ha/year
Description	Data on unplanned deforestation activities allocated to the UDef PA
Data source	Verra (through the AD Baseline Allocation report and VT0007)
Description of measurement methods and procedures to be applied	Approach determined and applied by CTress
Frequency of monitoring/recording	Must be renewed in 2027. Subsequent baselines will be renewed at least every six years
Applied value	215.68
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	• Baseline emissions calculation

Calculation method	Application of Appendix 1 (VMD0055) by Verra
Comments	None

Data/parameter	AD _{BSL, LB-UDef}
Data unit	ha/yr
Description	Data on unplanned deforestation activities allocated to UDef LB
Data source	Verra (VT0007 and Appendix 1 VMD0055)
Description of measurement methods and procedures to be applied	VT0007 and Appendix 1 (VMD0055)
Frequency of monitoring/recording	Must be renewed in 2027. Subsequent baselines will be renewed at least every six years
Applied value	480.30
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Leakage calculation
Calculation method	Application of Appendix 1 (VMD0055) and VT0007 by Verra
Comments	None

Data/parameter	$C_{AB_tree,i}$
Data unit	tCO _{2e} /ha
Description	Forest carbon stock in aboveground pool in stratum i
Data source	The average carbon stock in above-ground tree biomass per unit area is estimated based on field measurements
Description of measurement methods	The carbon stock in the above-ground pool was estimated using field data collected through forest inventory, allometric equations, and carbon conversion factors, in accordance with the criteria of modules

and procedures to be applied	VMD0001 and VMD0055. All procedures used are described in the Carbon Stock Report.
Frequency of monitoring/recording	Every ten years, in accordance with Section 4.1 of Module VMD0001
Applied value	559.86
Monitoring equipment	See the Carbon Stock Report (to be provided as evidence)
QA/QC procedures to be applied	See VMD0001
Purpose of the data	• Baseline emissions calculation • Calculation of project's emissions • Leakage calculation
Calculation method	See VMD0001
Comments	None

Data/parameter	$C_{BB_tree,i}$
Data unit	tCO _{2e} /ha
Description	Forest carbon stock in the below-ground reservoir in stratum i
Data source	The average carbon stock in above-ground tree biomass per unit area is estimated based on field measurements
Description of measurement methods and procedures to be applied	The carbon stock in the above-ground reservoir was estimated using field data collected through forest inventory, allometric equations, and carbon conversion factors, in accordance with the criteria of modules VMD0001 and VMD0055. All procedures used are described in the Carbon Stock Report.
Frequency of monitoring/recording	Every ten years, in accordance with Section 4.1 of Module VMD0001
Applied value	134.37
Monitoring equipment	Refer to the Carbon Stock Report (to be provided as evidence)
QA/QC procedures to be applied	See MD0001

Purpose of the data	• Baseline emissions calculation • Calculation of project's emissions • Leakage calculation
Calculation method	See VMD0001
Comments	None

Data/parameter	$C_{DW,i}$
Data unit	tCO _{2e} /ha
Description	Forest carbon stock in dead wood reservoir in stratum i
Data source	The average carbon stock in dead wood per unit area is estimated based on field measurements
Description of measurement methods and procedures to be applied	The carbon stock in the dead wood reservoir was estimated using field data collected through forest inventory, allometric equations, and carbon conversion factors, in accordance with the criteria of modules VMD0002 and VMD0055. All procedures used are described in the Carbon Stock Report.
Frequency of monitoring/recording	Every ten years, in accordance with Section 4.1 of Module VMD0002
Value applied	67.74
Monitoring equipment	Refer to the Carbon Stock Report (to be provided as evidence)
QA/QC procedures to be applied	See VMD0002
Purpose of the data	• Baseline emissions calculation • Calculation of project's emissions • Leakage calculation
Calculation method	See VMD0002
Comments	None

Data/parameter	$\Delta C_{cAB-LI,i}$
Data unit	tCO _{2e} /ha
Description	Conservatively estimated emissions resulting from changes in carbon stocks in the UDef PA, in above-ground reservoirs and litter, in the forest stratum i
Data source	Average carbon stock estimated based on field measurements
Description of measurement methods and procedures to be applied	Emissions resulting from changes in carbon stocks in above-ground reservoirs were estimated using field data collected through forest inventory and, where necessary, applying a discount factor in accordance with the criteria of modules VMD0001 and VMD0055. All procedures used are described in the Carbon Stock Report.
Frequency of monitoring/recording	Every ten years, in accordance with Section 4.1 of Module VMD0001
Value applied	545.51
Monitoring equipment	See the Carbon Stock Report (to be provided as evidence)
QA/QC procedures to be applied	See VMD0001
Purpose of the data	• Baseline emissions calculation • Calculation of project's emissions
Calculation method	See VMD0001
Comments	- Since the discount factor is 0, the value applied is the same as for UDef LB - The carbon stock in the litter reservoir was not accounted for in the Project

Data/parameter	$\Delta C_{cBB-DW,i}$
Data unit	tCO _{2e} /ha
Description	Conservatively estimated emissions resulting from changes in carbon stocks over a 10-year period in the UDef PA, in below-ground biomass and dead wood reservoirs, in the forest stratum i

Data source	Average carbon stock estimated based on field measurements
Description of measurement methods and procedures to be applied	Emissions resulting from changes in carbon stocks in below-ground and dead wood reservoirs were estimated using field data collected through forest inventory and, where necessary, applying a discount factor in accordance with the criteria of modules VMD0001, VMD0002, and VMD0055. All procedures used are described in the Carbon Stock Report.
Monitoring/recording frequency	Every ten years, in accordance with Section 4.1 of Module VMD0001/VMD0002
Value applied	179.94
Monitoring equipment	Refer to the Carbon Stock Report (to be provided as evidence)
QA/QC procedures to be applied	See VMD0001 and VMD0002
Purpose of the data	• Baseline emissions calculation • Calculation of Project's emissions
Calculation method	See VMD0001 and VMD0002
Comments	- Since the discount factor is 0, the value applied is the same as for UDef LB

Data/parameter	$\Delta C_{SOC-WP,i}$
Data unit	tCO _{2e} /ha
Description	Conservatively estimated emissions resulting from changes in carbon stocks over a 20-year period in the UDef PA, in organic carbon soil and wood product reservoirs, in the forest stratum i
Data source	N/A
Description of measurement methods and procedures to be applied	Carbon stocks in soil organic carbon and wood products were not accounted for in the Project
Frequency of monitoring/recording	N/A

Value applied	0
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	• Baseline emissions calculation • Calculation of Project's emissions
Calculation method	N/A
Comments	None

Data/parameter	$C_{p,post,i}$
Data unit	tCO _{2e} /ha
Description	Estimated carbon stock in the post-deforestation reservoir p in forest stratum i
Data source	Published and/or peer-reviewed study: Fourth National Communication on Brazil's Biennial Update Report to the United Nations Framework Convention on Climate Change – Reference Report: Land Use, Land-Use Change, and Forestry Sector, 2020
Description of measurement methods and procedures to be applied	The estimated carbon stock after deforestation was calculated according to the criteria of module VMD0055
Frequency of monitoring/recording	Every ten years, in accordance with Section 4.1 of Module VMD0001
Value applied	179.94
Monitoring equipment	Refer to the Carbon Stock Report (to be provided as evidence)
QA/QC procedures to be applied	See VMD0001 and VMD0002
Purpose of the data	• Baseline emissions calculation • Calculation of project's emissions

Calculation method	See VMD0001 and VMD0002
Comments	None

Data/parameter	$\Delta C_{MP,PA-UDef}$
Data unit	tCO _{2e}
Description	Cumulative net GHG emissions in the UDef PA
Data source	Calculated by the project proponent
Description of measurement methods and procedures to be applied	See Equation 39 of Module VMD0055
Monitoring/recording frequency	Monitoring must be carried out at least every five years or before each verification event, when this occurs more frequently
Applied value	0
Monitoring equipment	N/A
QA/QC procedures to be applied	Described in the topic "Estimation of ex post changes in net carbon stock and greenhouse gas emissions"
Purpose of the data	Calculation of project's emissions
Calculation method	See Equation 39 of Module VMD0055
Comments	None

Data/parameter	$\Delta C_{LK-ME,t}$
Data unit	tCO _{2e}
Description	Cumulative net GHG emissions due to leakage from market effects in year t
Data source	Calculated by the project proponent using national databases on firewood production

Description of measurement methods and procedures to be applied	Emissions due to leakage from market effects were estimated using firewood production data obtained from a national database and applying a standard leakage factor of 0.4, in accordance with the criteria of module VMD0011
Frequency of monitoring/recording	Monitoring must be carried out at least every five years or before each verification event, when this occurs more frequently
Applied value	1,428
Monitoring equipment	N/A
QA/QC procedures to be applied	See VMD0011
Purpose of data	Leakage calculation
Calculation method	See VMD0011
Comments	None

Data/parameter	PROP _{MIG}
Data unit	Ratio
Description	Proportion of households in the project's activity area that are recent migrants and are engaging in land use activities identified as a baseline driver of unplanned deforestation
Data source	N/A
Description of measurement methods and procedures to be applied	PROP _{MIG} was assigned a conservative value in accordance with Section 5.3.4.4 Module VMD0055
Frequency of monitoring/recording	Must be renewed in 2027. Subsequent baselines will be renewed at least every six years
Applied value	1.0
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A

Purpose of data	Leakage calculation
Calculation method	N/A
Comments	None

Data/parameter	$\Delta C_{OLB,t}$
Data unit	tCO _{2e} /ha
Description	Emissions resulting from changes in carbon stocks due to land use transition in areas available for displacement of activities outside the UDef LB, as calculated for year t
Data source	Verra
Description of measurement methods and procedures to be applied	See Appendix 2 A2.2 of Module VMD0055
Frequency of monitoring/recording	Must be renewed in 2027. Subsequent baselines will be renewed at least every six years
Applied value	424.99
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Leakage calculation
Calculation method	See Appendix 2 A2.2 of Module VMD0055
Comments	None

Data/parameter	$\Delta C_{LK-AS,t}$
Data unit	tCO _{2e}
Description	Cumulative net GHG emissions from leakage due to displacement of unplanned deforestation activities in year t

Data source	Calculated by the project proponent
Description of measurement methods and procedures to be applied	See Equation 47 of Module VMD0055
Frequency of monitoring/recording	Monitoring must be carried out at least every five years or before each verification event, when this occurs more frequently
Applied value	0
Monitoring equipment	N/A
QA/QC procedures to be applied	Described in the topic "Estimation of ex post changes in net carbon stock and greenhouse gas emissions," Section 3.3.3 of the PDD
Purpose of the data	Leakage calculation
Calculation method	Equation 47 of Module VMD0055
Comments	None

Data/parameter	$\Delta C_{LK-Udef}$
Data unit	tCO ₂ e
Description	Cumulative net GHG emissions due to leakage from the project's activity in year t
Data source	Calculated by the project proponent
Description of measurement methods and procedures to be applied	See Equation 49 of Module VMD0055
Frequency of monitoring/recording	Monitoring must be carried out at least every five years or before each verification event, when this occurs more frequently
Applied value	1.428
Monitoring equipment	N/A

QA/QC procedures to be applied	Described in the topic “Estimation of ex post changes in net carbon stock and greenhouse gas emissions” Section 3.3.3 of the PDD
Purpose of the data	Leakage calculation
Calculation method	Equation 49 of Module VMD0055
Comments	None

Data/parameter	E _{BSL,BiomassBurn,i,t}														
Data unit	tCO _{2e}														
Description	Non-CO ₂ emissions due to biomass burning as part of the project's activities in forest stratum i in year t														
Data source	Calculated by the project's proponent using the parameters and criteria defined in Module VMD0013														
Description of measurement methods and procedures to be applied	See VMD0013														
Frequency of monitoring/recording	Must be renewed in 2027. Subsequent baselines will be renewed at least every six years														
Applied value	<table><tr><td>Start</td><td>End</td><td>E_{BSL,BiomassBurn,i,t}</td></tr><tr><td>1/21/2024</td><td>12/31/2024</td><td>11,303</td></tr><tr><td>1/1/2025</td><td>12/31/2025</td><td>11,303</td></tr><tr><td>1/1/2026</td><td>12/31/2026</td><td>11,303</td></tr></table>			Start	End	E _{BSL,BiomassBurn,i,t}	1/21/2024	12/31/2024	11,303	1/1/2025	12/31/2025	11,303	1/1/2026	12/31/2026	11,303
Start	End	E _{BSL,BiomassBurn,i,t}													
1/21/2024	12/31/2024	11,303													
1/1/2025	12/31/2025	11,303													
1/1/2026	12/31/2026	11,303													
Monitoring equipment	N/A														
QA/QC procedures to be applied	See VMD0013														
Purpose of data	Baseline emissions calculation														

Calculation method	Application of Equation 1 presented in Module VMD0013 for the calculation of non-CO ₂ GHG emissions. In the baseline scenario, the burned area ($A_{burn,i,t}$) is considered equal to $AD_{BSL,PA-UDEF,i,t}$
Comments	None

Data/parameter	$E_{MP,BiomassBurn,i,t}$											
Data unit	tCO _{2e}											
Description	Non-CO ₂ emissions due to biomass burning as part of the project's activities in forest stratum i in year t											
Data source	Calculated by the project's proponent using the parameters and criteria defined in Module VMD0013											
Description of measurement methods and procedures to be applied	See VMD0013											
Frequency of monitoring/recording	Monitoring must be carried out at least every five years or before each verification event, when this occurs more frequently											
Applied value	<table><tr><td>Start</td><td>End</td><td>$E_{MP,BiomassBurn,i,t}$</td></tr><tr><td>1/21/2024</td><td>12/31/2024</td><td>0</td></tr><tr><td>1/1/2025</td><td>6/30/2025</td><td>0</td></tr></table>			Start	End	$E_{MP,BiomassBurn,i,t}$	1/21/2024	12/31/2024	0	1/1/2025	6/30/2025	0
Start	End	$E_{MP,BiomassBurn,i,t}$										
1/21/2024	12/31/2024	0										
1/1/2025	6/30/2025	0										
Monitoring equipment	N/A											
QA/QC procedures to be applied	See VMD0013											
Purpose of data	● Calculation of Project's emissions											
Calculation method	Application of Equation 1 presented in Module VMD0013 for the calculation of non-CO ₂ GHG emissions											
Comments	None											

Data/parameter	AD _{MP,PA-UDEF,i,t}											
Data unit	ha											
Description	Deforestation in UDef PA in forest stratum i in year t											
Data source	Refer to Section 5.3.3.2 of Module VMD0055 and the Standard Operating Procedure (SOP) mentioned in the Monitoring Plan											
Description of measurement methods and procedures to be applied	Refer to Section 5.3.3.2 of Module VMD0055 and the Standard Operating Procedure (SOP) mentioned in the Monitoring Plan											
Frequency of monitoring/recording	Monitoring must be performed at least every five years or before each verification event, when this occurs more frequently											
Applied value	<table><tr><td>Start</td><td>End</td><td>AD_{MP,PA-UDEF,i,t}</td></tr><tr><td>1/21/2024</td><td>12/31/2024</td><td>0</td></tr><tr><td>1/1/2025</td><td>6/30/2025</td><td>0</td></tr></table>			Start	End	AD _{MP,PA-UDEF,i,t}	1/21/2024	12/31/2024	0	1/1/2025	6/30/2025	0
Start	End	AD _{MP,PA-UDEF,i,t}										
1/21/2024	12/31/2024	0										
1/1/2025	6/30/2025	0										
Monitoring equipment	N/A											
QA/QC procedures to be applied	Refer to Section 5.3.3.2 of Module VMD0055 and the Standard Operating Procedure (SOP) mentioned in the Monitoring Plan											
Purpose of the data	● Calculation of Project's emissions											
Calculation method	Refer to Section 5.3.3.2 of Module VMD0055 and the Standard Operating Procedure (SOP) mentioned in the Monitoring Plan											
Comments	None											

Data/parameter	AD _{MP,LB-UDEF,i,t}		
Data unit	ha		
Description	Deforestation in UDef LB in forest stratum i in year t		
Data source	Calculated by the project's proponent using the parameters and criteria defined in Section 5.3.3.2 of Module VMD0055		

Description of measurement methods and procedures to be applied	Refer to Section 5.3.3.2 of Module VMD0055 and the Standard Operating Procedure (SOP) mentioned in the Monitoring Plan									
Frequency of monitoring/recording	Monitoring must be carried out at least every five years or before each verification event, when this occurs more frequently									
Applied value	<table><tr><td>Start</td><td>End</td><td>AD_{MP, LB-UDEF, i, t}</td></tr><tr><td>1/21/2024</td><td>12/31/2024</td><td>94</td></tr><tr><td>1/1/2025</td><td>6/30/2025</td><td>47</td></tr></table>	Start	End	AD _{MP, LB-UDEF, i, t}	1/21/2024	12/31/2024	94	1/1/2025	6/30/2025	47
Start	End	AD _{MP, LB-UDEF, i, t}								
1/21/2024	12/31/2024	94								
1/1/2025	6/30/2025	47								
Monitoring equipment	N/A									
QA/QC procedures to be applied	Refer to Section 5.3.3.2 of Module VMD0055 and the Standard Operating Procedure (SOP) mentioned in the Monitoring Plan									
Purpose of the data	● Calculation of Project’s emissions									
Calculation method	Refer to Section 5.3.3.2 of Module VMD0055 and the Standard Operating Procedure (SOP) mentioned in the Monitoring Plan									
Comments	None									

Data/parameter	Buffer%
Data unit	%
Description	Percentage of retention in the buffer reserve
Data source	Calculated by the project's proponent using the VCS AFOLU Non-Permanence Risk Tool
Description of measurement methods and procedures to be applied	See VCS AFOLU Non-Permanence Risk Tool
Frequency of monitoring/recording	Monitoring must be carried out at least every five years or before each verification event, when this occurs more frequently

Value applied	17
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	VCU calculation
Calculation method	See VCS AFOLU Non-Permanence Risk Tool
Comments	None

Data/parameter	$NER_{UDef,t}$
Data unit	tCO _{2e}
Description	Ex post cumulative net reductions in GHG emissions from the project's activity in year t
Data source	Calculated by the project's proponent using Equation 52 of Module VMD0055.
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	Monitoring must be carried out at least every five years or before each verification event, when this occurs more frequently.
Applied value	199.208
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	VCU calculation
Calculation method	See Equation 52 of Module VMD0055
Comments	None

3.1.3 Monitoring Plan (VCS, 3.16, 3.20)

The monitoring plan applied is described in detail in item 4, "Estimation of Ex Post Net Carbon Stock Changes and Greenhouse Gas Emissions," in Section 3.3.3 of the PDD and includes the following information:

- Technical description of the monitoring task: details of the procedures and methodologies to be applied;
- Data to be collected: list of all data and parameters to be collected as described in the project;
- Overview of data collection procedures: explanation of how the collection will be carried out, including methods and tools;
- Quality assurance and quality control (QA/QC) procedures: measures to ensure the reliability, accuracy, and consistency of the data;
- Data archiving: organization and secure storage of collected data;
- Organization and responsibilities: clear definition of the responsibilities of the parties involved in the execution of all the above steps.

3.1.4 Disclosure of the Monitoring Plan and Results (VCS, 3.18; CCB, CL4.2)

The Climate Monitoring Plan and all its results were publicly disclosed on the VERRA registration page and on the Biofilica Ambipar Environmental Investments S/A website. The validated and verified monitoring results will be available in the public version of the report on the Verra website, ensuring free access to any interested party.

Summaries of the main results will be prepared in accessible language and presented in person to the communities through adapted printed and digital materials. Copies will also be made available at strategic locations (farm headquarters and community centers). The results will be disclosed after each official verification of the project, with regular updates following the monitoring cycles defined in the PDD.

For stakeholders who do not have internet access, a printed copy was provided at strategic locations (farm headquarters). Information about the Project was also disseminated via email and WhatsApp (with a pre-established contact list of stakeholders) and on the project proponent's social media, such as Instagram and LinkedIn.

In this way, it is hoped that all necessary efforts to provide full transparency and distribute key Project information and documents will be achieved.

3.2 Quantification of GHG Emissions Reductions and Removals

3.2.1 Project's Emissions (VCS, 3.15)

Deforestation data from UDef PA and UDef LB for the Monitoring Period were estimated by applying Steps 1-5 presented in item 4-c "Estimation of ex post changes in net carbon stock and greenhouse gas emissions" in Section 3.3.3 of the PDD.

Step 1: Analyse changes in land use types

In this step, the area of unplanned deforestation between the beginning and end of the monitoring period was estimated. To do this, reference data collected from a stratified sampling of the project area and the leakage belt, the so-called sampling frame, was used. Section 3.3.3 of the PDD summarises the reference data collection procedure, while the standard operating report provided to the VVB contains the detailed description of the methodology employed in full, including quality assessment and quality control (QA/QC) procedures. Here, the results and specific methodological details that are important for your understanding are emphasised.

Stratification of the sampling domain

Stratification aimed to partition the domain into regions with varying levels of deforestation risk and increase sampling efficiency. To this end, the following strata were defined (Figure 5): a) Detected deforestation; b) Detected degradation; c) Stable forest; d) Forest in an area at risk. These classes have different deforestation probabilities and are therefore useful for increasing sampling efficiency.

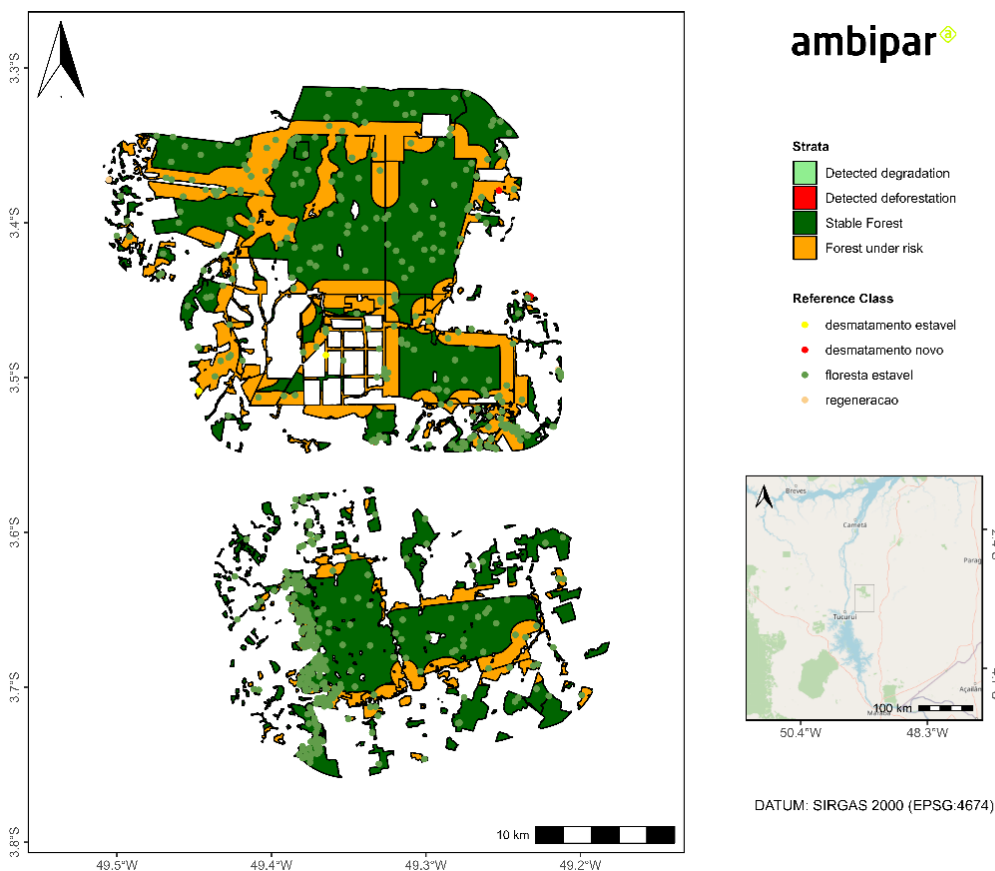


Figure 5. Stratification of the sampling domain with overlapping reference data (points).

The stratification was carried out with the support of public data from deforestation monitoring systems. The *Detected Deforestation* and *Detected Degradation* strata correspond to the areas classified as such in the period from 2024 to 2025, based on the land cover products and fire alerts in Collection 9 of the MapBiomas Project.

The *Stable Forest* stratum comprises all areas of forest cover where no sign of deforestation or degradation was detected by the systems used during the monitoring period.

In turn, the *Forest at Risk* stratum covers areas initially classified as stable forest, but which are located in regions with greater vulnerability to deforestation. This vulnerability is determined based on spatial criteria, such as proximity to roads and other access routes, the edges of the project area boundaries, and the neighbourhood of areas in the Detected Deforestation and Detected Degradation classes. A 1km buffer was drawn around these areas and all stable forests within this buffer were included in the *Forest at risk* stratum.

Figure 5 shows the result of the stratification, while Table 6 shows the area of each stratum.

Table 6. Area of each stratum.

Stratum name	Area (ha)	%	Area (ha)	%	Area (ha)	%
Detected Deforestation	4	0.0%	0	0.0%	4	0.0%
Detected Degradation	213	0.3%	22	0.1%	191	0.6%
Forest	39,529	63.7%	24,917	76.4%	14,612	49.7%
Forest under risk	22,309	36.0%	7,686	23.6%	14,622	49.7%
Total	62,054	100.0%	32,625	100.0%	29,429	100.0%

Area of each land use and land cover class

A sample size of 711 was distributed among the strata in proportion to a pre-estimate of the variance of the probability of deforestation in each stratum. The land use and land cover class at the beginning and end of the monitoring period was assessed by three interpreters trained in land use and land cover classification from Sentinel satellite images at 10 metre resolution. The images were downloaded from Google Earth Engine and table 7 shows the identification and date of each image. It is worth noting that, due to the presence of clouds, images with tolerable coverage were used, captured on close dates - but before the start and after the end of the monitoring period were used for discarding - in order to avoid underestimates in the detection of monitored deforestation.

Table 7. Identification of the satellite images used to interpret the land use and land cover reference data classes.

Resolution	Coverage	Acquisition date	Scene identifier
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Vector (Satellite or airplane)	Sens or	Spatial (m)	Spectral	(km2)	(DD/MM/YY)	Tile ID
Sentinel 2	MSI	10	0.49 – 0.84 µm	2.650	07/01/2025	"22MFA","22MFB ","22MGA","22M GB"
Sentinel 2	MSI	10	0.49 – 0.84 µm	1.850	01/09/2023 - 31/12/2023	"22MFA","22MFB ","22MGA","22M GB"

The three interpreters carried out their classifications completely independently, after training and an initial alignment section. The level of total agreement (all 3 interpreters) and partial agreement (at least 2 interpreters) was 92% and 98% respectively. The land use change class was determined when at least two interpreters had matching classifications. The results of each interpreter's classifications of the use and cover class at the beginning and end of the monitoring period, the respective use and cover change classes and the final use and cover class, for each sample, together with the analysis of agreement between interpreters was submitted to the VVB.

Table 8, Table 9 and Table 10, referring to Table 6 of Module VMD0055, show the classification results for sampling in the domain, in the project area and in the leakage belt.

Table 8. Reference classifications in the field of sampling (table 6 of module VMD0055).

Stratum name	Sampling Stratum (ss)	Change Categories (CHC)				Stratum total
		Unplanned deforestation (UDEP)	Regeneration (R)	Stable forest (SF)	Stable deforestation (SD)	
Detected Deforestation	1	2	0	3	0	5
Detected Degradation	2	0	0	9	0	9
Forest	3	1	22	408	16	447
Forest under risk	4	0	0	248	2	250
Total	10	3	22	668	18	711

Table 9. Reference classifications at UDef-PA (table 6 of module VMD0055).

Stratum name	Sampling Stratum (ss)	Change Categories in the UDef-AP (CHC)				Stratum total
		Unplanned deforestation (UDEP)	Regeneration (R)	Stable forest (SF)	Stable deforestation (SD)	
Detected Deforestation	1	0	0	0	0	0
Detected Degradation	2	0	0	2	0	2

Forest	3	0	0	170	0	170
Forest under risk	4	0	0	55	0	55
Total		0	0	227	0	227

Table 10. Reference classifications at UDef-LB (table 6 of module VMD0055).

		Change Categories in the Udef-LB (CHC)				
Stratum name	Sampling Stratum (ss)	Unplanned deforestation (UDEF)	Regeneration (R)	Stable forest (SF)	Stable deforestation (SD)	Stratum total
Detected Deforestation	1	2	3	0	0	5
Detected Degradation	2	0	7	0	0	7
Forest	3	1	238	16	22	277
Forest under risk	4	0	193	2	0	195
	Total	3	441	18	22	484

Table 11, Table 12 and Table 13, referring to table 7 of Module VMD0055, shows the result of calculating the sample weights for each stratum.

Table 11. Sample weights of the strata and estimated proportions of each land use and land cover change class (LCC) in the monitoring period in the sampling domain.

		Change Categories (CHC)				
Stratum name	Sampling Stratum (ss)	Unplanned deforestation (UDEF)	Regeneration (R)	Stable forest (SF)	Stable deforestation (SD)	Stratum total (W%)
Detected Deforestation	1	>0.01%	0.00%	0.00%	>0.01%	>0.1%
Detected Degradation	2	>0.01%	0.00%	0.34%	>0.01%	0.3%
Forest	3	0.14%	3.14%	58.14%	2.29%	63.9%
Forest under risk	4	>0.01%	0.00%	35.66%	0.29%	35.7%
Total		0.14%	3.14%	94.15%	2.57%	100.0%

Table 12. Sample weights of the strata and estimated proportions of each land use and land cover change class (LCC) in the monitoring period at UDef-PA.

		Change Categories (CHC)				
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Stratum name	Sampling Stratum (ss)	Unplanned deforestation (UDEF)	Regeneration (R)	Stable forest (SF)	Stable deforestation (SD)	Stratum total (W%)
Detected Deforestation	1	0.00%	0.00%	0.00%	0.00%	0.00%
Detected Degradation	2	0.00%	0.00%	0.07%	0.00%	0.07%
Forest	3	0.00%	0.00%	76.37%	0.00%	76.37%
Forest under risk	4	0.00%	0.00%	23.56%	0.00%	23.56%
	Total	0.00%	0.00%	100.00%	0.00%	100%

Table 13. Sample weights of the strata and estimated proportions of each land use and land cover change class (LCC) during the monitoring period at UDef-LB.

Stratum name	Sampling Stratum (ss)	Change Categories (CHC)				Stratum total (W%)
		Unplanned deforestation (UDEF)	Regeneration (R)	Stable forest (SF)	Stable deforestation (SD)	
Detected Deforestation	1	0.01%	0.01%	0.00%	0.00%	0.01%
Detected Degradation	2	>0.01%	0.65%	0.00%	0.00%	0.65%
Forest	3	0.18%	42.66%	2.87%	3.94%	49.65%
Forest under risk	4	>0.01%	49.18%	0.51%	0.00%	49.69%
	Total	0.18%	92.49%	3.38%	3.94%	100%

Only 3 of the 711 pixels sampled in the domain were from the deforestation class, resulting in an estimated proportion of unplanned deforestation (CHC = UDEF) of **0.14%** for the sampling domain.

For the project area, this proportion was **0%**, as no deforestation pixels were found. For the leakage belt, this proportion was **0.18%**.

Step 2: Calculation of the total area of each category

Using the data from the tables above with equations 24 and 25 from the VMD0055 module, the estimated areas of unplanned deforestation in UDef-PA and UDef-LB were **0 and 52 hectares** respectively.

Step 3: Calculate the uncertainty of the estimated areas of each category of change

The values calculated were **U%(AUDef-PA) = 0%** for UDef-PA and **U%(AUDef-PA) = 159%** for UDef-LB. The high uncertainty in UDef-LB is due to the small amount and unpredictability of deforestation during the monitoring period, which resulted in low effectiveness of the stratification used, even using the best deforestation drivers available as auxiliary variables in the stratification.

Step 4: Conservatively inflate the estimated area of unplanned deforestation

Para reduzir a incerteza para 100% do valor estimado, seriam necessárias 1400+ amostras. Ao invés de coletar mais amostras de referência, aplicamos um acréscimo de **159%** às estimativas de desmatamento da UDef-LB, correspondendo ao valor superior do intervalo de confiança da estimativa. Este valor é conservador em relação ao desconto que seria aplicado de acordo com as equações 29 e 30 do VMD0055.

Therefore, the increase values applied were $IF_{UDef-PA} = 0$ and $IF_{UDef-LB} = 83.53$, resulting in total values of $A_{PA, inflated, UDef} = 0$ e $A_{LB, inflated, UDef} = 140.9$ ha. To reduce the uncertainty to 100 per cent of the estimated value, 1400+ samples would be needed. Instead of collecting more reference samples, we applied an increase of 159% to the UDef-LB deforestation estimates, corresponding to the upper value of the estimate's confidence interval. This value is conservative compared to the discount that would be applied according to equations 29 and 30 of VMD0055.

Step 5: Determine the AD for unplanned deforestation

The values are broken down between the years of the monitoring period in the table below. The values in years 1 and 2 correspond to the total deforestation value for the deforestation period, distributed proportionally to the number of days in each year (365 and 181, respectively):

Table 14. AD values allocated to the Project Area (UDef PA), in hectares.

		ADMP,PA- UDEP,i,t (ha)
		Stratum i
BVP	t	Forest
1	1	0
1*	2	0

Table 15. AD values allocated to the Leakage Belt (UDef LB), in hectares.

		ADMP,LB- UDEP,i,t (ha)
		Stratum ID
BVP	t	Forest
1	1	94
1*	2	47

Estimated Annual Emissions from Unplanned Deforestation

Emissions from changes in carbon stocks in the Project Area and for the Leakage Belt were calculated respectively by equations 34 and 35 of Module VMD0055. The calculation procedures were presented

in Step 5 of item 4-c "Estimation of ex post changes in net carbon stock and greenhouse gas emissions" in Section 3.3.3 of the PDD.

Table 16. Carbon emissions from unplanned deforestation in UDef-PA.

BVP	t	Start	End	Total	$\Delta C_{MP,PA-UDef,1,t}$ (tCO _{2e})
1	1	21/01/2024	31/12/2024	0	0
1	2	01/01/2025	30/06/2025	0	0

Table 17. Carbon emissions from unplanned deforestation in UDef LB.

BVP	t	Start	End	Total	$\Delta C_{MP,LB-UDEF,1,t}$ (tCO _{2e})
1	1	21/01/2024	31/12/2024	53,077	53,077
1	2	01/01/2025	30/06/2025	27,161	27,161

Ex-post estimate of other GHG emissions at UDef-PA

Baseline emissions from the burning of fossil fuels and the use of nitrogen fertilisers were omitted for the period monitored. Emissions from biomass burning (Table 18) were calculated using the VMD0013 Estimation of Greenhouse Gas Emissions from Biomass and Peat Burning (E-BPB) module. Finally, other GHG emissions were estimated according to equations 38 in Module VMD0055 (Table 19). The calculation procedures were presented in item 4-c "Estimation of ex-post changes in net carbon stock and greenhouse gas emissions" in Section 3.3.3 of the PDD.

Table 18. GHG emissions from biomass burning in year t resulting from unplanned deforestation within the UDef PA in the monitoring period (t CO_{2e}).

BVP	t	Start	End	Total	$E_{MP,biomassBurn,1,t}$ (tCO _{2e})
1	1	21-Jan-2024	31-Dec-2024	0	0
1	2	1-Jan-2025	30-Jun-2025	0	0

Table 19. Other GHG emissions in year t resulting from unplanned deforestation UDef-PA in the monitoring period (t CO_{2e})

BVP	t	Start	End	Total	$GHG_{MP,PA-UDef,E,t}$ (tCO _{2e})
1	1	21-Jan-2024	31-Dec-2024	0	0
1	2	1-Jan-2025	30-Jun-2025	0	0

Estimated Net Emissions from Unplanned Deforestation and Degradation

The estimate of net emissions caused by unplanned deforestation and degradation was carried out using equations 39 (estimate for UDef PA) and equation 40 (estimate for UDef LB) of Module VMD0055. The calculation procedures were presented in item 4-c "Estimation of ex post changes in net carbon stock and greenhouse gas emissions" in Section 3.3.3 of the PDD.

Table 20. Estimated net emissions for UDef PA.

BVP	t	Start	End	$\Delta C_{MP,PA-UDef,1,t}$ (tCO _{2e})	GHG _{MP,PA-UDef,E,t} (tCO _{2e})	$\Delta C_{MP,PA-UDef,t}$ (tCO _{2e})
1	1	21-Jan-2024	31-Dec-2024	0	0	0
1	2	1-Jan-2025	30-Jun-2025	0	0	0

Table 21. Estimated net emissions for UDef LB.

BVP	t	Start	End	Total	$\Delta C_{MP,LB-UDEF,1,t}$ (tCO _{2e})
1	1	21-Jan-2024	31-Dec-2024	53,077	53,077
1	2	1-Jan-2025	30-Jun-2025	27,161	27,161

3.2.2 Leakage Emissions (VCS, 2.5, 3.2, 3.6, 3.15, 4.3)

Ex-post leakage monitoring was conducted to quantify unintended displacement of deforestation and associated greenhouse gas emissions outside the project area. Four types of leakage were assessed:

(i) Activity displacement by geographically restricted agents, monitored within the Leakage Belt (LB)

The difference in carbon stocks between the baseline scenario and the monitoring period within the UDef LB was calculated using Equation 41 of Module VMD0055. Other GHG emissions occurring in the UDef LB were also accounted for using Equations 42 and 43 of the Module.

Table 22. Emission factor for other GHG emissions at UDef LB during the monitoring period.

BVP	t	Start	End	Total	GHG _{LB,E,1,t} (tCO _{2eha⁻¹})
1	1	21-Jan-2024	31-Dec-2024	52	52
1	2	1-Jan-2025	30-Jun-2025	52	52

Table 23. Other GHG emissions at UDef LB during the monitoring period.

BVP	t	Start	End	Total	GHG _{MP,LK-UDEF,E,t} (tCO _{2e})
1	1	21-Jan-2024	31-Dec-2024	20,235	20,235
1	2	1-Jan-2025	30-Jun-2025	10,138	10,138

Finally, cumulative net leakage emissions were calculated for UDef LB using Equation 44 of Module VMD0055. The calculation procedures were presented in item 4-c “Estimation of ex post changes in net carbon stock and greenhouse gas emissions” in Section 3.3.3 of the PDD.

Table 24. Net leakage emissions in the monitoring period.

BVP	t	Start	End	Total	$\Delta C_{LK, LB}$ (tCO _{2e})
1	1	21-Jan-2024	31-Dec-2024	-197,335	-197,335
1	2	1-Jan-2025	30-Jun-2025	-102,346	-102,346

The value of $\Delta C_{LK, LB}$ for all years of the monitored period was negative, demonstrating that forest carbon stocks in UDef LB during the period remained higher than those projected in the baseline.

(ii) The displacement of activities by geographically mobile agents was estimated considering:

a) The net reduction in deforestation and emissions during the monitoring period within the UDef PA and possible displacement outside the UDef LB;

b) The proportion of geographically mobile deforestation agents projected in the baseline. This value is calculated considering the Proportion of Land Cover Transition Agents Migrated in the Baseline (PROPMIG). For the first baseline, a conservative value of 1.0 will be assigned in accordance with Section 5.3.4.4 of Module VMD0055.

c) Area-weighted emission factors for areas accessible to geographically mobile deforestation drivers outside the UDef LB, calculated and provided by Verra. For the first baseline, the value will be 424.99 tCO_{2e}.

The cumulative area outside the UDef LB that underwent land cover conversion due to displacement of activities was calculated using Equation 45 of Module VMD0055. The total cumulative emissions resulting from the change in carbon stock due to the displacement of activities to available areas outside the UDef LB were calculated by Equation 46 of the Module. The calculation procedures were presented in item 4-c “Estimation of ex post changes in net carbon stock and greenhouse gas emissions” in Section 3.3.3 of the PDD.

Table 25. Emissions from changes in carbon stocks due to the displacement of activities to available zones outside UDef LB in year t.

BVP	t	Start	End	Total	$\Delta C_{LK,OLB}$ (tCO _{2e})
1	1	Jan 21, 2024	Dec 31, 2024	91,661	91,661
1	2	Jan 1, 2025	June 30, 2025	45,454	45,454

Finally, total leakage emissions from activity displacement are the sum of leakage occurring within ($\Delta C_{LK,LB}$) and outside the UDef LB ($\Delta C_{LK,OLB}$) and were calculated using Equation 47 of Module VMD0055. As the sum is less than zero, total leakage emissions from activity displacement is zero. The calculation procedures were presented in item 4-c “Estimation of ex post changes in net carbon stock and greenhouse gas emissions” in Section 3.3.3 of the PDD.

Table 26. Net GHG leakage emissions due to displacement of unplanned deforestation activities in year t.

BVP	t	Start	End	Total	$\Delta C_{LK-AS,LB}$ (tCO _{2e})
1	1	Jan 21, 2024	Dec 31, 2024	0	0
1	2	Jan 1, 2025	June 30, 2025	0	0

(iii) Market effect leakage

This occurs when, in the baseline scenario, commodities were produced for regional, national, or international markets, related to the extraction of wood for timber, firewood or charcoal purposes. The collection of wood from native forests for charcoal production was identified through socioeconomic studies conducted in the project activity zone.

Municipal data provided by the Brazilian Institute of Geography and Statistics (IBGE) on plant extraction and forestry production were used. Based on data available between 2015-2023 in the municipalities within the project's area of activity (Table 27), the average annual production of charcoal was obtained and used as the main data for a conservative estimate of carbon emissions due to the displacement of firewood.

Table 27. Quantity produced in plant extraction (tons).

Municipality	2015	2016	2017	2018	2019	2020	2021	2022	2023
Baião (PA)	68	70	80	75	77	78	80	79	80
Breu Branco (PA)	2800	2987	3000	-	-	-	-	13822	-
Goianésia do Pará (PA)	3500	3000	3100	2329	2117	1198	1148	1000	-
Maju (PA)	500	30	32	33	30	30	30	30	30

Average

1381

Emissions were calculated using the equations in Module VMD0011: Estimation of Emissions from Market Effects (LK-ME), Section 5.2.

Table 28. Net GHG emissions due to leakage by market effects in year t

BVP	t	Start	End	Total	$\Delta C_{LK-ME,t}$ (tCO _{2e})
1	1	Jan 21, 2024	Dec 31, 2024	952	952
1	2	Jan 1, 2025	June 30, 2025	476	476

Finally, total leakage emissions ($\Delta C_{LK-Udef}$) are equal to the sum of emissions from activity displacement, market effects, and GHG emissions associated with leakage mitigation measures, and were calculated using Equation 49 of Module VMD0055.

Table 29. Net GHG emissions due to leakage from project activities in year t.

BVP	t	Start	End	$\Delta C_{LK-Udef}$ (tCO _{2e})	$\Delta C_{LK-AS, LB}$ (tCO _{2e})	GHG _{LK,E,t} (tCO _{2e})	$\Delta C_{LK-ME,t}$ (tCO _{2e})
1	1	Jan 21, 2024	Dec 31, 2024	952	0	0	952
1	2	Jan 1, 2025	June 30, 2025	476	0	0	476

3.2.3 GHG emissions reductions and carbon dioxide removals (VCS, 3.15, 4.1)

The total net GHG emission reductions from the unplanned deforestation prevention project activity ($NER_{Udef,t}$) were calculated using Equation 50 of Module VMD0055.

Table 30. Total net GHG emissions reductions.

BVP	t	Start	End	$\Delta C_{BSL,PA-Udef,t}$ (tCO _{2e})	$\Delta C_{MP,PA-Udef,t}$ (tCO _{2e})	$\Delta C_{LK-Udef,t}$ (tCO _{2e})	$NER_{Udef,t}$ (tCO _{2e})
1	1	Jan 21, 2024	Dec 31, 2024	132,838	0	952	131,886
1	2	Jan 1, 2025	June 30, 2025	67,798	0	476	67,322

The non-permanence risk analysis was applied to the Project using the AFOLU Non-Permanence Risk Tool v4.2, resulting in a score of 17 points. Therefore, it is considered that the Caiarara REDD+ Project presents a non-permanence risk of 17% for the monitored period.

Indicate the non-permanence risk classification (%)	17
Was the non-permanence risk report attached as an appendix or a separate document?	☒ Yes No ☐
For ARR and IFM projects with harvesting, state the Long-Term Average (LTA) in tCO ₂ e.	
Has the LTA been updated based on monitored data, if applicable?	☐ Yes No ☐
Indicate, in tCO ₂ e, the total expected GHG benefit to date.	140,104
If a loss has occurred (including a loss or reversal event), indicate the amount of tCO ₂ e lost:	There was no loss event in the monitored period

The number of credits to be held in the AFOLU collective reserve account is determined as a percentage of the total carbon stock benefits. The value of buffer credits for the monitored period was calculated using Equation 51 of Module VMD0055. The number of Verified Carbon Units (VCUs) for the years in which monitoring was carried out and submitted for verification was calculated using Equation 52 of the Module.

Table 31. Estimated buffer pool allocation and estimated VCU reduction.

BVP	t	Period		Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated VCU reduction (tCO ₂ e)
		Start	End					
1	1	Jan 21, 2024	December 31, 2024	132,838	0	952	22,583	109,303
1	2	Jan 1, 2025	June 30, 2025	67,798	0	476	11,526	55,796

Verified period	Estimated reductions/removals ex ante	Reductions/removals achieved	Percentage difference	Explanation for the difference
Jan 21, 2024 to Dec 31, 2024	92,588	131,886	-42.4%	During the monitored period, deforestation in the UDef PA

Jan. 1, 2025 to June 30, 2025	47,516	67,322	-41.7%	and leakage emissions were lower than estimated in the baseline scenario, indicating the project's effectiveness in reducing emissions.
Total	140,104	199,208	-42.2%	

3.3 Optional Criterion: Benefits of Adaptation to Climate Change

3.3.1 Activities and/or processes implemented for Adaptation (CCB, GL1.3)

During this monitoring period, the following activities were carried out to help communities and biodiversity adapt to the impacts of climate change:

Activity	Climate change risks	Project activities designed for adaptation to climate change	Project activities carried out for adaptation to climate change	Indicators
Strengthening of property surveillance and monitoring of deforestation and hot spots	* Increase in deforestation. * Damage caused by forest fires. * Increase in CO2 emissions. * Loss of biodiversity.	* Monitoring and surveillance in the project's area to prevent the area from being invaded for deforestation or illegal hunting, avoiding greenhouse gas emissions and biodiversity loss. * Satellite monitoring allows for the immediate identification of possible fire spots, generating a rapid response in the field to verify and mitigate possible damage. * The preservation of these areas helps maintain ecosystem resilience, especially for biodiversity, in the face of climate change challenges.	* During this monitoring period, a new surveillance company was hired to improve monitoring and recording processes, ensuring the traceability of the information generated. As a result, 323 hectares were conserved (of the 323 hectares planned for deforestation).	Deforested area Reference: Section 0
Biodiversity Conservation and Monitoring	* Population decline. * Species extinction.	* Monitoring of endangered and endemic species, enabling the implementation of a species conservation plan.	* During this monitoring period, a diagnosis of fauna and flora was carried out in relation to biodiversity conservation and monitoring activity.	Macaco-caiarara (<i>Cebus kaapori</i>). Reference: section 5.4.1

	* Loss of biodiversity.			
Development and strengthening of value chains	* Food insecurity. * Decrease in income generation. * Decreased productivity due to increased periods of drought.	* Through technical assistance and rural extension, it is possible to provide knowledge of technical and sustainable practices to address the consequences of climate change, such as the extension and intensification of the dry season already observed by community members.	* During this monitoring period, no activities were carried out to develop and strengthen value chains.	Number of education/training actions carried out
Promotion of education actions	* Environmental degradation. * Forest fires. * Loss of biodiversity due to illegal hunting.	* Raising awareness of the importance of wild flora and fauna, reducing illegal deforestation, hunting, and fishing practices. * Community members trained to take action on various environmental fronts, such as techniques to prevent the spread of forest fires and how to fight them.	* During this monitoring period, a lecture was given on the importance of drinking water consumption, water purification methods, and water pollution protection, which was an important action in terms of health, citizenship, and awareness.	Number of educational/training actions carried out. Reference: Section 4.3.1.

4 COMMUNITY

4.1 Net positive impacts on the community

4.1.1 Impacts on the Community (CCB, CM2.1)

The expected impacts were identified through the Socioeconomic Diagnosis conducted in the Project's region, involving local communities and farm workers. The social impact assessment was carried out using the methodology developed by Richards and Panfil (2011), Richards (2011), and Pitman (2011), which presents a matrix with a descriptive summary of the impacts mapped during field research in the

communities and with other agents in the project's surroundings. The matrix shows the group directly involved, the impact, the benefit, the costs and risks associated with each impact, and the likely changes in the group's well-being. The results of the mapping are presented below.

Community group	Communities located near the Project's Area that use the forest area.
Impact(s)	Positive - Development and strengthening of value chains through training and coordination to obtain technical assistance to encourage environmental practices and sustainable and adaptive agricultural production models. - Environmental awareness and permanence of families on their land; - Strengthening of historically marginalized groups. - Improvement in the well-being of stakeholders. Negatives - Loss of access to areas for subsistence hunting and fishing. - Loss of access to areas for extractive products in general.
Type of benefit/cost/risk	• Expected or existing impacts: - All impacts are considered expected, as they may occur as project's activities are implemented. • Direct and/or indirect: - All impacts are considered direct because they directly influence the knowledge and practices of community members for maintaining their families in their communities, protecting biodiversity, and applying productive techniques and social organization. • Benefits: - Increased technical training for community members to develop sustainable and adaptive practices that add value and diversity to their production and income generation; - Improvement in the living conditions of communities; - Greater environmental and social awareness of the environment in which they live;

	- Maintenance of families in the territory through the use of sustainable practices, access to technical training, social organization, and knowledge of biodiversity protection. • Costs: - Human and financial resources related to training, education, and technical assistance. • Risks: - Low participation of community groups in training processes; - Few families practicing sustainable production techniques; - Low engagement of families in collective and community activities in the pursuit of public policies and services. - Possible illegal and uncontrolled entry into forest areas of the Caiarara REDD+ Project to search for water, extractive products in general, and to hunt and fish.
Change in well-being	- Families remaining in the communities; - Gradual and/or sustainable decrease in the use of the project's area; - Improvement in family income; - Negative impact on the maintenance and subsistence system due to the use of some forest and non-forest products by families.

Community group	Workers at the Reflorestamento Água Azul I and II Farms (RAAI and RAAII).
Impact(s)	Positive - Strengthening of human capital through access to professional training and capacity building; - Increased efficiency in field surveillance operations.
Type of benefit/cost/risk	• Expected or existing impacts: - All impacts are considered expected, as they will occur as the project's activities are implemented. • Direct and/or indirect: - All impacts are positive and direct on workers, based on the application of training and capacity building, which directly influence their conduct, dynamics, and quality of work.

	• Benefits: - Increased effectiveness and productivity of farm activities related to the protection and monitoring of forest areas; - Training and capacity building to improve the daily activities of workers. • Costs: - Human and financial resources related to training and capacity building actions. • Risks: - Low engagement of workers during the project's activities.
Change in well-being	- Enhance the professional quality of workers; - Improvement in the quality and working conditions of employees.

4.1.2 Mitigation of Negative Impact on the Community (VCS, 3.19; CCB, CM2.2)

In the Socioeconomic Diagnosis conducted in the communities surrounding the Project, it was identified that some families accessed the project area for hunting, fishing, and extraction of forest products, such as wood, vines, and medicinal plants. Restricted access resulting from the protection of the area may impact the way of life and livelihoods of these families. To mitigate these impacts, the Caiarara REDD+ Project will implement sustainable productive alternatives through coordination with public institutions and local organizations, promoting the development of value chains, technical assistance, and adaptive agricultural practices, with the aim of reducing dependence on natural resources in the project area.

In addition, to ensure a transparent and collaborative relationship between the project's proponent and the communities, a communication and engagement plan has been implemented that includes channels for receiving feedback and a conflict resolution mechanism, with special attention to the safety and equitable participation of women and vulnerable groups. These measures follow the principles of equity and respect for local diversity, ensuring safe and accessible environments for all, in order to strengthen dialogue and stakeholder engagement, as well as encouraging community involvement in both the activities carried out and the preservation of forests.

The project does not employ or release hazardous substances, does not use pesticides or chemical fertilizers, and does not generate air pollutants, solid waste, or industrial effluents that affect air, water, or soil quality. If any environmental risk is identified, preventive actions will be taken in accordance with the precautionary principle, even in the face of scientific uncertainty.

During the monitoring period, the project took the first steps to mitigate the negative impacts identified. Recognizing the importance of addressing these issues, actions were initiated to establish a collaborative and transparent relationship with the affected communities, while planning sustainable alternatives to mitigate long-term impacts. As part of this effort, a communication plan was implemented, with the opening of official project channels and direct channels with stakeholders (see Section 2.3.10 for more

details). These channels aim to facilitate dialogue, clarify doubts, and promote stakeholder engagement in the Project. In addition, planning for a work plan to offer sustainable alternatives has begun, seeking to reconcile the needs of communities with conservation objectives.

These actions are aligned with the precautionary principle, adopting a preventive approach to avoid conflicts and minimizing socio-environmental impacts. In addition, they reinforce the project's commitment to maintaining and improving the identified High Conservation Value (HCV) attributes, promoting both environmental protection and dialogue with communities.

The project reaffirms its commitment to conducting these initiatives in an ethical and responsible manner, seeking to build trusting relationships with communities and generate mutual benefits over time. Additionally, mitigation measures will be evaluated periodically and, if necessary, adjusted and adapted according to new needs and conditions identified, thus ensuring the continued effectiveness of these actions in the well-being of the community and in environmental preservation on an ongoing basis.

4.1.3 Net positive community well-being (VCS 3.19; CCB, CM2.3, GL1.4)

The activities proposed by the Caiarara REDD+ Project encourage socioeconomic and sustainable development for the stakeholders involved, focusing on education and assistance actions to promote more sustainable agricultural practices and habits, to develop and strengthen value chains, and to engage through mechanisms of transparency, communication, and incentives for collective participation in the actions developed by the Project.

In the scenario without the Project, the context of low income and lack of technical assistance and rural extension services leads families to seek alternatives to increase their income from economic and subsistence activities practiced in an unsustainable and unplanned manner that can promote deforestation and forest degradation in the region.

The project provides for the coordination of technical assistance necessary for community members to adapt their agricultural and productive practices to climatic conditions, with an emphasis on sustainable production techniques that allow for greater efficiency in the use of water resources and adaptation to more severe periods of drought. The development and strengthening of value chains allows communities to diversify their sources of income, reducing their dependence on a single economic activity and thus increasing their capacity to adapt. Strengthening local governance ensures communication between stakeholders and the project, promoting activities that bring effective solutions that meet the needs of all stakeholders. Environmental education and the empowerment of historically marginalized groups are also essential to ensure that all segments of the community have the knowledge and tools to adapt to change.

These activities will seek to promote awareness of the impacts of climate change, as well as empower vulnerable groups to contribute meaningfully to decision-making and the implementation of adaptive practices, and are essential for raising awareness among communities about environmental challenges and providing them with the tools they need to adopt more sustainable practices, which directly contributes to the well-being of stakeholders by ensuring the protection of the natural resources on which

they depend. The project ensures that all community groups benefit positively, promoting improvements in their well-being and strengthening their resilience in the face of climate challenges.

During this monitoring period, the project initiated social activities and held a lecture on water and health with students from the Vila Mamorana community. The aim of the initiative was to raise awareness about the importance of drinking clean water, methods for cleaning and purifying water for consumption, and protecting water from pollution, thereby improving the well-being of communities, raising awareness about the importance of clean water, and educating families about adapting their daily habits.

4.1.4 Protection of High Conservation Values (CCB, CM2.4)

No High Conservation Value Attributes (HCVs) related to community well-being were identified in the Caiarara REDD+ Project area, as described in Section *Erro! A origem da referência não foi encontrada.*, due to the fact that the farms that make up the Project Area are private properties and are marked as off-limits. The assessment was conducted based on the participatory socioeconomic diagnosis carried out by the project, which considered criteria such as dependence on ecosystem services, use of the forest for subsistence, and cultural importance.

Nevertheless, the project ensures that no planned activity will be able to negatively affect any HCVs related to community well-being, should they be identified in the future. The project's actions are focused on forest conservation, community strengthening, environmental education, and sustainable production practices, and do not involve removals, forced territorial restrictions, or changes in access to vital resources.

The project maintains channels for ongoing dialogue with communities and participatory monitoring mechanisms, ensuring that any potential HCVs are promptly identified and protected, in accordance with CCB principles.

4.2 Other impacts on stakeholders

4.2.1 Mitigation of negative impacts on other stakeholders (VCS, 3.18, 3.19; CCB, CM3.2)

No negative impacts on other actors are expected in the context of the Project, a premise supported by socio-environmental safeguards already in place and the direct involvement of stakeholders in the project design. To reinforce this premise, participatory strategies have been implemented, ensuring the inclusion of stakeholders in the design of activities and decision-making. In addition, the project has a communication plan that includes clear procedures for conflict resolution. If this plan proves ineffective, adjustments will be made to the forms of communication and conflict referral processes, always seeking the best solution.

If unforeseen adverse effects are identified, the project provides for the application of corrective measures based on participatory monitoring mechanisms. The project continuously monitors social impacts, focusing on the early identification of possible risks to the well-being of stakeholders. The

communication plan includes formal protocols for receiving manifestations, handling complaints, and mediating conflicts, with periodic reviews to ensure effectiveness.

4.2.2 Net impacts on other stakeholders (VCS, 3.18, 3.19; CCB, CM3.3)

No other negative impacts on the well-being of other local stakeholder groups were identified, as the activities carried out in relation to the surrounding communities are based mainly on coordination with government agencies and other local institutions precisely to promote improved living conditions, greater access to public policies, and activities related to improving existing practices. The activities outlined and proposed for the Project seek only impacts that promote development, inclusion, and well-being for communities and other parties involved.

4.3 Community impact monitoring

4.3.1 Community Monitoring Plan (CCB, CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

This section presents the variables monitored in relation to the community, considering the scope of the Project. The monitoring variables are directly linked to the Project's objectives for the surrounding communities and property workers and to the expected products, results, and impacts identified in the project's causal model and related to community well-being. Monitoring the Project's impacts on communities and other stakeholders is an essential management tool, allowing for the evaluation of the effectiveness of activities in achieving the proposed objectives. The development of a monitoring system is based on the planned goals, supported by the construction of indicators to be collected, verification tools, and procedures for analyzing and evaluating results. When necessary, the system will indicate corrective measures or improvements to achieve the desired progress.

The Community Impact Monitoring Plan is based on process and outcome indicators, described below. In addition, this plan will be continuously reviewed, with evaluation and validation by stakeholders, ensuring its adaptability to the dynamics and needs of the communities involved.

During this monitoring period, three meetings and two educational activities were held as part of the activities within the scope of the Caiarara REDD+ Project Community.

- Meetings/gatherings held

The meetings held by the project involved the communities and farm workers, each with specific objectives and a fundamental role in the construction and planning of activities.

The first interaction took place between December 2023 and January 2024, during the socioeconomic diagnosis, an essential step in understanding the conditions and characteristics of the communities and workers involved. At that time, the project team held explanatory meetings, in which it presented the general objectives and proposals of the project, in addition to collecting data through interviews with the 11 mapped communities and farm workers, allowing for greater knowledge of the local context. The main purpose of this activity was to obtain a comprehensive view of the realities and needs of the community,

ensuring that the design of the project's activities was sensitive to local demands and reflected the specificities of each group.

Figure 6. Records of meetings and interviews conducted during Socioeconomic Diagnosis in the mapped communities. Source: STA internal archive.



Figure 7. Records of meetings and interviews conducted during Socioeconomic Diagnosis with workers from the RAI and RAI farms. Source: STA internal archive.



The second meeting was held in July 2024, during the feedback event with stakeholders, which consisted of meetings aimed at sharing the results and analyses obtained during the diagnosis and providing more information about the project. These meetings again involved the 11 mapped communities and farm workers and aimed not only to present the information collected, but also to reinforce the project's commitment to transparency and ongoing dialogue. During this stage, the team highlighted the planned activities and offered important additional information, seeking to strengthen the community's trust and engagement in the process. In addition, a questionnaire was administered to consult the community about the project's activities, suggest a name, and provide space for comments and other contributions. This stage was essential to ensure that all stakeholders understood the progress of the project and could actively participate in its development.

Figure 8. Photos of feedback meetings with stakeholders held with workers from the RAAI and RAAII farms. Source: Ambipar internal archive.



Figure 9. Photos of feedback meetings with stakeholders held with the mapped communities. Source: Ambipar internal archive.



Finally, in August 2024, a specific meeting was held with the community of Vila Mamorana, which stood out as a crucial moment for establishing a closer bond between the project team and local leaders. The main objective of this initial meeting was to understand the community's most urgent demands and deepen knowledge about its internal organization and priorities. In addition, the meeting was an opportunity to align expectations, establish a direct communication channel, and begin planning future activities in a collaborative manner. This moment was marked by an effort to build a solid foundation of trust and cooperation, fundamental elements for the success of the actions to be implemented.

Figure 10. Record of the meeting between the project team and leaders of Vila Mamorana in August 2024. Source: Dow internal archive.



These meetings played a strategic role in strengthening the relationship between the project and the communities, ensuring that activities were carried out in a participatory and transparent manner, aligned with local needs and realities.

Data/Parameter	Number of meetings/gatherings held
Data unit	Number
Description	Measurement of the number of meetings/gatherings held to contribute to the development and improvement of actions and activities related to the Project's social activities.
Description of measurement methods and procedures to be applied	All meetings or gatherings held were identified and recorded, including minutes, reports, and audiovisual recordings. In addition, stakeholder participation was recorded through signed lists and registration forms. The data and reports produced by Caiarara were stored in digital files throughout the Project's life cycle. Original (physical) reports, meeting minutes, and field notes produced during the execution of social activities were also stored.
Monitoring frequency	Annual.
Purpose of the data	The indicator provides data on the relationship between stakeholders and the project's activities. In addition, the data allows for the analysis and identification of trends, such as the most popular topics, most participatory groups, etc. This information can be used to adjust the strategy to improve future activities.
Monitored value	3 meetings/gatherings.

Comments

- **Education/training actions carried out**

On March 23, 2025, a lecture on water and health was held with students, family members, and teachers from the Paraíso Municipal Preschool and Elementary School in the Vila Mamorana community. The activity aimed to inform about the importance of drinking water consumption, methods of cleaning and purifying water for consumption, and protecting water from pollution, which is an important action for health, citizenship, and awareness. A total of 60 people, including 40 women, participated in the activity, which promoted improvements in the well-being of communities, awareness of the importance of clean water and education, and adaptation of families' daily habits. In addition, two water filters were raffled off to students.

Figure 11. Records of the lecture on water and health at the Vila Mamorana Community School on March 23, 2025.

Source: Dow internal archive.



During the monitoring period, training was also provided to form forest fire brigades, with the aim of enabling farm workers to prevent and fight fires in forest areas. The training was divided into two classes, providing participants with theoretical and practical knowledge on fire control techniques, safe handling of equipment, and safety measures. In addition, the activity sought to promote greater awareness of the environmental and social impacts caused by forest fires, highlighting the importance of preventive action as a tool for environmental preservation and property safety.

A total of 54 people participated in and benefited from the training, including two women, expanding their professional qualifications and strengthening their emergency response capabilities. Among the participants, two women stood out, whose presence demonstrates the importance of encouraging inclusion and diversity in activities traditionally associated with predominantly male audiences. Training these individuals not only contributes to the safety of farms and forest areas but also strengthens stakeholders by creating a group capable of dealing with risks that directly affect their daily lives and their environment.

Figure 12. Fire brigade training records in 2024. Source: Dow internal archive.



Data/Parameter	Number of education/training activities conducted
Data unit	Number
Description	Measurement of the number of actions carried out by the Project, contributing to the technical training and environmental awareness of the Project's stakeholders.

Description of the measurement methods and procedures to be applied	All education and training actions carried out were identified and recorded, including detailed documentation of these actions, such as activity reports, materials distributed, audiovisual records, and the topics covered. In addition, participants' interactions during the activities were recorded and feedback on the content and dynamics of the actions was collected using signed lists and registration forms. The data and reports produced by Caiarara were stored in digital files throughout the project's life cycle. Original (physical) reports, meeting minutes, and field notes produced during the execution of social activities were also stored.
Monitoring frequency	Annual.
Purpose of the data	To count how many actions were taken and what their themes were, in order to evaluate what is being done by the Project. This information can be used to adjust the strategy to improve future activities.
Monitored value	2 education/training actions
Comments	-

Data/Parameter	Number of people benefiting from the project
Data unit	Number
Description	Counting of any person who has benefited during the implementation and monitoring of the Project, through the planned actions and activities related to the social scope
Description of the measurement methods and procedures to be applied	The people who benefited during the project's actions were identified. This includes members of local communities, workers involved in the activities, and other impacted groups, such as training and workshop participants. The record was made through attendance lists, registration forms, and reports of participation in socio-productive and capacity building activities. Thus, it was possible to account for the people involved in the project's activities. The data and reports produced by Caiarara were stored in digital files throughout the project's life cycle. Original (physical) reports, meeting minutes, and field notes produced during the execution of social activities were also stored.
Monitoring frequency	Annual.
Purpose of the data	To monitor the evolution of participation and benefits generated by the project's activities that reach stakeholders.
Amount monitored	114 people benefited.

Comments

-

4.3.2 Disclosure of the Monitoring Plan (CCB, CM4.3)

The Community Monitoring Plan and all its results were publicly disclosed on the VERRA registration page and on the Biofilica Ambipar Environmental Investments S/A website. The verified monitoring results will be available in the public version of the report on the Verra website, ensuring free access to anyone interested.

Summaries of the main results were prepared in accessible language and presented to the communities via WhatsApp, the channel selected and validated by the community for communications. Printed materials were also made available at strategic locations (farm headquarters and community centers). Information about the Project was also disclosed on the project proponent's social media accounts, such as Instagram and LinkedIn.

In this way, it is hoped that all necessary efforts to provide full transparency and distribute key Project information and documents will be achieved.

4.4 Optional Criterion: Exceptional Benefits for the Community

Not applicable, the Caiarara REDD+ Project does not have the Community Gold Seal.

4.4.1 Short- and Long-Term Community Benefits (CCB, GL2.2)

Not applicable.

4.4.2 Marginalized and/or Vulnerable Community Groups (CCB, GL2.4)

Not applicable.

4.4.3 Net impacts on women (CCB, GL2.5)

Not applicable.

4.4.4 Benefit Sharing Mechanisms (CCB, GL2.6)

Not applicable.

4.4.5 Governance and Implementation Structures (CCB, GL2.8)

Not applicable.

4.4.6 Capacity Development of Small Producers/Community Members (CCB, GL2.9)

Not applicable.

5 BIODIVERSITY

5.1 Net positive impacts on biodiversity

5.1.1 Changes in biodiversity (VCS, 3.19; CCB, B2.1)

Change in Biodiversity	Reduction in unplanned deforestation.																						
Monitored Change	During the monitored period, the deforestation predicted in the baseline was greater than what actually occurred, meaning that there was a reduction in unplanned deforestation, both in the project area and in the leakage belt for the entire monitored period, which means that the REDD+ Project was effective in protecting forest cover. <table> <tr> <th rowspan="2">Year</th><th colspan="2">Project's Area</th><th colspan="2">Leakage Belt</th></tr> <tr> <th>Projected Deforestation (ha)</th><th>Deforestation occurred (ha)</th><th>Projected deforestation (ha)</th><th>Deforestation occurred (ha)</th></tr> <tr> <td>Jan/2024–Dec/2025</td><td>216</td><td>0</td><td>480</td><td>94</td></tr> <tr> <td>Jan/2025–Jun/2025</td><td>107</td><td>0</td><td>240</td><td>47</td></tr> </table> A total change of 323 ha in the Project Area and 720 ha in the Leakage Belt due to deforestation was predicted in the baseline (Section Erro! A origem da referência não foi encontrada.). However, according to the deforestation monitoring data collected over the years and already mentioned in this report (Sections 3.2.1 3.2.3), the total change during this monitoring period was zero hectares in the Project's Area and 141 ha in the Leakage Belt, representing 0% and 19.5% of the expected change, respectively. Deforestation below the baseline forecast consequently led to qualitative changes, such as the conservation and maintenance of forest integrity and existing ecological processes, which reduces soil erosion , contributes to water quality conservation, and maintains the genetic variability of plant and				Year	Project's Area		Leakage Belt		Projected Deforestation (ha)	Deforestation occurred (ha)	Projected deforestation (ha)	Deforestation occurred (ha)	Jan/2024–Dec/2025	216	0	480	94	Jan/2025–Jun/2025	107	0	240	47
Year	Project's Area		Leakage Belt																				
	Projected Deforestation (ha)	Deforestation occurred (ha)	Projected deforestation (ha)	Deforestation occurred (ha)																			
Jan/2024–Dec/2025	216	0	480	94																			
Jan/2025–Jun/2025	107	0	240	47																			

	animal species, important for resilience to pests and diseases, pollination, and climate regulation, thus reducing the risk of extinction and conserving forest cover, which acts as a protective shield against winds and storms. In addition, mitigation actions are proposed in the section Erro! A origem da referência não foi encontrada .so that biodiversity remains stable over time.
Justification for Change	The collection of deforestation data and verification of accuracy, via satellite images and in the field, are described in the section Erro! A origem da referência não foi encontrada .. In addition to monitoring forest cover via satellite imagery, other activities influenced the results obtained during the monitoring period: asset surveillance, which, in addition to inspection rounds on Dow farms, also includes fence maintenance, firebreak maintenance, and firefighting training; and the implementation of socio-environmental actions with communities (which permeate the "overarching theme" of the importance of sustainable use and maintenance of natural resources), such as the initial project meetings to consult and engage surrounding communities and workers, and the educational lecture on water and health held at a school in the Vila Mamorana community.

5.1.2 Mitigation actions (VCS, 3.19; CCB, B2.3)

To date, in this first monitoring period, no negative impacts on biodiversity resulting from project's activities have been identified. To monitor and improve existing knowledge about the species that inhabit the project's area, regular monitoring of fauna and flora will be implemented. These actions aim to provide comparative data between studies conducted in different periods, allowing for analysis of species dynamics, population abundance, and possible population growth or decline. Such monitoring is consistent with the precautionary principle, as it ensures the early identification of possible changes in ecosystems and guarantees the adoption of corrective measures, if necessary, to preserve the region's biodiversity.

Furthermore, through environmental education activities planned for stakeholders and the promotion of sustainable practices, it is hoped to achieve community involvement and awareness of the importance of these species in maintaining ecological balance, reducing potential threats. Finally, increased surveillance will prevent third-party activities such as illegal hunting and forest degradation, contributing to the effective protection of key species.

Therefore, no significant negative impacts on biodiversity are expected as a result of the project's activities. If, at any time, a negative impact on biodiversity resulting from the project's activities is identified, mitigation measures will be immediately adopted.

5.1.3 Net Positive Impacts on Biodiversity (VCS, 3.19; CCB, B2.2, GL1.4)

Species and habitat	The endemic or threatened species of fauna and flora found in the project area are: Fauna Macaco-caiarara (<i>Cebus kaapori</i>); Onça-pintada (<i>Panthera onca</i>); Tiriba-pérola (<i>Pyrrhura lepida</i>); Inhambu-galinha (<i>Tinamus guttatus</i>); Bacacu-preto (<i>Xipholena lamellipennis</i>); Ararajuba (<i>Guaruba guarouba</i>); Atelopus spumarius; Podocnemis unifilis; Chelonoidis denticulata; Flora Acapu (<i>Vouacapoua americana</i>); Cedro (<i>Cedrela fissilis</i>); Sumauma-da-terra-firme (<i>Eriotheca longipedicellata</i>); Maçaranduba (<i>Manilkara elata</i>); Itauba (<i>Mezilaurus itauba</i>); Louro canela (<i>Ocotea fragrantissima</i>). The main objective of the Caiarara REDD+ Project is to prevent deforestation. To this end, it implements actions to strengthen surveillance on farms and social activities that promote the preservation of fauna and flora species in the project's area. During the monitoring period, activities carried out included strengthening property surveillance and initiating coordination with communities. These initiatives contribute significantly to biodiversity conservation in the project's area, ensuring habitat integrity and the protection of endangered species. To date, no negative impacts on biodiversity have been identified, as the project's activities aim to prevent deforestation, indirectly benefiting local biodiversity.
Areas necessary for habitat connectivity	The project's area contains areas of Campinaranas, areas considered at risk of extinction due to sand and timber extraction. During the monitoring period, property surveillance activities contributed to the protection of these areas, preventing third parties from accessing the sites and damaging the habitat in any way. Thus, the project did not have negative impacts on the Campinarana areas, since its activities strengthened their existence and protected their biodiversity.

In terms of climate change adaptation benefits, the project contributes to maintaining the local climate, minimizing global climate change, and maintaining habitat connectivity, especially considering a larger

spatial scale such as the Project's Zone; in addition to protecting ecosystem services such as pollination, seed dispersal, gene flow, soil protection, hydrographic basins, among others.

5.1.4 High Conservation Values Protected (CCB, B2.4)

The project identified the presence of HCV 1, which represents the existence of concentrations of biological diversity, including endemic, rare, threatened, or endangered species that are significant at the global, regional, or national level; and HCV 3, which represents the existence of areas containing rare or threatened ecosystems.

The activities carried out by the Caiarara REDD+ Project during this monitoring period have had a positive impact on key species (high conservation value attribute 1), as they include activities to intensify patrolling of the project area, promoting the protection of forests and, consequently, the conservation of fauna and flora species, including those threatened with extinction. In addition, the activity also has positive impacts on the Campinaranas areas, considered threatened ecosystems (high conservation value attribute 3), since these are located on Dow's property, within the project's area.

Thus, no negative effects related to biodiversity were identified as a result of the activities proposed by the project on High Conservation Value Attributes. However, the project has adaptive response mechanisms in place in case unforeseen negative impacts are identified throughout the project's life cycle, such as reviewing the project's activities and strengthening mitigation actions to prevent, detect, and respond to potential risks to biodiversity, ensuring an adaptive management structure.

5.1.5 Species used (VCS, 3.19; CCB, B2.5, 2.6)

No species of fauna or flora were introduced as part of the activities of the Caiarara REDD+ Project.

5.1.6 Invasive species (VCS, 3.19; CCB, B2.5)

There are no invasive species in the Caiarara REDD+ Project area.

5.1.7 Exclusion of GMOs (CCB, B2.7)

No GMO species were used to generate GHG emission reductions or carbon dioxide removal in the Caiarara REDD+ Project.

5.1.8 Justification of Entries (VCS, 3.19; CCB, B2.8)

The Caiarara REDD+ Project did not use any fertilizers, chemical pesticides, biological control agents, or other inputs.

5.2 Impacts on external biodiversity

5.2.1 Negative impacts on external biodiversity (CCB, B3.1) and mitigation actions (CCB, B3.2)

Negative external impact	Mitigation measure(s)
Displacement of illegal activities outside the Project Area	To prevent the displacement of illegal hunting, fishing, and deforestation activities, the mitigation measure established is to engage rural communities surrounding the Project's Area through activities proposed by the Theory of Change, namely: Strengthening local governance; Developing and strengthening value chains; Improving the well-being of stakeholders; Promoting educational activities; and Strengthening inclusion, diversity, and equity. Through these activities, the Project seeks to offer sustainable production alternatives, encourage new environmentally responsible practices in the daily lives of communities, raise awareness about the protection of biodiversity and ecosystem services, and provide training and improve the well-being of surrounding residents, seeking greater engagement with the <i>Caiarara REDD+ Project</i>. The long-term goal is to promote changes in cultural habits, integrate sustainable practices, and prevent illegal deforestation and hunting, thereby strengthening environmental preservation and fostering a greater understanding of the importance of standing forests, protected local biodiversity, and the benefits of a balanced environment. During this monitoring period, the Project carried out its first engagement activities with the communities in the Project's area, starting with initial diagnostics, a feedback event with stakeholders, the opening of project communication channels, and an educational initiative on the importance of water protection for health.

5.2.2 Net benefits of external biodiversity (VCS, 3.19; CCB, B3.3)

The main positive impact outside the Project Area is the promotion of biodiversity through the maintenance of large forest remnants that can serve as a refuge and protection for threatened species and ecosystems. Thus, despite the possible displacement of activities and disturbances outside the Project's Area, which may cause occasional negative impacts, the connectivity of the landscape, the promotion of gene flow between forest areas, the protection of ecosystem services such as pollination, seed dispersal, and protection of soils and hydrographic basins, the maintenance of the local climate, the maintenance of natural habitats, and the *in situ* monitoring of threatened and endemic species are benefits that justify the presence of the Caiarara REDD+ Project.

If any negative impact on external or internal biodiversity is identified throughout the project's life cycle, the Caiarara REDD+ Project is committed to proposing measures to mitigate the impact in the best possible way.

5.3 Biodiversity Impact Monitoring

5.3.1 Biodiversity Monitoring Plan (CCB, B4.1, B4.2, GL1.4, GL3.4)

This section presents the monitored variables related to biodiversity, the types of measurement, the sampling methods related to the Project's activities within the scope of biodiversity, and the values monitored during this monitoring period. The monitoring variables are directly linked to the Project's objectives for biodiversity conservation and to the expected outputs, outcomes, and impacts identified in the Project's causal model related to biodiversity conservation.

Flora

In January 2024, a flora diagnosis was carried out in the project's area by STA Soluções Técnicas e Serviços de Engenharia Ambiental LTDA. The characterization of floristic groups was conducted through an inventory in the project's area and consultation of bibliographic data. The inventory was carried out in conjunction with the carbon stock study, using random plots set up to identify the individuals present at the site. The species recorded, both in primary and secondary sources, were listed with their scientific name, common name, and respective family. In addition, the presence of endangered, rare, and/or endemic species was analyzed, considering global (IUCN) and national (MMA; CNCFLORA) interests.

The indicators presented below were developed based on the results of the diagnosis, of which the objective was to support the project's activities, understand the dynamics of the region, and map existing conditions. This diagnosis represents the first period of identification and serves as an initial study to establish a baseline, which will be used as a reference for comparisons in future monitoring periods.

Data / Parameter	Species richness with DBH > 10 cm and Shannon-Weaver indices
Area	Project's Area and Campinarana
Data unit	Number
Description	Number of species with DBH > 10 cm in the Caiarara REDD+ project area
Description of measurement methods and procedures to be applied	To monitor the number of species with DBH > 10 cm in the project's area, all trees present in the selected transects were identified in the field. For identification, five plots were used in the forest and three plots in the Campinarana. In the forest, the plots were divided into five transects, measuring 20 x 100 m. In the Campinarana, five 10 x 10 m subplots were installed in each plot, totaling 15 subplots where the inventory was carried out. All species with DBH > 10 cm were recorded on the field form to later calculate the species richness in the area.

Monitoring frequency	At each verification				
Purpose of the data	The data will allow for the analysis and identification of the richness of floristic species existing in the area, in addition to comparison with each data collection, making it possible to identify a decrease or increase in biodiversity.				
Monitored value	<table border="1"> <thead> <tr> <th>Year</th><th>Monitored value</th></tr> </thead> <tbody> <tr> <td>2024</td><td>172 species of flora</td></tr> </tbody> </table>	Year	Monitored value	2024	172 species of flora
Year	Monitored value				
2024	172 species of flora				
Comments	-				

Data / Parameter	Number of flora species with some degree of threat or endemic to Pará.				
Area	Project's area and Campinarana				
Data unit	Number				
Description	Estimate the number of species with some degree of threat or endemic to Pará based on the identification of species carried out in the field during the campaign.				
Description of measurement methods and procedures to be applied	Identification was carried out by cross-referencing species information obtained in the field with the IUCN lists for threatened species and the Pará lists for endemic species. This made it possible to verify which species identified in the project's area are classified as threatened or endemic to the region. The information obtained was transferred to an Excel document for storage and comparison.				
Monitoring frequency	At each verification				
Purpose of the data	The data allows us to identify the number of species with some degree of threat and/or endemic species existing in the project's area.				
Monitored value	<table border="1"> <thead> <tr> <th>Year</th><th>Monitored value</th></tr> </thead> <tbody> <tr> <td>2024</td><td>Critically endangered (CR) - 1 species Endangered (EN) - 2 species Vulnerable (VU) - 4 species Endemic – 1 species</td></tr> </tbody> </table>	Year	Monitored value	2024	Critically endangered (CR) - 1 species Endangered (EN) - 2 species Vulnerable (VU) - 4 species Endemic – 1 species
Year	Monitored value				
2024	Critically endangered (CR) - 1 species Endangered (EN) - 2 species Vulnerable (VU) - 4 species Endemic – 1 species				
Comments	-				

Data / Parameter	Density of individuals of <i>Vouacapoua americana</i> (Acapu).
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Area	Project's area				
Data unit	Number				
Description	Density of individuals of the species <i>Vouacapoua americana</i> (Acapu) identified in each monitoring campaign in the Project's Area				
Description of the measurement methods and procedures to be applied	To monitor the density of the species, the density of individuals of <i>Vouacapoua americana</i> (Acapu) identified in the respective monitoring campaign was highlighted from the results of the flora monitoring.				
Monitoring frequency	At each verification				
Purpose of the data	The data allows the density of individuals of the species <i>Vouacapoua americana</i> (Acapu) to be identified, assisting in the analysis of population trends in the Project's Area. This indicator is related to the Gold Biodiversity Seal.				
Monitored value	<table border="1"> <thead> <tr> <th>Year</th><th>Monitored value</th></tr> </thead> <tbody> <tr> <td>2024</td><td>33.80 individuals/ha</td></tr> </tbody> </table>	Year	Monitored value	2024	33.80 individuals/ha
Year	Monitored value				
2024	33.80 individuals/ha				
Comments	-				

Data/Parameter	Density of individuals of <i>Ocotea fragrantissima</i> (Louro-canela).				
Area	Project's area				
Data unit	Number				
Description	Density of individuals of the species <i>Ocotea fragrantissima</i> (Louro-canela) identified in each monitoring campaign in the Project's Area				
Description of the measurement methods and procedures to be applied	To monitor the density of the species, the density of individuals of <i>Ocotea fragrantissima</i> (Louro-canela) identified in the respective monitoring campaign was highlighted from the results of the flora monitoring.				
Monitoring frequency	At each verification				
Purpose of the data	The data allows the density of individuals of the species <i>Ocotea fragrantissima</i> (Louro-canela) to be identified, assisting in the analysis of population trends in the Project's Area. This indicator is related to the Gold Biodiversity Seal.				
Monitored value	<table border="1"> <thead> <tr> <th>Year</th><th>Monitored value</th></tr> </thead> <tbody> <tr> <td>2024</td><td>0.40 individuals/ha</td></tr> </tbody> </table>	Year	Monitored value	2024	0.40 individuals/ha
Year	Monitored value				
2024	0.40 individuals/ha				
Comments	-				

Data / Parameter	Density of individuals of <i>Manilkara elata</i> (Maçaranduba).	
Area	Project's area	
Data unit	Number	
Description	Density of individuals of the species <i>Manilkara elata</i> (Maçaranduba) identified in each monitoring campaign in the Project's Area	
Description of the measurement methods and procedures to be applied	To monitor the density of the species, the density of <i>Manilkara elata</i> (Maçaranduba) individuals identified in the respective monitoring campaign was highlighted from the flora monitoring results.	
Monitoring frequency	At each verification	
Purpose of the data	The data allows the density of individuals of the <i>Manilkara elata</i> (Maçaranduba) species to be identified, assisting in the analysis of population trends in the Project's Area. This indicator is related to the Gold Biodiversity Seal.	
Monitored value	Year	Monitored value
	2024	5.40 individuals/ha
Comments	-	

Fauna

In January 2024, a fauna diagnosis was conducted in the project area to identify the species that inhabit the forest. This was carried out by Ambiens Resolve Inovação Sustentabilidade e Meio Ambiente LTDA, which conducted the study for the avifauna, mastofauna and herpetofauna groups.

Each fauna group was identified using available bibliography and field identification, which was carried out using different methodologies for each class. For the avifauna group, field sampling was carried out using MacKinnon's quantitative list technique, which consists of drawing up successive lists of species numbers during walks along predefined routes¹⁵. In this way, it is possible to carry out an inventory of species and make comparisons of spatial differences.

For medium and large mammals, the active search technique was used. Visual records were made for species that have diurnal habits, such as arboreal species. For species with nocturnal habits, indirect

¹⁵ RIBON, R. 2010. Bird sampling using Mackinnon's checklist method. In: Ornithology and Conservation. Applied Science. Research and Survey Techniques. (eds. Von Matter, S., Straube, F., Accordi, I., Piacentini, V. & Cândido-Junior, J.). Technical Books, Rio de Janeiro, pp. 33-44

sampling techniques are appropriate, with identification based on footprints, feces, burrows, and scavenge marks¹⁶. In addition, camera traps were also installed to record species present at the site.

Finally, for the herpetofauna group, active searching was also used as a methodology for identification, conducting visual inspections in locations with the greatest potential to harbor species representative of this group, covering pre-existing trails and bodies of water. In addition to visual inspections, auditory detection of anuran amphibians was also carried out.

The following indicators were constructed based on the diagnosis. The results of this initial identification serve as an initial study to establish a baseline, which will be used as a reference for comparisons in future monitoring periods.

Data / Parameter	Fauna species richness				
Data unit	Number				
Description	Number of species per fauna group (avifauna, mastofauna, and herpetofauna)				
Data source	Data collection in field campaigns				
Description of measurement methods and procedures to be applied	The richness of fauna species has different methodologies for each group, which are: Avifauna: data collection can be done using MacKinnon's quantitative list technique. This technique was developed for rapid surveys in tropical environments rich in bird species. Mastofauna: data collection can be done through camera trap records, sightings, and identification of footprints and excrement. Herpetofauna: data collection can be done by active search, i.e., visual and auditory contacts.				
Monitoring frequency	At each verification				
Purpose of the data	The data allows for the identification of the diversity and abundance of species existing in the project's area.				
Monitored value	<table border="1"> <thead> <tr> <th>Year</th><th>Monitored value</th></tr> </thead> <tbody> <tr> <td>2024</td><td> Avifauna: 124 species Mastofauna: 24 species Herpetofauna: 23 species </td></tr> </tbody> </table>	Year	Monitored value	2024	Avifauna: 124 species Mastofauna: 24 species Herpetofauna: 23 species
Year	Monitored value				
2024	Avifauna: 124 species Mastofauna: 24 species Herpetofauna: 23 species				
Comments	-				

¹⁶ PARDINI, R., DITT, E.H., CULLEN, Jr. L., BASSI, C., RUDRAN, R., 2003. Rapid survey of medium and large mammals. p. 181-201. In: Cullen Jr., Laury, Rudran, Rudy, Valladares-Padua, Cláudio (Eds.). Methods of study in conservation biology and wildlife management. Ed. UFPR. 665 p.

Data / Parameter	Number of fauna species with some degree of threat or endemic to the Belém Endemism Center				
Area	Project's Area				
Data unit	Number				
Description	Estimate the number of species with some degree of threat or endemic to the Belém Endemism Center based on the identification of species carried out in the field during the campaign.				
Description of measurement methods and procedures to be applied	Identification was carried out by cross-referencing species information obtained in the field with the IUCN and Pará lists. This made it possible to verify which species identified in the project area are classified as threatened according to the IUCN or endemic to the region according to the state list. The information obtained will be transferred to an Excel document for storage and comparison.				
Monitoring frequency	At each verification				
Purpose of the data	The data allows us to identify the number of species with some degree of threat or endemism existing in the project's area.				
Monitored value	<table border="1"> <thead> <tr> <th>Year</th><th>Monitored value</th></tr> </thead> <tbody> <tr> <td>2024</td><td> Critically endangered (CR) - 1 species Vulnerable (VU) - 9 species Endemic to the Belém Endemism Center - 2 species </td></tr> </tbody> </table>	Year	Monitored value	2024	Critically endangered (CR) - 1 species Vulnerable (VU) - 9 species Endemic to the Belém Endemism Center - 2 species
Year	Monitored value				
2024	Critically endangered (CR) - 1 species Vulnerable (VU) - 9 species Endemic to the Belém Endemism Center - 2 species				
Comments	-				

Data / Parameter	Number of records of <i>Cebus kaapori</i> (Macaco-caiarara)
Area	Project's area
Data unit	Number
Description	Number of records of the <i>Cebus kaapori</i> (Macaco-caiarara) identified in each monitoring campaign in the Project's Area
Description of the measurement methods and procedures to be applied	To monitor the species indicator, the number of records of <i>Cebus kaapori</i> (Macaco-caiarara) individuals identified in the respective monitoring campaign was highlighted from the fauna monitoring results.
Monitoring frequency	At each verification

Purpose of the data	The data allows the number of individuals of the <i>Cebus kaapori</i> (Macaco-caiarara) species to be identified, assisting in the analysis of population trends in the Project's Area. This indicator is related to the Gold Biodiversity Seal.	
Monitored value	Year	Monitored value
	2024	2 records
Comments	-	

5.3.2 Disclosure of the Biodiversity Monitoring Plan (CCB, B4.3)

The Biodiversity Monitoring Plan and all its results were publicly disclosed on the VERRA registration page and on the Biofilica Ambipar Environmental Investments S/A website. The verified monitoring results will be available in the public version of the report on the Verra website, ensuring free access to anyone interested.

Summaries of the main results were prepared in accessible language and presented to the communities via WhatsApp, the channel selected and validated by the community for communications. Printed materials were also made available at strategic locations (farm headquarters and community centers). Information about the Project was also disclosed on the project proponent's social media accounts, such as Instagram and LinkedIn.

In this way, it is hoped that all necessary efforts to provide full transparency and distribute key Project information and documents will be achieved.

5.4 Optional Criterion: Exceptional Benefits for Biodiversity

5.4.1 Population trends of trigger species (CCB, GL3.2, GL3.3)

Trigger Species	<i>Cebus kaapori</i> (Macaco-caiarara)
Scenario with project	The Macaco-caiarara, classified as "Critically Endangered" (CR) by the IUCN (2023), faces a continuous decline in the population of mature individuals. This primate is endemic to the Belém Endemism Center and is among the most threatened in the world due to habitat loss and pressure from human activities such as deforestation and hunting. Despite this, the population present in the Caiarara Project's area was not known before monitoring activities began.

	The first fauna and flora diagnosis conducted as part of the project identified the occurrence of this species in the project's area for the first time. This detailed survey provided an initial record of the population, which will serve as a reference for future studies. In addition, the diagnosis provided valuable information about the habitats used by the species, which is essential data for guiding the project's actions.
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Trigger Species	<i>Vouacapoua americana</i> (Acapu)
Project scenario	<i>Vouacapoua americana</i>, known as acapu, is classified as "Critically Endangered" (CR) by the IUCN (2023) and has a continuously declining population trend. This tree, endemic to the tropical forests of the Amazon region, is highly valued for its hard and resistant wood, which makes it a target for intense logging. In addition, habitat loss and deforestation are additional threats to the species. Before the project began, there was no detailed data on the occurrence of the species in the project's area. The initial diagnosis of fauna and flora carried out by the project revealed the presence of the acapu in the area for the first time, establishing a starting point for future monitoring and studies. In addition to recording the existing population, the survey generated important information about the ecological conditions and preferred habitats of the species, data that will support the project's conservation strategies.

Trigger Species	<i>Ocotea fragrantissima</i> Ducke (Louro-canela)
Project scenario	<i>Ocotea fragrantissima</i> Ducke (Louro-canela), a tree species widely distributed in the tropical forests of the Brazilian Amazon, is classified as "Endangered" (EN) by the IUCN (2023), with a population trend of continuous decline. Known for its ecological value, the species contributes to the dynamics of tropical forests, serving as a food source for local fauna through its aromatic fruits. However, it is strongly threatened by logging and habitat loss due to deforestation.

	The initial diagnosis of fauna and flora carried out by the project revealed the presence of the species in the area for the first time, establishing a starting point for future monitoring and studies. In addition to recording the existing population, the survey generated important information about the ecological conditions and preferred habitats of the species, data that will support the project's conservation strategies.
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Trigger Species	<i>Manilkara elata</i> (Maçaranduba)
Project scenario	<i>Manilkara elata</i>, a large tree widely known for its dense and durable wood, is classified as "Endangered" (EN) by the IUCN (2023), showing a trend of continuous population decline. In addition to its wood, Maçaranduba plays a crucial ecological role, providing fruit that serves as food for various animal species. Despite its wide distribution in the Amazon region, excessive logging and rapid deforestation have significantly reduced its populations. The initial diagnosis of fauna and flora carried out by the project revealed the presence of the species in the area for the first time, establishing a starting point for future monitoring and studies. In addition to recording the existing population, the survey generated important information about the ecological conditions and preferred habitats of the species, data that will support the project's conservation strategies.